



MACRO INVESTING ASSOCIATION

FALL Lecture Guide



Agenda



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Introduction to Macro Investing Association

MIA is an investment club focused on investment research, financial analysis, and macroeconomic strategy across various asset classes

- **Consistently Attend Meetings:** Attendance is crucial, as each session builds on prior knowledge
- **Engage with Content:** Read assigned materials, track market trends, and develop insights
- **Ask Thoughtful Questions:** Challenge ideas, seek clarity, and contribute to discussions
- **Demonstrate Initiative:** Go beyond the basics, bring forward your own ideas, and engage with club leadership

How to Stand Out & Get Promoted:

- **Consistent Attendance:** Engage actively in our meetings and discussions
- **Showing Interest:** Keep up with assigned readings, market updates, and research tasks
- **Asking Questions & Contributing:** Participate in discussions, challenge ideas, and bring fresh perspectives
- **Demonstrating Initiative:** Go beyond the material, propose trade ideas, and take on small leadership tasks
- The best-performing analysts will be promoted to PM roles where they will lead research efforts oversee investment theses

Format and Goals



Club Goals

- Help analysts develop real-world investment skills
- Gain hands-on experience in investment research, financial analysis, and macroeconomic strategy
- Connect with industry professionals and like-minded students
- Learn and explore career opportunities in finance

Source:

Note:



Introduction to Macro Investing

Macro Investing

What is Macro Investing?

Macro Investing is type of investing which involves having a **top-down approach** towards the financial markets.

It mainly involves looking at **large scale economies and geo-political trends** to make investment decisions across a wide range of asset classes including equities, fixed-income, commodities and currencies.

Most macro investors allocate capital across

- Equities(stocks)
- Bonds(bonds, interest rate swaps)
- Currencies(FX markets)
- Commodities(oil, gold, wheat etc)
- Alternative Assets(real estate, crypto)



Discretionary Macro

Discretionary Macro involves relying on **fundamental analysis, news and judgement of human decisions.**

Ex George Soros

Systematic Macro

Systematic Macro involves using **quantitative models, algorithms and data-driven strategies**

Ex Ray Dalio

Event Driven and Thematic Macro

Thematic Macro focuses on a specific theme such as **inflation trades, emerging markets boom etc.**

Event-Driven Macro focuses on positioning for **major geo-political events, central banks etc.**

Source: Bloomberg, CapIQ, Factset

Note:



Macro Economic Indicators

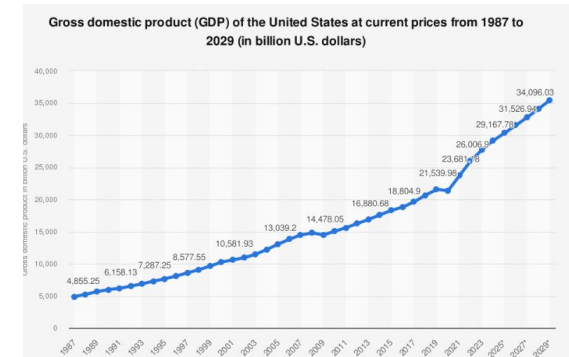
Macro Economic Indicators

Introduction

Macro economic indicators **are key economic indicators** that help measure the overall health and performance of the economy.

They are widely used by **investors, policy makers and analysts** to assess economic trends to make more informed decision on the broader economy.

USA GDP



GDP

The total **value of goods and services produced in a country.**

GDP = Consumption + Investment + Government Spending + Net Exports

CPI

Measures inflation by **tracking the price changes of consumer goods and services**

Markets preferred measure of inflation

CPI = Cost of market goods in current period / Cost of market goods in previous year * 100

PCE

Measures how much US households spend on goods and services.

FED's **preferred measure of inflation**

Expenditure = Price x Consumption

Unemployment Rate

The unemployment rate measures **the share of workers who do not currently have a job** but are actively looking for a job.

Rate = Unemployed individuals / Labor Force * 100

Federal Reserve

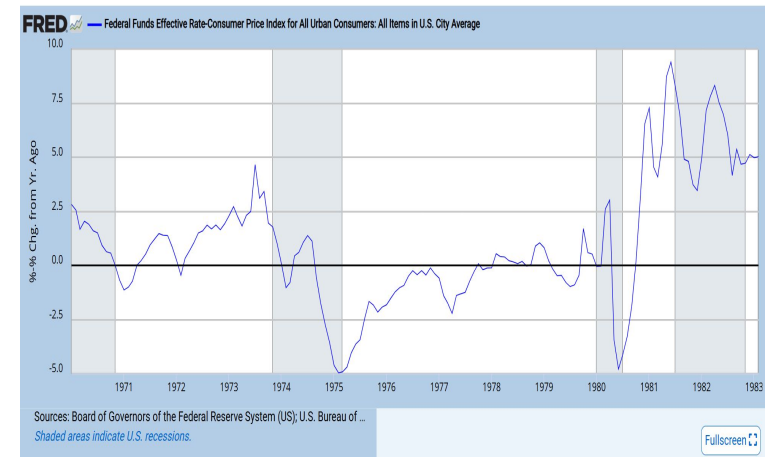
Introduction

- FED is the central banking system of the united states established in 1913 to help the country's financial system in response to the banking crisis.
- The FED runs with a dual mandate which is keeping unemployment and inflation rate stable.
- **The President of the United States nominates candidates for the Board of Governors and then the nominees are questioned and approved by the Senate Banking Committee**

Interest Rates

- FED controls the cost of capital in the country though interest rates. The FED either hikes or cuts rates based on how the economy is performing.
- **1 bps = 0.10%**
- The FED interest rates have a broader impact across wide range of asset classes. As the FED impacts the cost of capital which directly and indirectly affects the spending and consumptions patters which have a strong impact on the economy

FED Interest Rates



Source: FRED

Note:

FED Tools



FED has many tools at its disposal to control the money supply in the markets

- **FFR(Federal Funds Rate)** : The Fed's main policy rate which influences short term interest rates.
 - High Interest Rates -> **Slow borrowing** & Spending
- Low Interest Rates -> **Encourages borrowing** & Spending
- **Open Market Operations** : The FED buys or sells US Treasury securities to control the money supply and influence interest rates
 - Buying Bonds -> **Injects money** into the banking system
 - Sells Bonds -> **Drains money supply** from the markets
- **Balance Sheet Policy** : The FED influences the long term interest rates by adjusting its balance sheet. The two methods are called Quantitative Easing and Quantitative Tightening
 - Quantitative Tightening
 - The FED **reduces its balance sheet** by letting bonds mature without reinvesting
 - **Reduces liquidity** and borrowing costs
 - Used to **fight inflation** and prevent asset bubbles
 - Quantitative Easing:
 - FED buys **large amount of Treasuries** and mortgage based securities
 - Increases the **money supply and lowers long term interest rates**
 - Widely used during **2008 and Covid-19**, also widely used in Japan

FED Current Interest Rates



FED Balance Sheet(Assets)



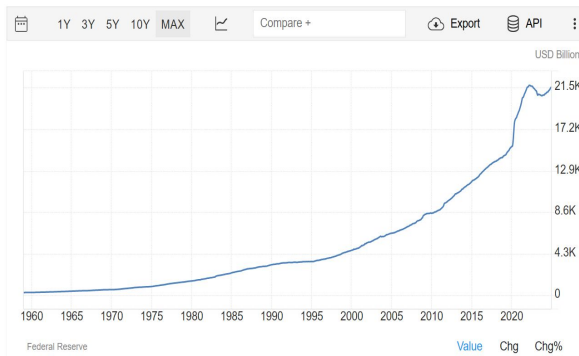
Source: FRED, Trading Economics

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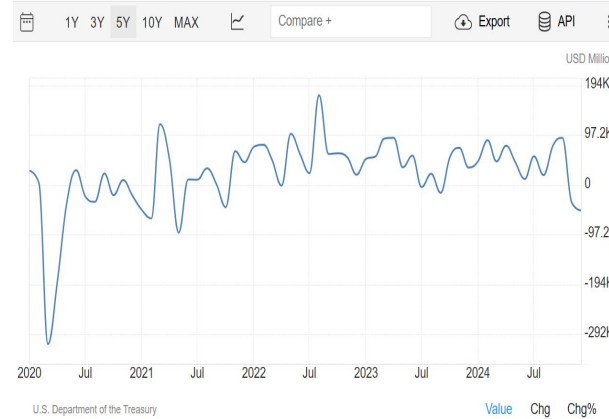
Examples

FED has strong impact on other financial indicators

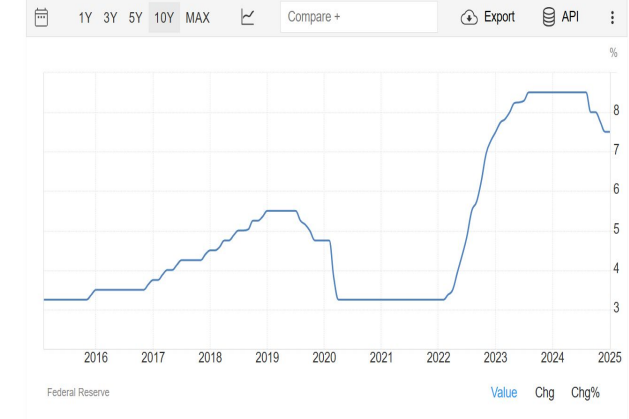
M2 Money Supply



Foreign Bond Purchases



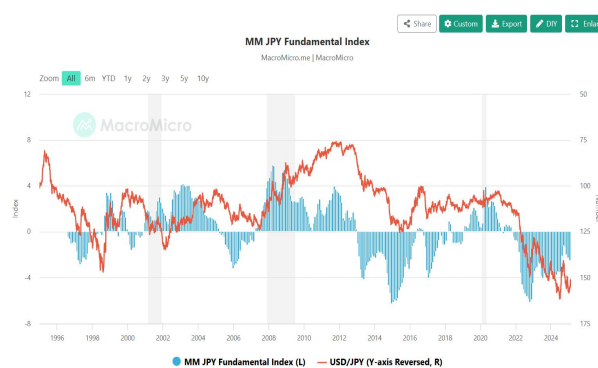
Bank Prime Lending Rate



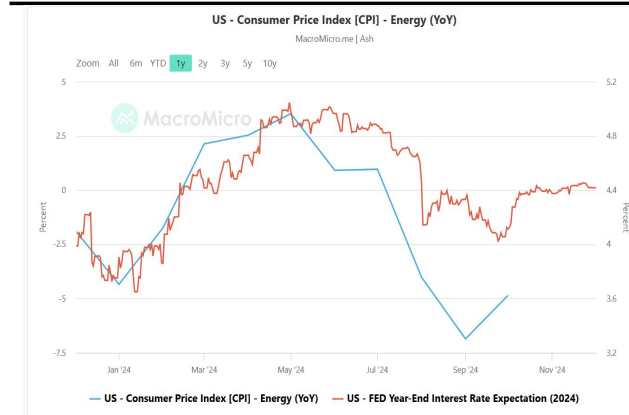
10 year bond yield



USD-JPY Currency Price



US Energy CPI vs FED Interest Rate Expectations



Source: FRED, Yahoo Finance, Macro Micro

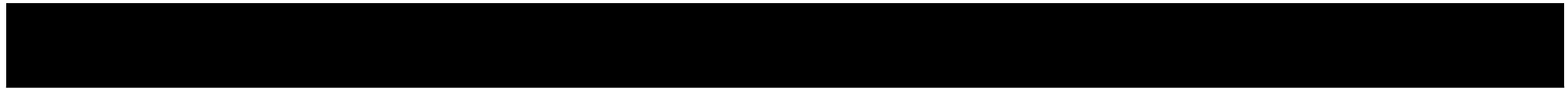
Note: Macro Micro is macroeconomic database for charts and info

MACRO INVESTING ASSOCIATION

Lecture- 2



Agenda



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The Structure of the FED	5
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Introduction to Central Banks

- The primary monetary authority of a nation (FED, BOJ, ECB, BOE, PBOC, RBI)
- Independent or semi-independent institution that manages the supply of money and credit
- Acts as a regulator of commercial banks (bank of bank)

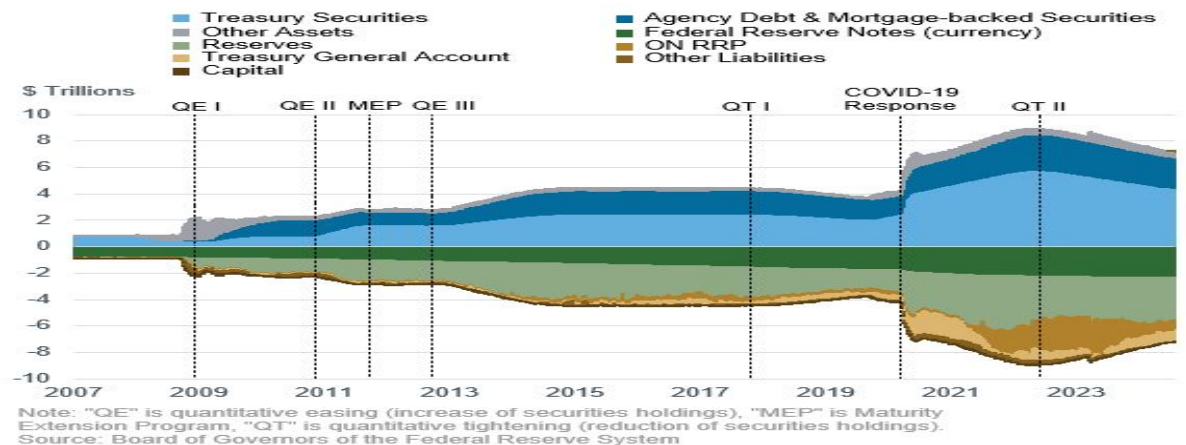
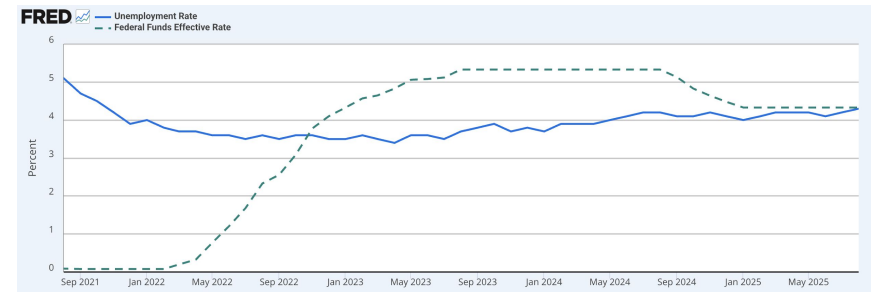


Mandates

- Price Stability
- Maximum Employment
- Financial Stability (Implicit)

Functions

- Conduct monetary policy
- Issue currency
- Lender of last resort
- Manage reserves & intervene in FX markets



The Structure of the Federal Reserve System

- The Federal Reserve System has a decentralized structure
- The U.S. was divided into 12 Districts, each with a separately incorporated Reserve Bank
- Each of the 12 Reserve Banks is separately incorporated and has a nine-member board of directors

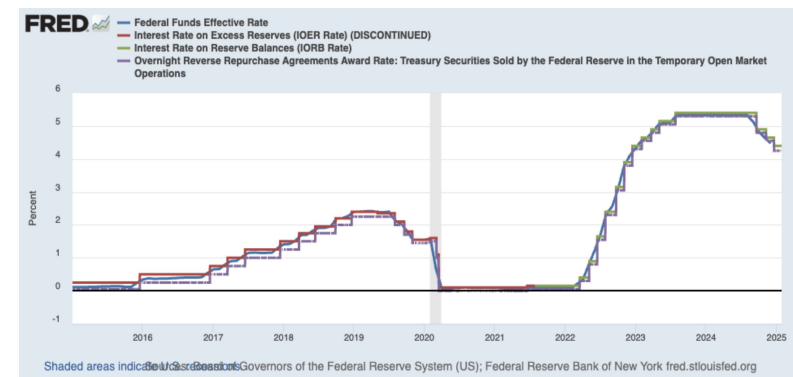


Federal Open Market Committee(FOMC)

- FOMC has 12 members: 7 Governors, the New York Fed President and 4 rotating Reserve Bank Presidents
- FOMC meets eight times per year to determine the stance of monetary policy
- Fed manages policy by targeting the federal funds rate through mainly open market operations

Federal Funds Rate (FFR)

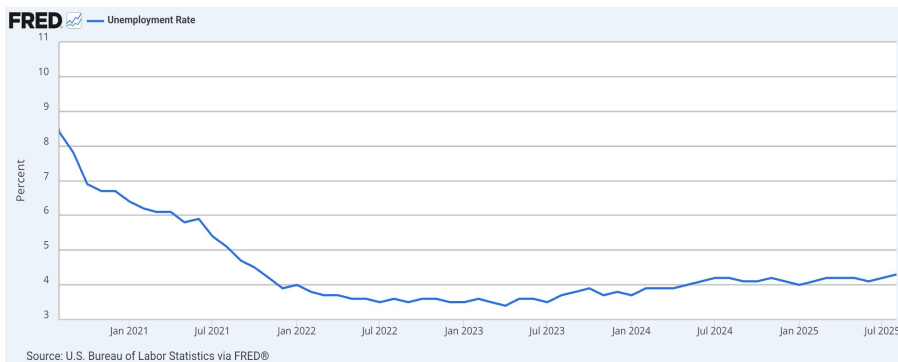
- Overnight interbank lending rate for reserves
- FOMC announces a target range to guide liquidity conditions
- Interest on Reserve Balances (IORB) establishes effective floor for banks
- Overnight Reverse Repo (ON RRP) extends floor to nonbank institutions
- Discount rate to set the upper bound of FFR



Key Tools and Indicators

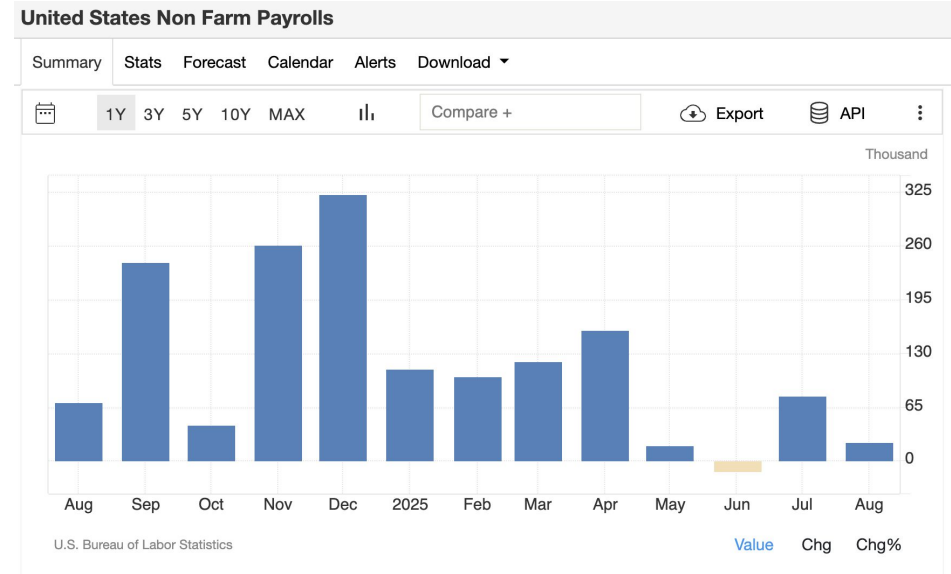
Policy Tool

- Federal Funds Rate: main short-term policy rate
- Open Market Operations: buy/sell Treasuries to adjust reserves
- Balance Sheet Policies (QE/QT): influence long-term yields & liquidity
- QE: Central bank buys long-term government securities and Inject Liquidity , QT: Central bank reduces bond holdings to withdraw liquidity
- Forward Guidance: shape market expectations via communication



Indicators

- CPI: Headline measure of consumer inflation based on a fixed basket of goods and services
- PCE: inflation gauge that adjusts for changing spending patterns; the Fed's preferred 2% target measure
- Unemployment Rate: Share of the labor force actively seeking but without jobs
- NFP (Nonfarm Payrolls): Monthly BLS jobs report covering total employment change
- ADP National Employment Report: Private payroll estimate released before NFP



Transmission of Policy to Financial Markets

Policy Rate

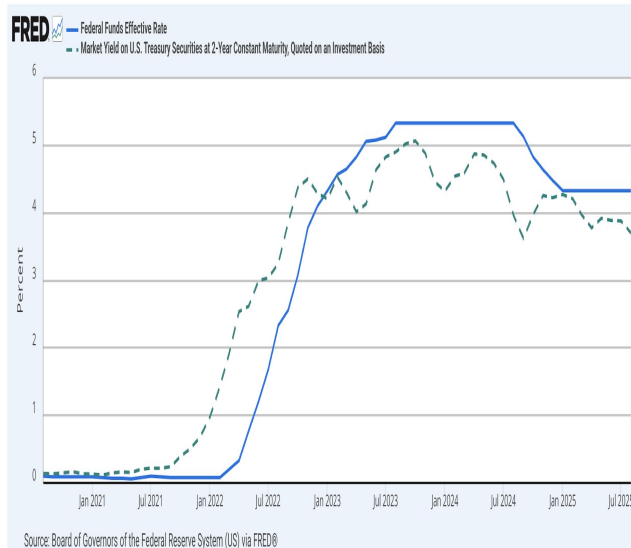
Adjustments in the FFR transmit through money markets to the **yield curve**, recalibrating borrowing costs economy-wide

Discount Rate

Shifts in policy rates alter discount factors, compressing present values of cash flows and reshaping equity valuations

Liquidity (QE/QT)

Balance sheet interventions modulate reserve availability, suppressing or elevating term premia and influencing risk appetite

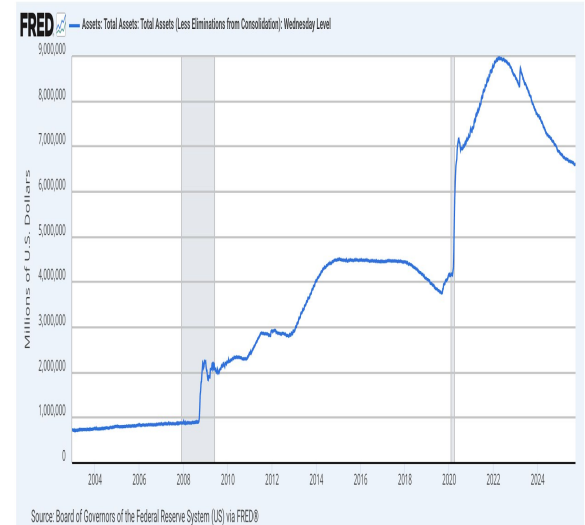


S&P500 Earnings Yield vs 10Y Treasury Yield

www.currentmarketvaluation.com



CURRENT MARKET VALUATION



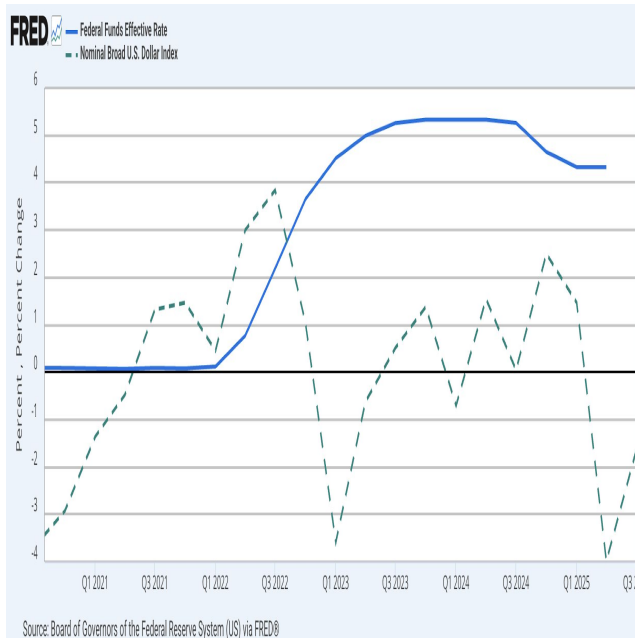
Source: FRED, Seeking Alpha

Note:

Transmission of Policy to Financial Markets

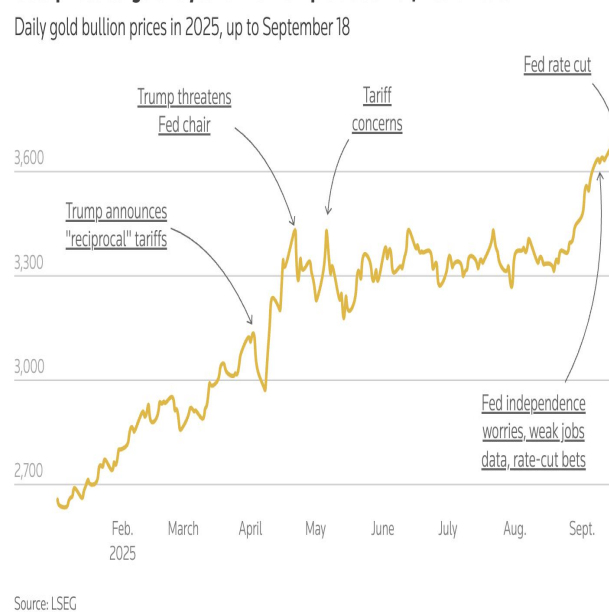
FX & Global Spillovers

Differential policy stances drive capital flows and exchange rate dynamics, with U.S. tightening reinforcing dollar dominance



Commodities & Real Assets

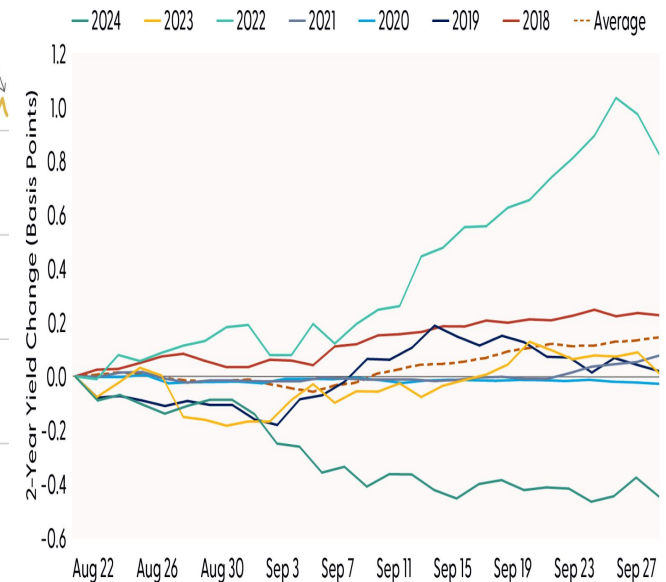
Dollar strength typically suppresses commodity prices, while excess liquidity and inflation risk elevate demand for hedge



Market Surprise

Unanticipated policy actions induce rapid repricing across asset classes as expectations and risk premia adjust

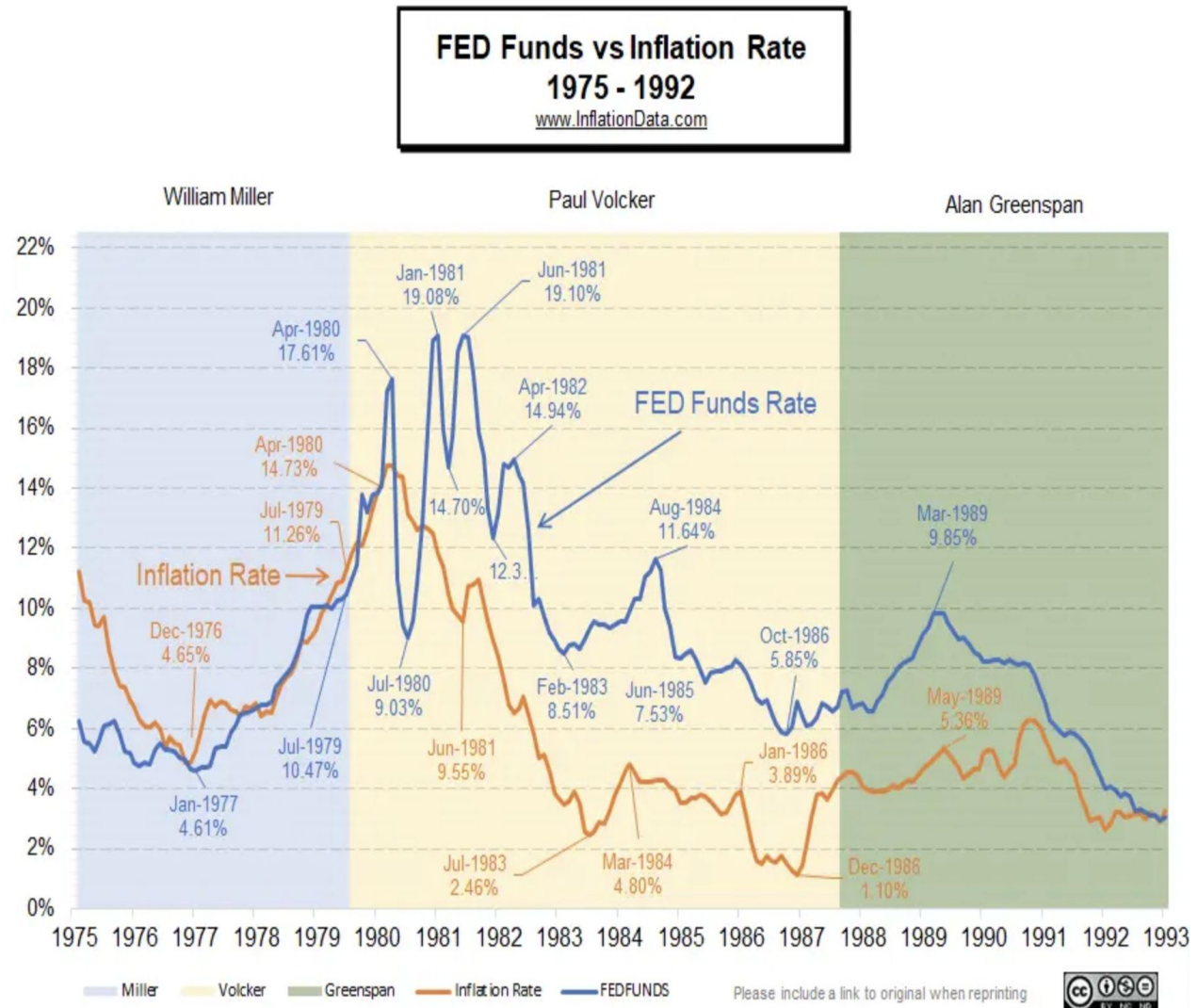
Two-Year Treasury Yields Tend to Drift Higher After the Jackson Hole Symposium



Historical Case Studies

Volcker Shock (1979–1981)

- U.S. faced stagflation: inflation >13%, unemployment ~7%, growth weak
- Oil shocks (1973, 1979) fueled inflation further
- Fed Chair Paul Volcker shifted to an aggressive anti-inflation stance
- Fed Funds Rate surged from nearly 11% (1979) to 20% (1981)
- 10Y Treasury yield peaked above 15%
- Equities entered deep bear market (S&P fell ~30%)

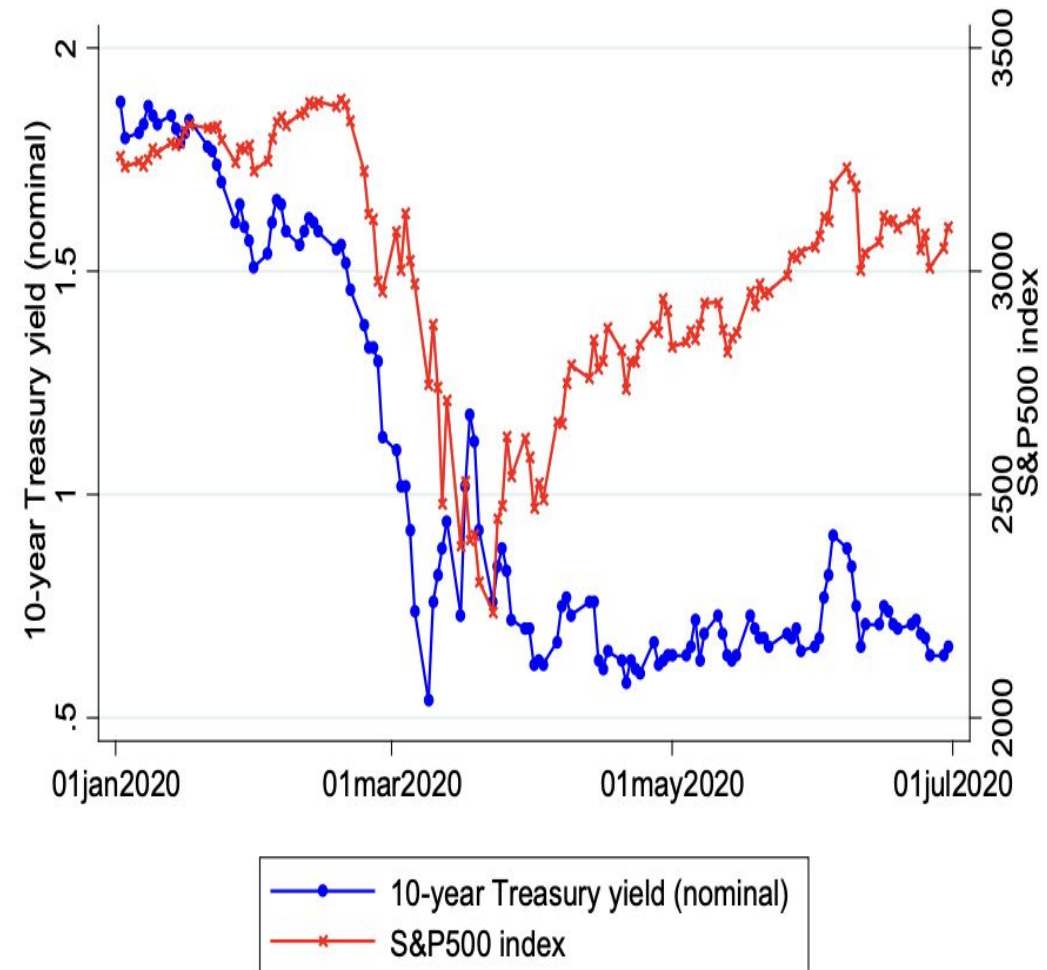


Historical Case Studies

COVID Emergency Cuts

- In March 2020, the Fed slashed rates to 0% in an emergency weekend meeting
- S&P 500 crashed ~30% in Feb–Mar but rebounded sharply after the Fed's emergency actions
- Treasury yields collapsed (10Y fell below 0.6%)
- Gold surged as a hedge

Panel A. Treasury yields spiked in mid-March as the S&P500 index kept falling



Forward Guidance & Communication Strategy

- Central banks use communication to shape market expectations, often moving asset prices without immediate policy action, could be time-based, state-based or Qualitative
- ECB 2012: Draghi's "*Whatever it takes*" calmed bond markets instantly.
- Fed Dot Plot: Maps expected rate path, anchoring bond yields
- COVID-19 (2020): Explicit zero-rate pledge stabilized markets before recovery.

Figure 2. FOMC participants' assessments of appropriate monetary policy: Midpoint of target range or target level for the federal funds rate

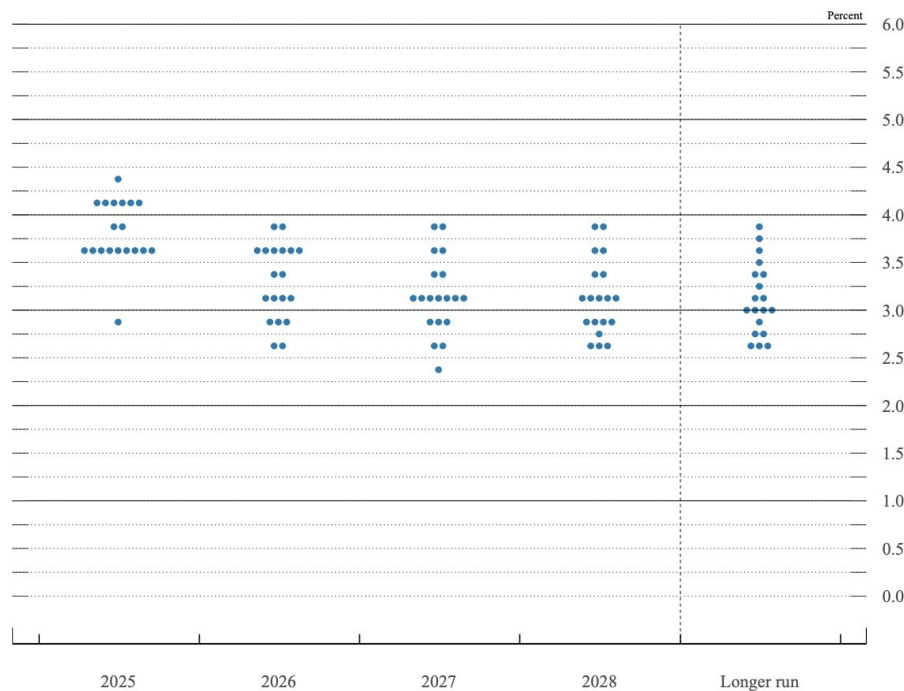
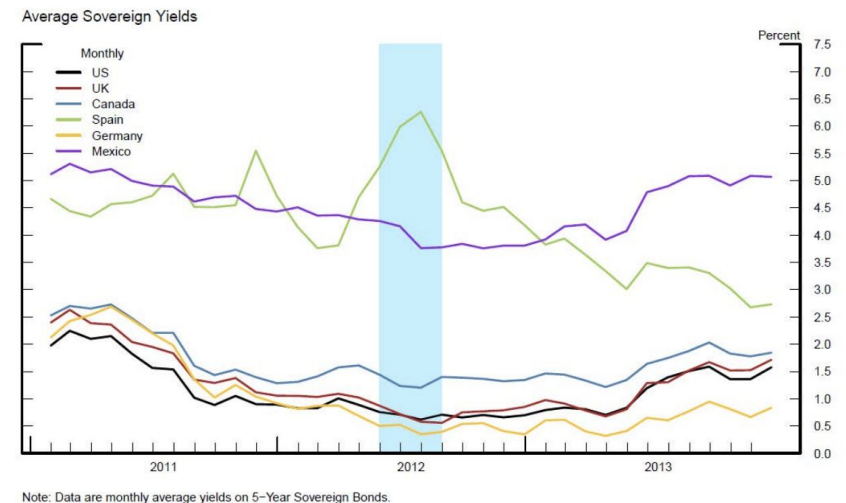
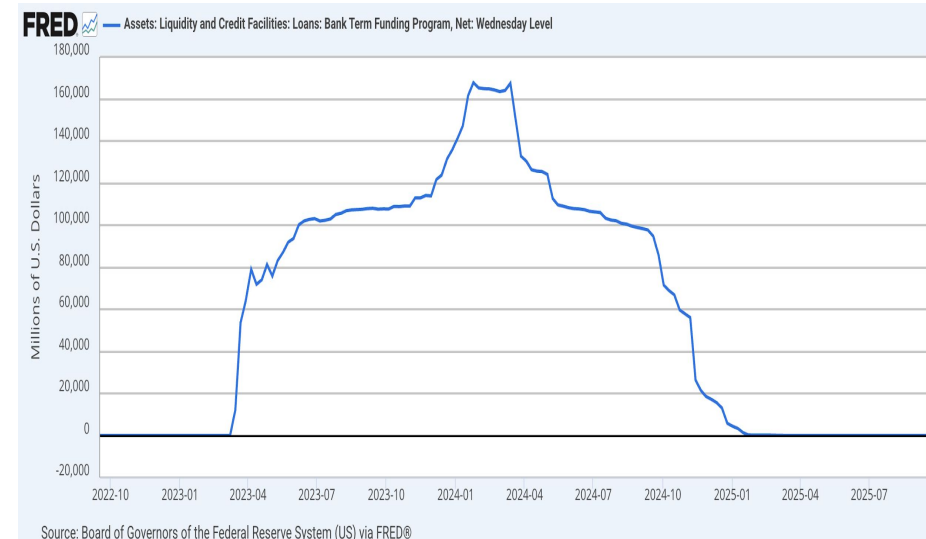
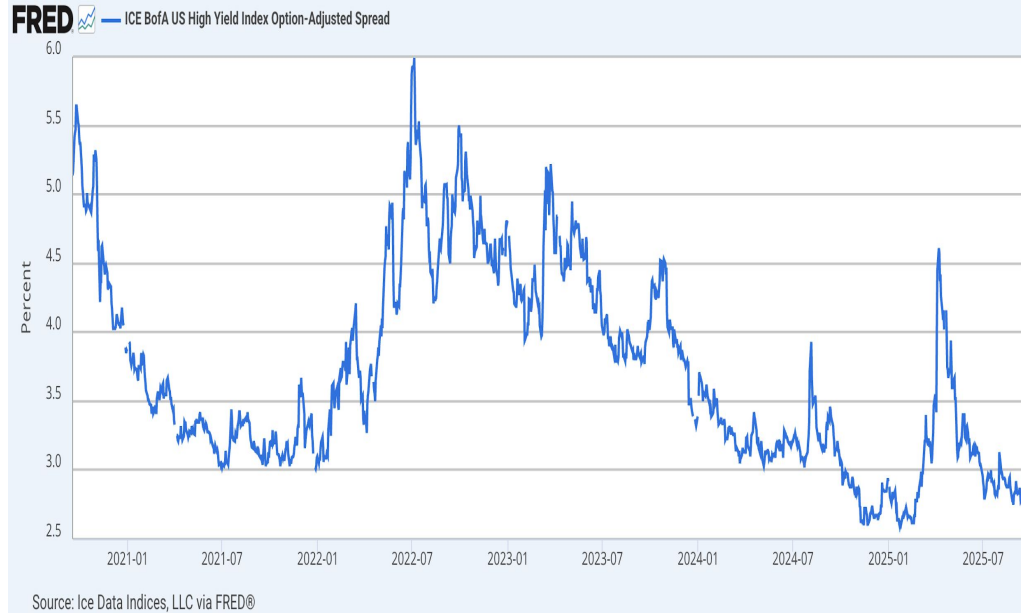


Figure 3: Sovereign yields



Financial Stability & Central Banks

- Lender of Last Resort: Provides liquidity to banks during crises to stop runs
- Crisis Interventions: QE, emergency lending, swap lines to restore confidence
- Macroprudential Tools: Stress tests, capital buffers, leverage ratios to curb systemic risk
- 2008 GFC: QE + emergency facilities stopped financial collapse
- March 2020 COVID: Trillions injected → stabilized Treasuries & credit spreads
- SVB 2023: BTFP prevented wider bank run



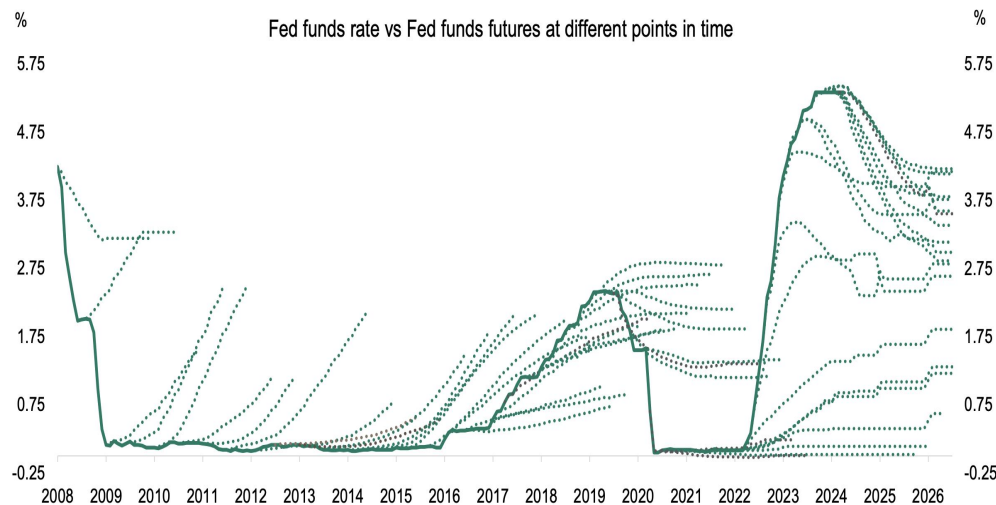
Market Expectations vs Policy Reality



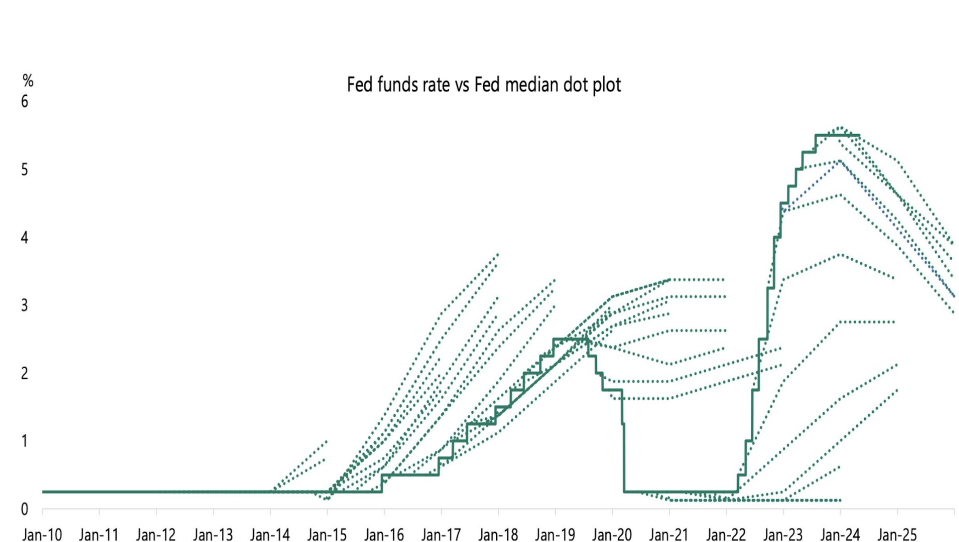
- Forward Expectations: Investors use Fed Funds Futures, SOFR, or OIS curves to price the expected path of policy
- Reality Check: Actual FOMC decisions often diverge

2023–24 Cuts Mispricing:

- Markets priced early Fed cuts on recession fears
- Fed pushed back with higher-for-longer stance
- Yield curve whipsawed, equities sold off, volatility in FX



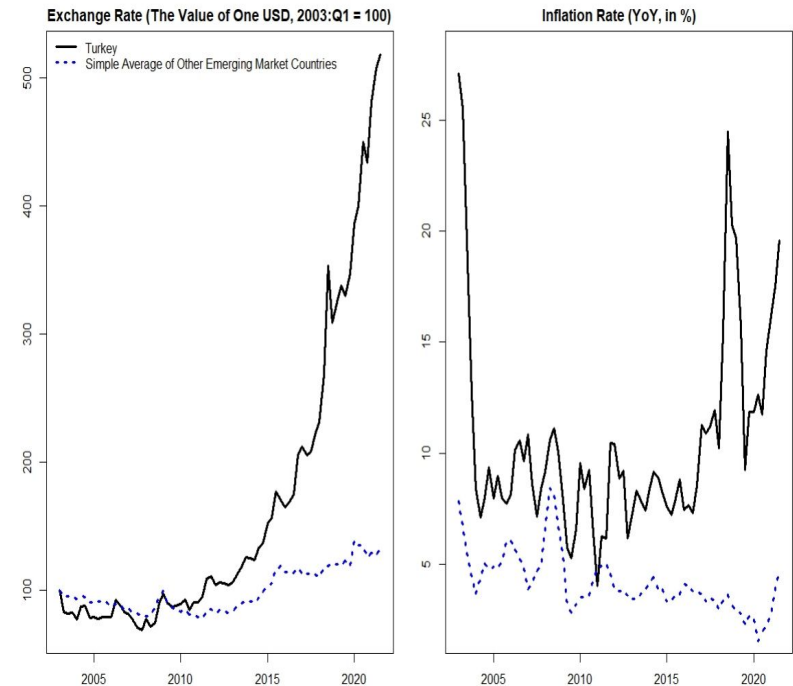
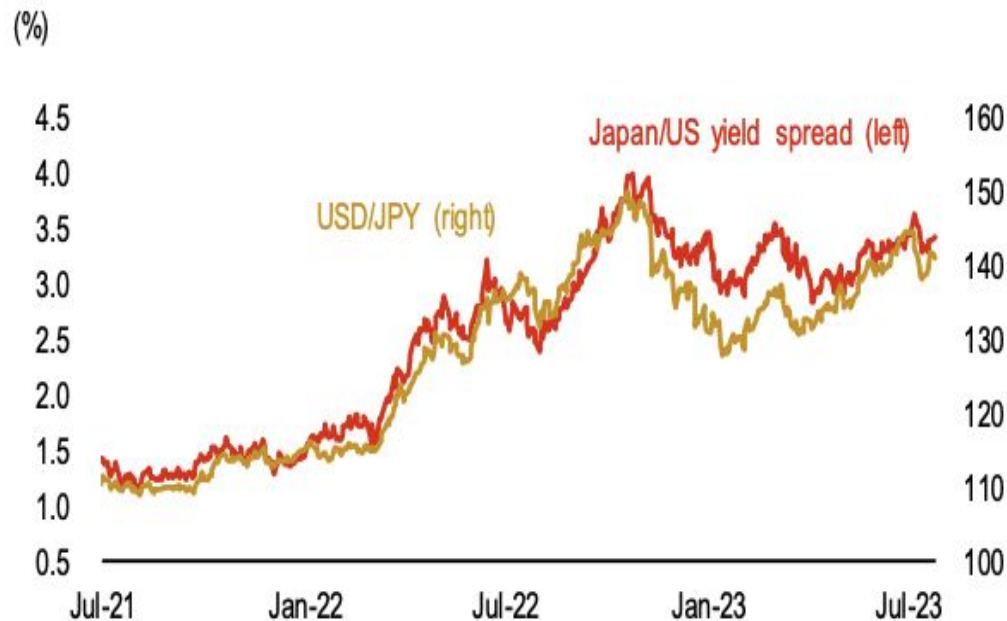
Fed forecasts are almost always wrong about the future path of the Fed funds rate



Global Divergence in Policy & Emerging Markets

- Global policy divergence drives FX moves, cross-border capital flows, and relative asset performance
- Aggressive U.S. tightening versus Japan's yield-curve control fueled a historic USD/JPY surge, highlighting interest-rate differentials as the key driver of FX flows
- Turkey (CBRT): Policy rate cuts amid >70% inflation destroyed credibility, triggering a lira collapse and capital flight
- China (PBoC): While the Fed tightened, China eased, leading to yuan depreciation and persistent outflow pressures

Chart 2: Japan/US Yield Spread (UST, JGB), USD/JPY Rate

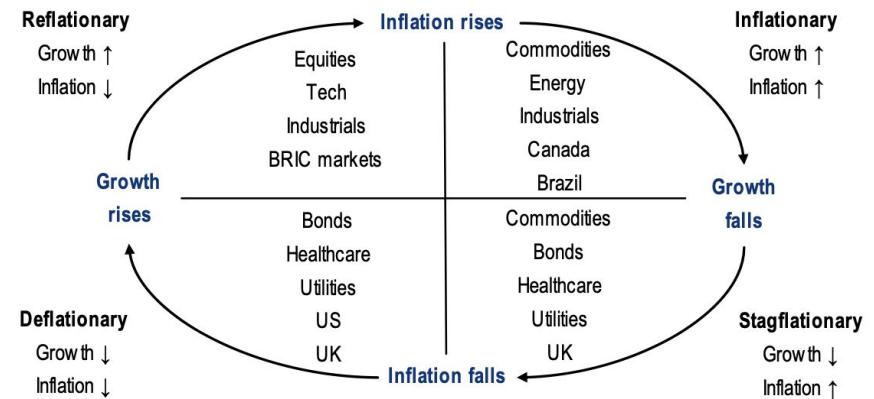


Source: Bloomberg; compiled by Daiwa Securities.

Portfolio Management Implications

- Hiking Cycle: Favor value, short-duration bonds, defensive sectors, strong USD exposure
- Easing Cycle: Growth equities, long-duration bonds, commodities (gold, oil), EM assets
- Surprise Channel: Portfolios must hedge tail risks from unexpected Fed shifts (2013 Taper, 2022 Jackson Hole)
- Diversification: Balance across equities, fixed income, commodities, and FX to manage policy shocks

Chart 1: The Investment Clock



Source: BofA Merrill Lynch Global Equity Strategy

Table 2: Investment clock and asset class performance since 1990

Reflationary phase		Inflationary phase	
Expansion to potential		Expansion above potential	
	Annualized returns		Annualized returns
Equities	17.1	Commodities	21.6
Bonds	2.9	Equities	4.7
Cash	1.1	Bonds	1.2
Commodities	0.4	Cash	0.2
Deflationary phase		Stagflationary phase	
Contracting below potential		Contracting from peak	
	Annualized returns		Annualized returns
Bonds	5.9	Commodities	21.1
Cash	1.8	Bonds	6.9
Equities	-2.7	Cash	0.7
Commodities	-19.0	Equities	-14.0

Source: BofA Merrill Lynch Global Equity Strategy



MACRO INVESTING ASSOCIATION

Lecture-3

Agenda



Market News

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Introduction to Fixed Income

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Government Bonds Overview

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Yield Curve and Term Structure , Factors Driving Interest Rates

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Bond Pricing Basics and Duration and Convexity

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Risks in Govt Bonds

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Case Study

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Portfolio Management Implications

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Introduction to Fixed Income

- Instruments promising fixed or contractual cash flows (coupons + principal)
- Distinct from equities: repayment priority and no ownership rights
- **Risk-free benchmark (sovereign bonds)** vs credit-bearing debt (corporates, EM)
- Valuation rooted in time value of money and discounting future cash flows

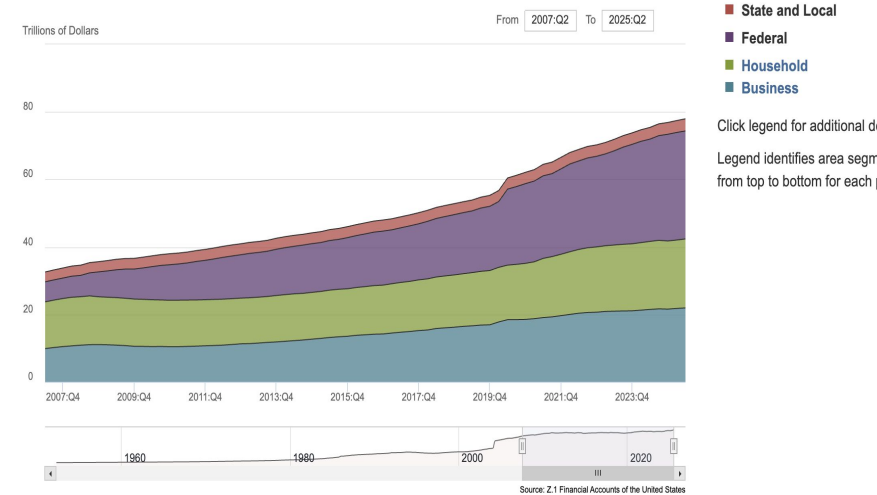
Economic & Policy Role

- **Primary tool for governments to finance deficits and manage intertemporal budgets**
- Anchor for monetary policy transmission: central banks steer short rates, bond market sets long rates
- Safe collateral for repo, derivatives, and global banking system liquidity
- Provides reference yield curve that underpins pricing of all other asset classes

Investor Perspective

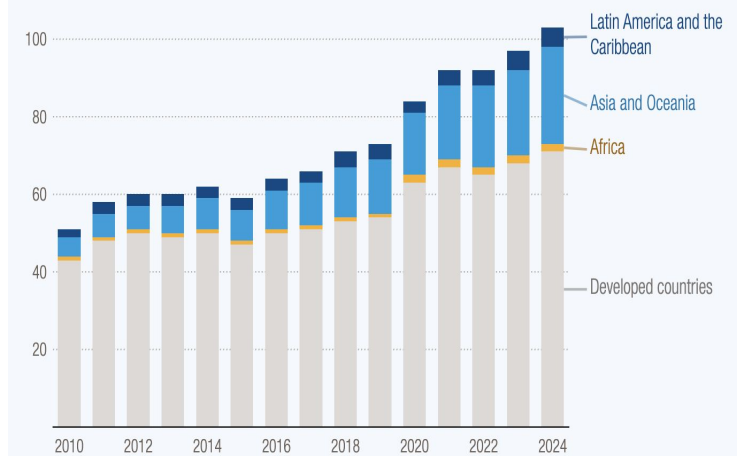
- Predictable income stream and contractual repayment (if sovereign is credible)
- Duration/convexity allow tactical positioning on rate expectations
- **Diversification vs equities: historically negative correlation (though regime-dependent)**
- Hedge in risk-off episodes (flight to quality) and inflation hedge via linkers (ex: TIPS)
- Strategic allocation: core holding for pensions, insurers, central banks, sovereign wealth funds

All nonfinancial sectors debt



Global public debt surpasses \$100 trillion in 2024.

Global public debt in US\$ trillion, by region



Government Bonds Overview

Breakdown of US Treasuries

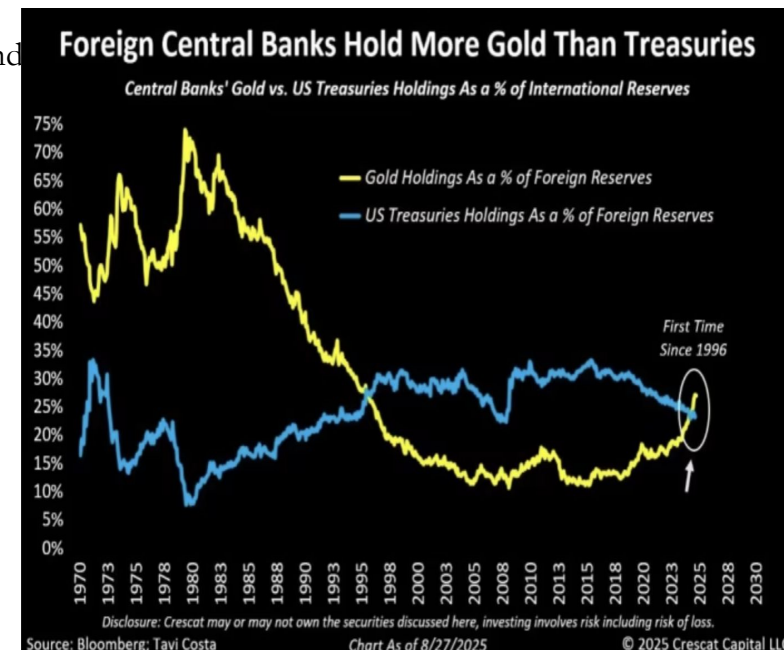
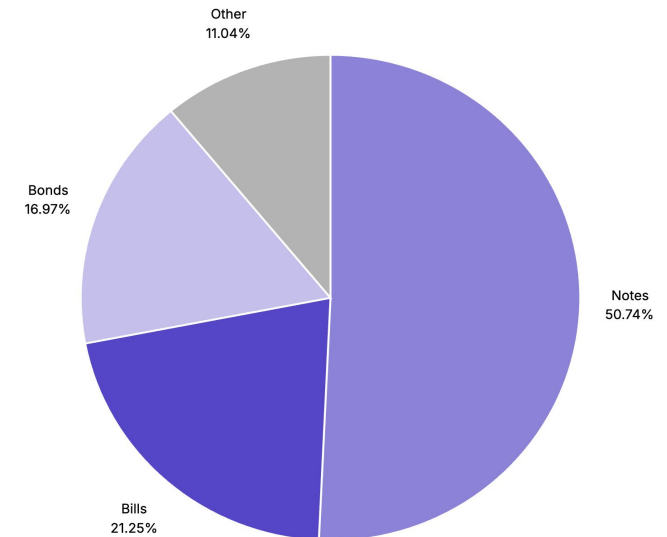
- Treasury Bills (T-Bills): maturities < 1 year, issued at discount, zero-coupon
- Treasury Notes (T-Notes): maturities 2–10 years, pay semiannual coupons
- Treasury Bonds (T-Bonds): maturities 20–30 years, long-duration instruments
- Zero-Coupon / STRIPS: principal & coupon cash flows traded separately
- Inflation-Linked Bonds (TIPS, UK Gilts, ILBs): principal indexed to inflation

Role in the Financial System

- Benchmark “risk-free” curve underpinning discount rates for all assets
- Safe collateral in repo & derivatives markets (USTs as global collateral base)
- Core holdings for central banks (reserves), pensions, insurers, and sovereign wealth funds
- Vehicle for monetary policy operations (open market ops, QE/QT)
- Facilitates government fiscal financing & debt management

Global Perspective

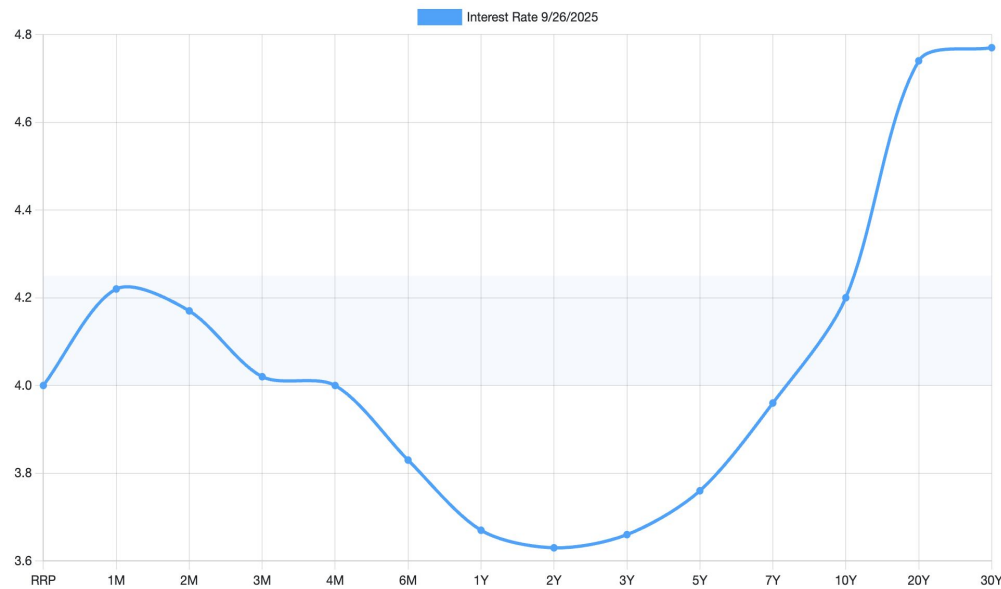
- U.S. Treasuries: world’s deepest & most liquid bond market (~\$27T outstanding)
- Eurozone: fragmented markets but German Bunds serve as benchmark
- Japan: JGBs dominate with ultra-low yields under yield-curve control (YCC)
- EM Sovereigns: higher yields, credit risk & currency mismatch (e.g., Turkey, Argentina)
- Growing share of inflation-linked and green sovereign bonds in issuance mix



Yield Curves & Term Structure

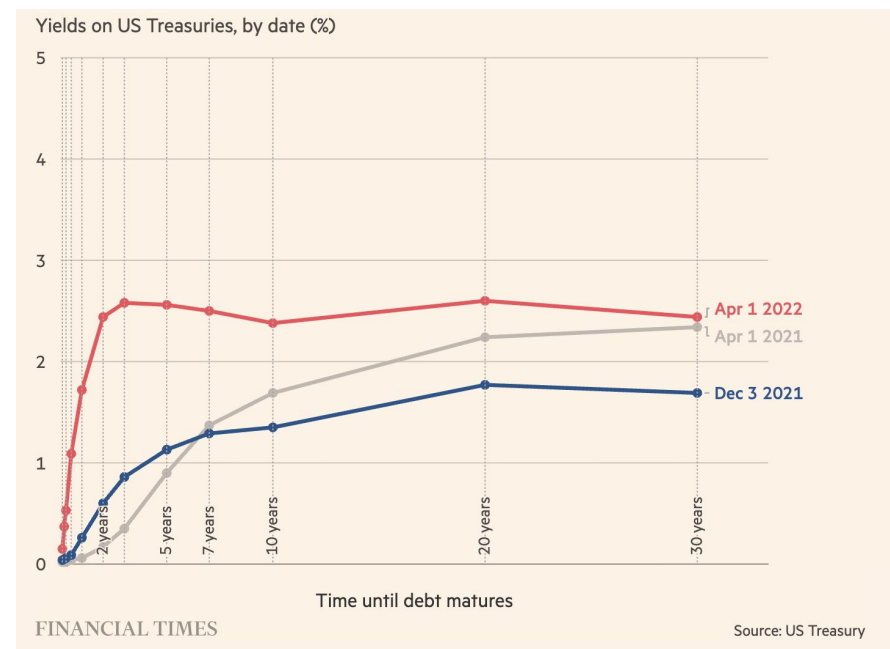
Yield Curve Shapes, Interpretation & Theories

- Normal (Upward Sloping): Long > short; signals growth, inflation risk, and positive term premium
- Flat: Transition phase; policy uncertainty
- **Inverted: Short > long; reliable recession predictor (2y–10y inversion precedes every U.S. recession since 1950s)**
- Expectations Hypothesis: Long rates = average of expected future short rates which reflects Fed path
- Liquidity Premium Theory: Investors demand extra yield for long maturities



Why the Yield Curve Matters

- Macro Signal: One of the best leading indicators for recessions, inflation, and growth expectations
- **Policy Gauge: 2y reflects Fed policy path; 10y anchors long-term credibility; curve slope signals policy stance.**
- **Valuation Anchor: Corporate bonds, mortgages, equities all priced relative to Treasuries**
- Risk Management: Used in steepener/flatteners and hedging portfolio duration
- Global Impact: Spillovers drive EM capital flows, FX trends, and global risk appetite



Factors Driving Interest Rates

Fundamentals

- Monetary Policy: Fed sets short rates
- Inflation: Higher inflation pushes nominal yields up
- Growth: Strong growth lifts real yields

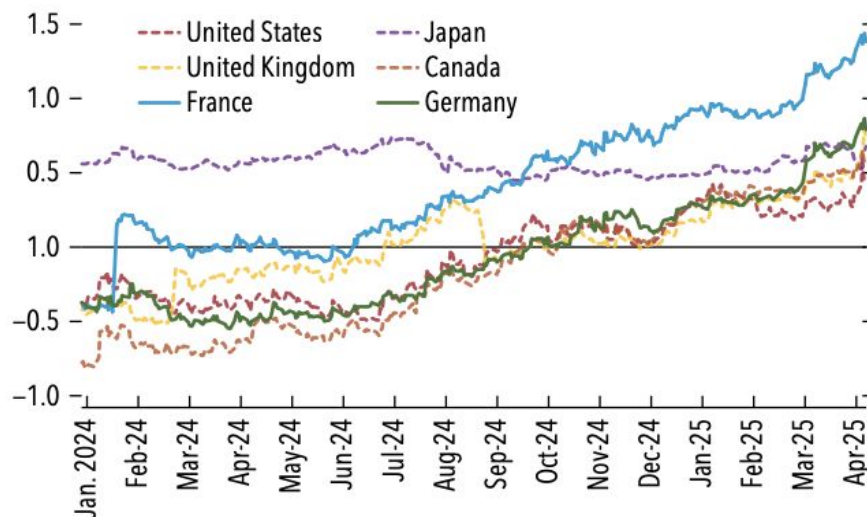
Market Forces

- Fiscal Supply: Large deficits require more issuance raising yields
- Foreign Demand: Central banks & SWFs buying lowers yields
- Technicals: Repo stress and hedging flows can distort yields

Risk Premiums

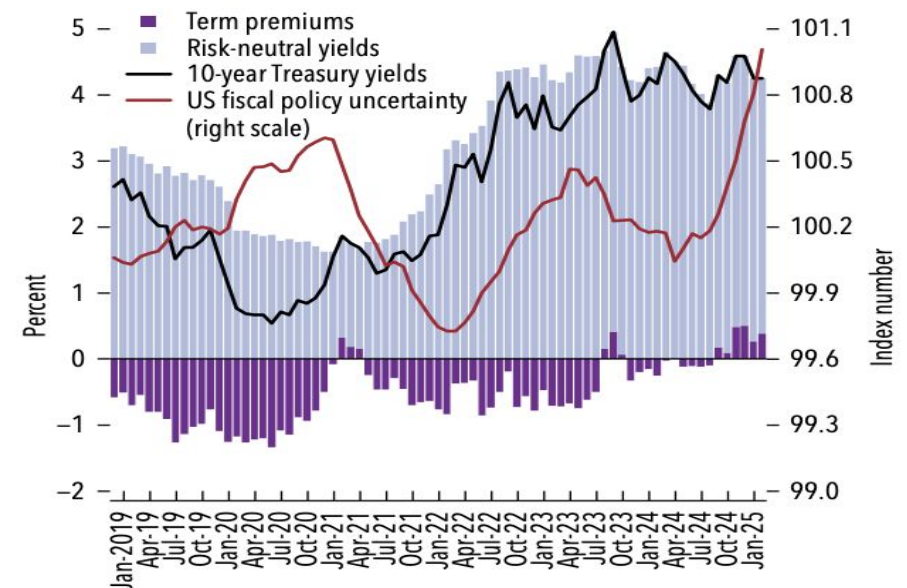
- **Term Premium: Extra yield demanded for long bonds under uncertainty**
- Geopolitical Risk: Crises cause safe-haven buying, temporarily lowering yields
- Volatility Premium: High uncertainty ,

1. Recent Evolution of Term Spreads for Select Advanced Economies



Sources: Bloomberg Finance L.P.; and IMF staff calculations.

Figure 1.6. US 10-Year Treasury Nominal Yields, Risk Premiums, and Fiscal Uncertainty



Source: Bloomberg, IMF,

Note:

Rising Sovereign Debt & Equity Market Stress

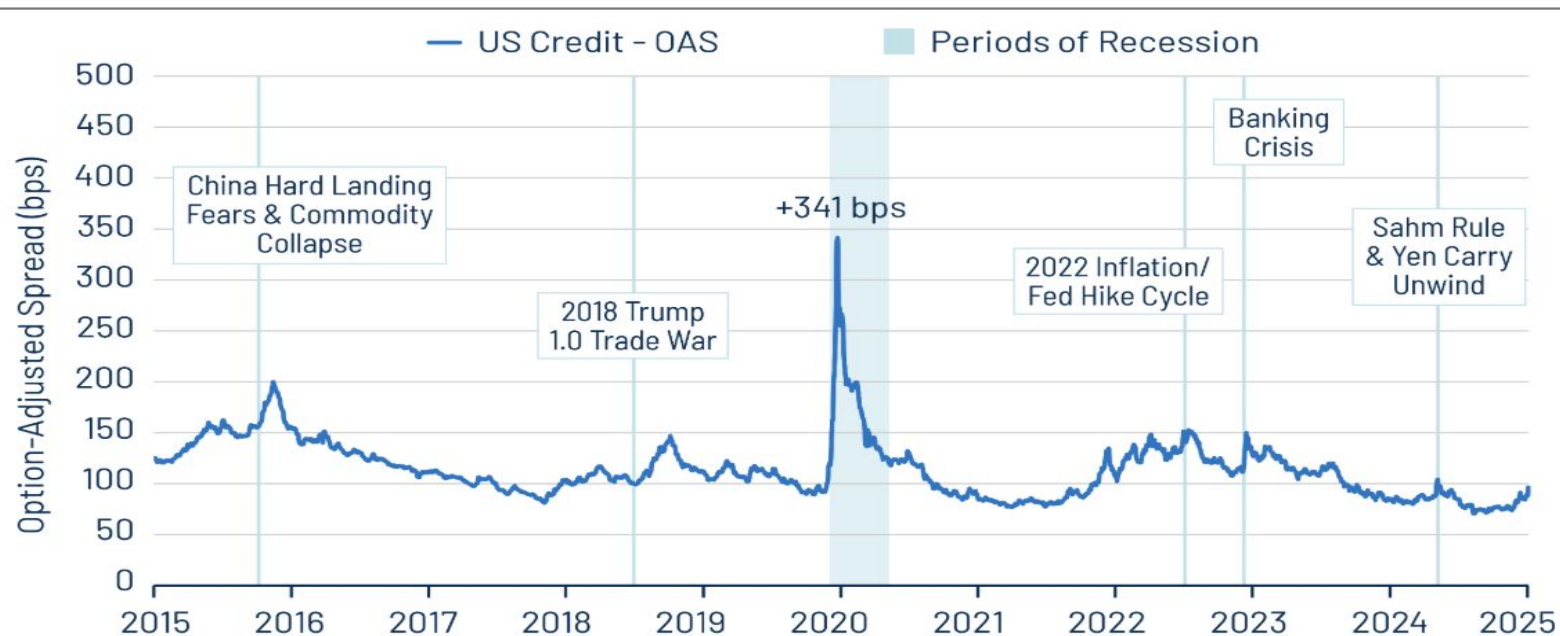
Sovereign Debt Surge in Developed Economies

- United States: Debt-to-GDP above 100% and climbing, with future projections much higher
- Japan: World's highest debt burden; even modest rises in yields challenge BOJ's yield-curve control
- France: Public debt elevated, fiscal space narrowing; debt ratios above many EU peers

Sovereign Debt Squeeze

- **Crowding Out:** Heavy government borrowing soaks up capital; Treasuries compete directly with corporate debt
- **Spread Gap:** In debt surges, the gap between UST 10Y yields and HY/small-cap yields widens, raising private cost of capital

Exhibit 1: Bloomberg US Credit Index—OAS, Last 10 years



Source: Bloomberg. As of 03 Apr 25. Select the image to expand the view.

Bond Pricing Basics



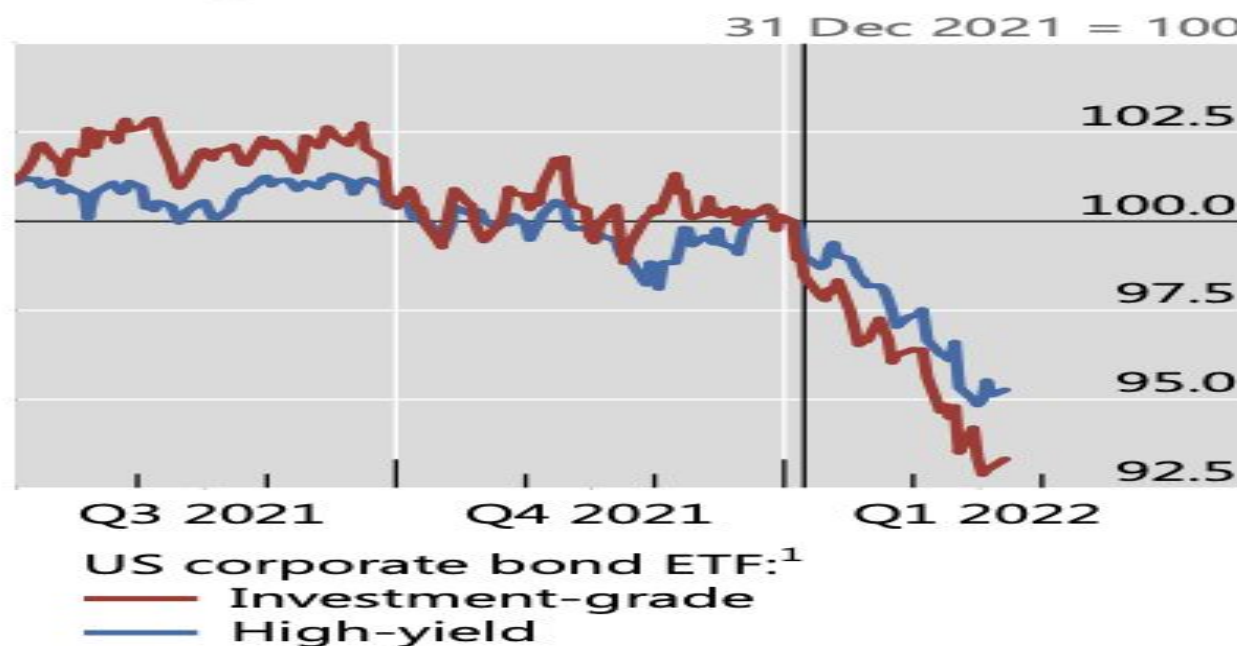
Price–Yield Trade-Off

- **Bond valuation rests on a simple law: as yields rise, prices fall**
- Investors demand discounts on older bonds when new issues carry higher coupons
- The relationship is convex, meaning losses accelerate as yields rise but gains are asymmetrically larger when yields fall

Structure of Cash Flows

- A bond's risk profile is shaped by its coupon size and maturity horizon
- Higher-coupon, shorter-maturity bonds return cash flows quickly, anchoring prices
- Longer-dated or zero-coupon bonds push all value far into the future, magnifying exposure to rate shifts
- This structural difference explains why 30-year zeros swing violently while 2-year bills remain stable

IG bond prices fell more than HY due to longer duration



Long-Term Rates and Housing Dynamics



Link Between Treasuries & Mortgages

- 30-year US mortgage rates are closely tied to the 10-year Treasury yield, which serves as the benchmark for long-term financing costs
- As the 10-year rises, mortgage rates climb almost in lockstep → borrowing for households becomes more expensive
- This transmission channel means bond market shifts spill directly into the real economy via housing affordability

Housing Market Impact

- Higher mortgage rates reduce affordability → fewer buyers qualify, demand softens
- Home prices either stagnate or decline as higher financing costs offset income gains
- Construction activity slows as financing for developers tightens
- **Wealth effect: housing is the largest asset for US households, so tighter mortgage conditions reduce consumer spending power**

Figure 1: Mortgage Rates and Treasury Yields



Source: Source: FRED and author's calculations.

Note:

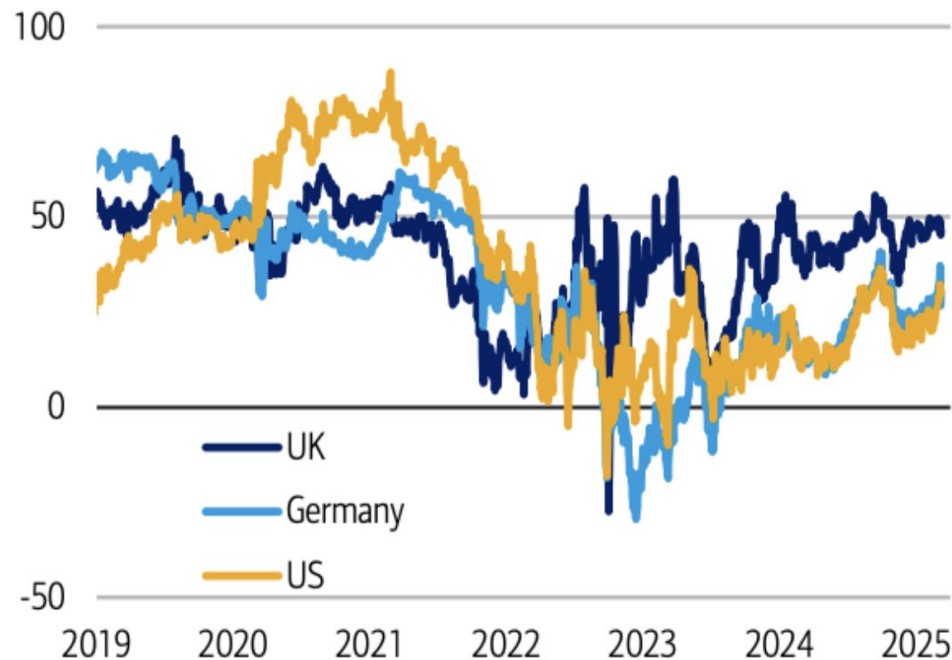
Duration (First-Order Risk)

Duration

- A bond's interest-rate sensitivity, not its maturity
- Captures the weighted average timing of discounted cash flows
- **Modified Duration: % price change for a 1% move in yields**
- Economically, the first derivative of bond pricing, the slope of the price–yield curve

Exhibit 8: ...and now it's steeper

As left, focusing on the more recent past.

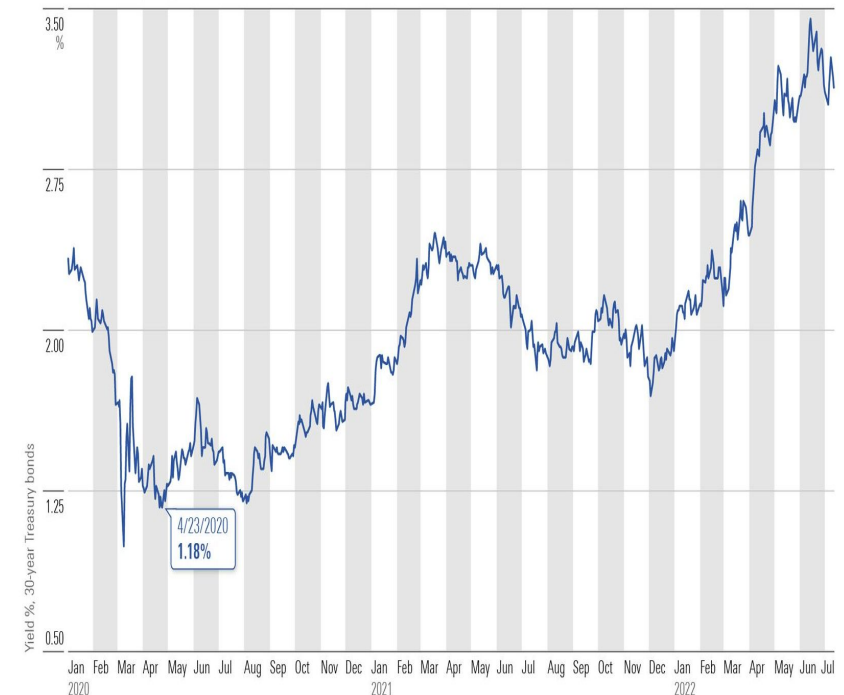


Source: BofA Global Research, LSEG Data & Analytics

Strategic Relevance

- The risk language of fixed income (pensions, insurers, central banks)
- High duration = high volatility (long-dated, low-coupon bonds)
- **2022 example: Long Treasuries (20Y+) down ~30%, Short Treasuries (1–3Y) remains flat, A pure duration shock, not credit risk**

Yield Has Nearly Tripled



Source: Morningstar, BOFA

Note:

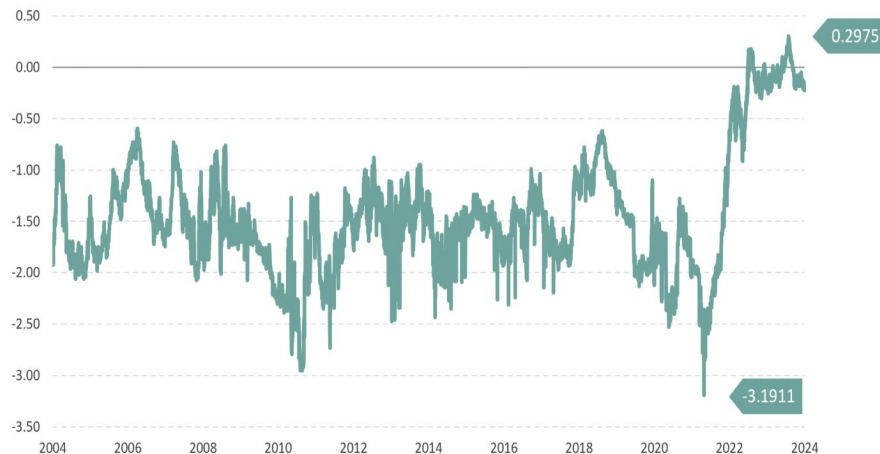
Convexity (Second-Order Risk)

Convexity

- Bond prices don't follow a straight line, the price–yield curve is curved (convex)
- Duration is first derivative (slope) but it ignores curvature
- **Convexity is second derivative, correcting duration's error when yields move a lot**

Exhibit 1: ICE BofA US Mortgage-Backed Securities Index Effective Convexity Trend

As of March 26, 2024



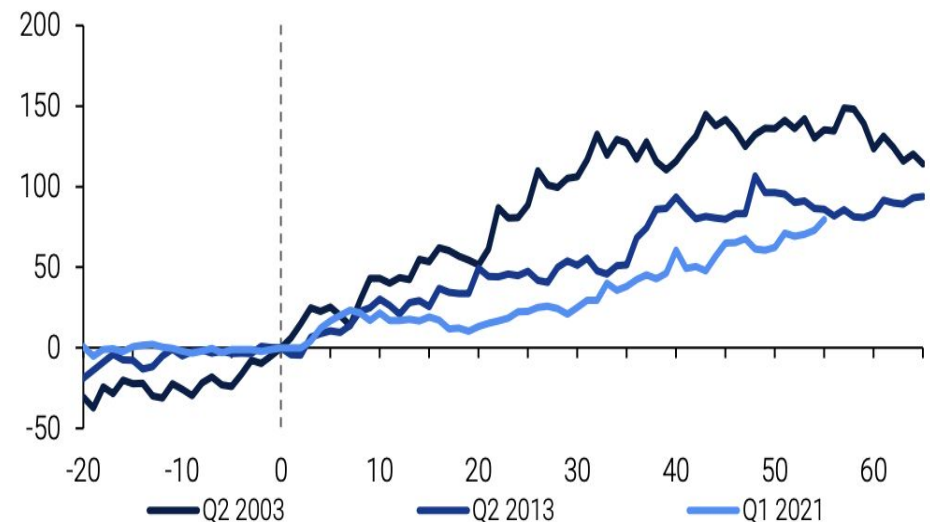
Source: Bloomberg (© 2022, Bloomberg Finance LP). Indexes are unmanaged and one cannot directly invest in them. They do not include fees, expenses or sales charges. **Past performance is not an indicator or a guarantee of future results.**

Market Consequences

- **Positive convexity (Treasuries, most sovereigns): Prices rise more when yields fall, fall less when yields rise**
- Negative convexity (MBS, callable debt): Refinancing/prepayment cuts off upside, adds downside risk
- Portfolio angle: Managers hedge both duration and convexity; “long convexity” exposure is prized in turbulent markets
- Historical proof: 1994 bond crash & 2003 MBS selloff

Interest rates have sold off similar to prior convexity episodes...

Cumulative change in 10-year yield, index from June 13th 2003, April 30th 2013, and December 31st 2020



Risks in Government Bonds

Interest Rate Risk

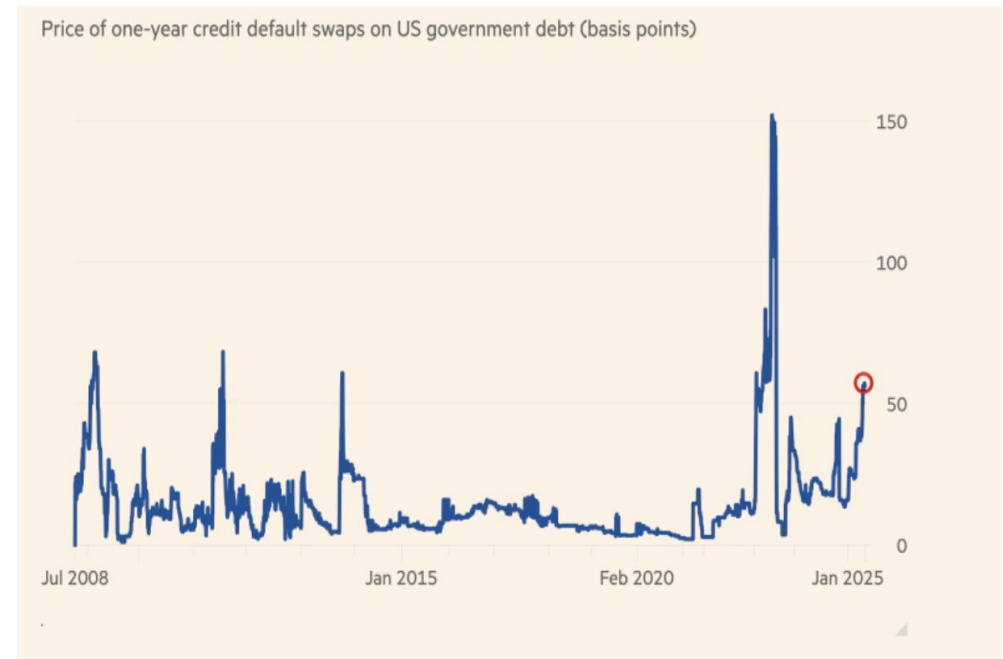
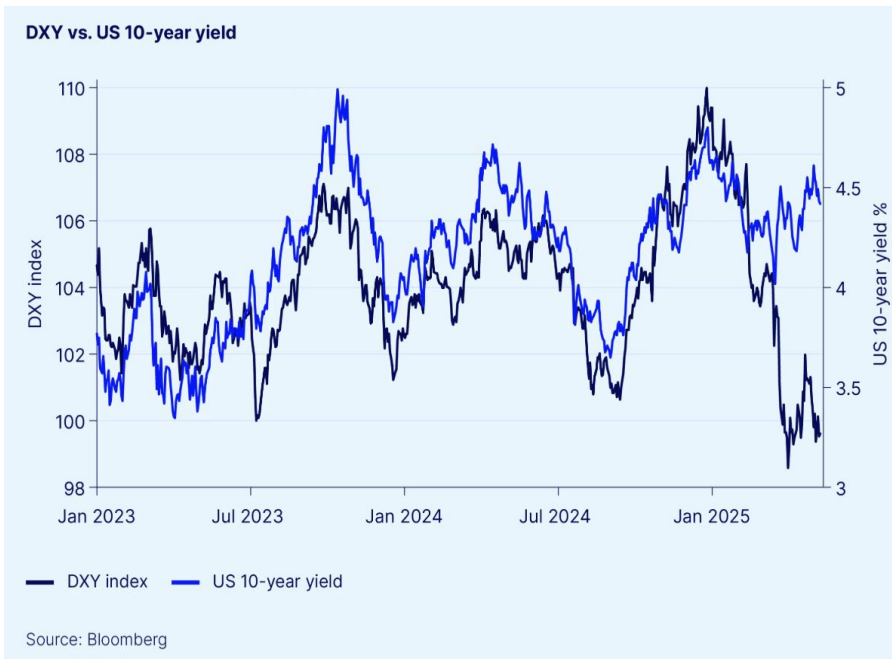
- **Long duration magnifies convexity, making bond prices acutely sensitive to curve shifts**
- Government bonds, often assumed to be stabilizers, can themselves become transmission channels for policy volatility

Inflation & Liquidity Risk

- Fixed nominal coupons are eroded when realized inflation exceeds expectations
- **Even the “risk-free” UST market depends on balance sheet intermediation**

Credit / Sovereign Risk

- Sovereign solvency rests on fiscal discipline and the credibility of monetary backstops
- “Risk-free” is not an absolute condition but a political construct. When credibility erodes, government bonds transition from anchor assets to volatility amplifiers



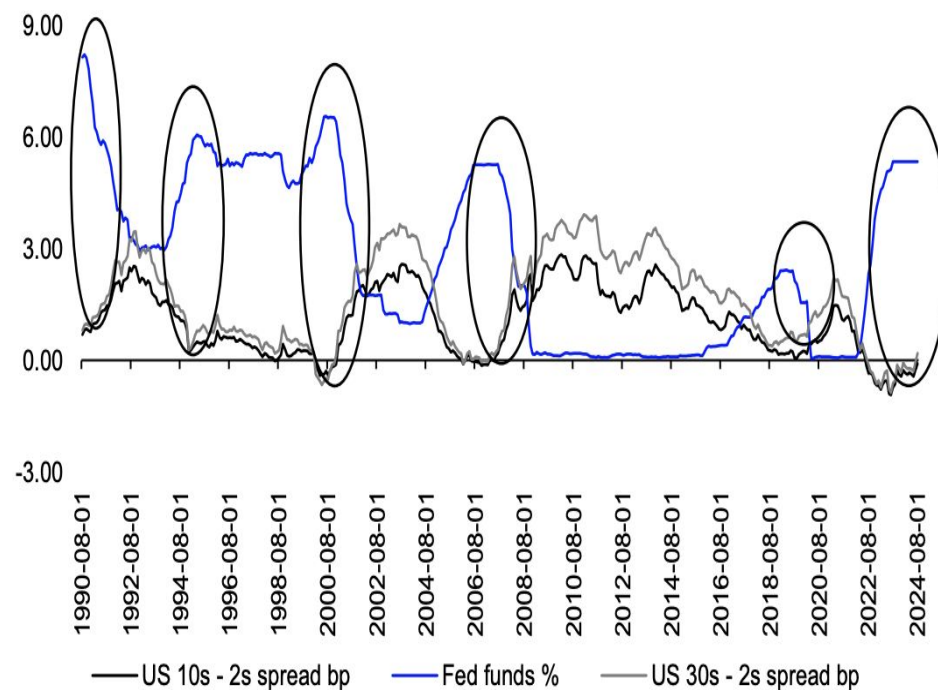
Source: Bloomberg, Financial Times

Note:

Market Pricing vs Central Bank Reality

- Forward curves as mirrors of belief : futures and OIS encapsulate collective expectations of policy, but these are probabilistic not promises
- **The “early cut” fallacy (2023–24) markets extrapolated disinflation into premature easing; the Fed/ECB held firm, forcing brutal front-end repricing**
- Volatility as the cost of misbelief, when expectations diverge from realized policy, yield curves convulse, convexity risk rises, hedging costs surge

Chart 2: US yield curve spreads and Fed funds rate



Source: US Federal Reserve. Monthly data to end-August 2024.

Case Study



DM 2022 Bond Rout

- **U.S. Treasuries (20Y+) and UK Gilts lost ~30% in market value as inflation surged to multi-decade highs (9%+ U.S. CPI, 11%+ UK CPI)**
- Central banks responded with the fastest hiking cycle since the 1980s (“higher-for-longer” stance)
- This was the worst sovereign bond drawdown in modern history — a duration shock rather than a default event
- Even “risk-free” assets behaved like equities, undermining the traditional 60/40 portfolio hedge

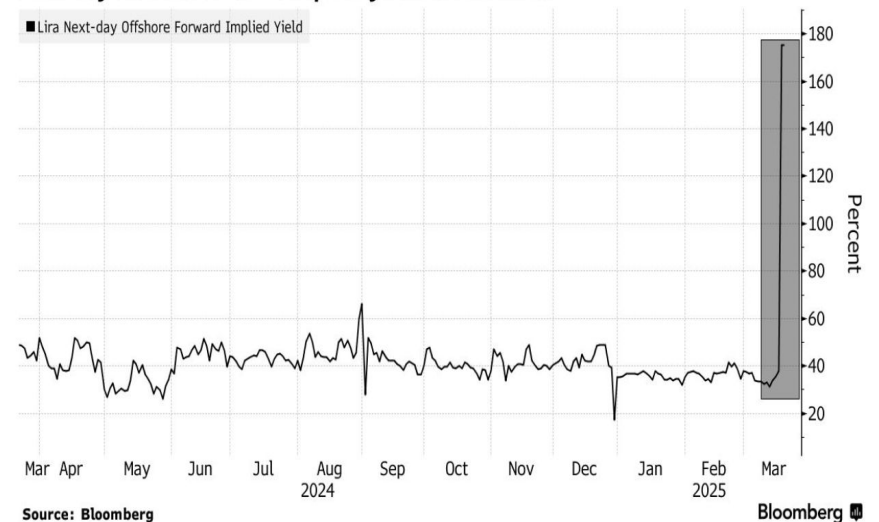


Emerging Market

- Argentina defaulted in 2020 (its 9th in history), restructured bonds again
- **Turkey experienced multiple currency crises (2018, 2021, 2023) driven by heterodox policy, high inflation (>70%), and loss of central bank credibility**
- In EM, the core risk is sovereign creditworthiness and FX credibility
- **Investors worry less about duration and more about debt sustainability, inflation, and IMF intervention**

Turkey's Lira Offshore Rates Still Elevated

Next-day offshore forward implied yields are at 175%



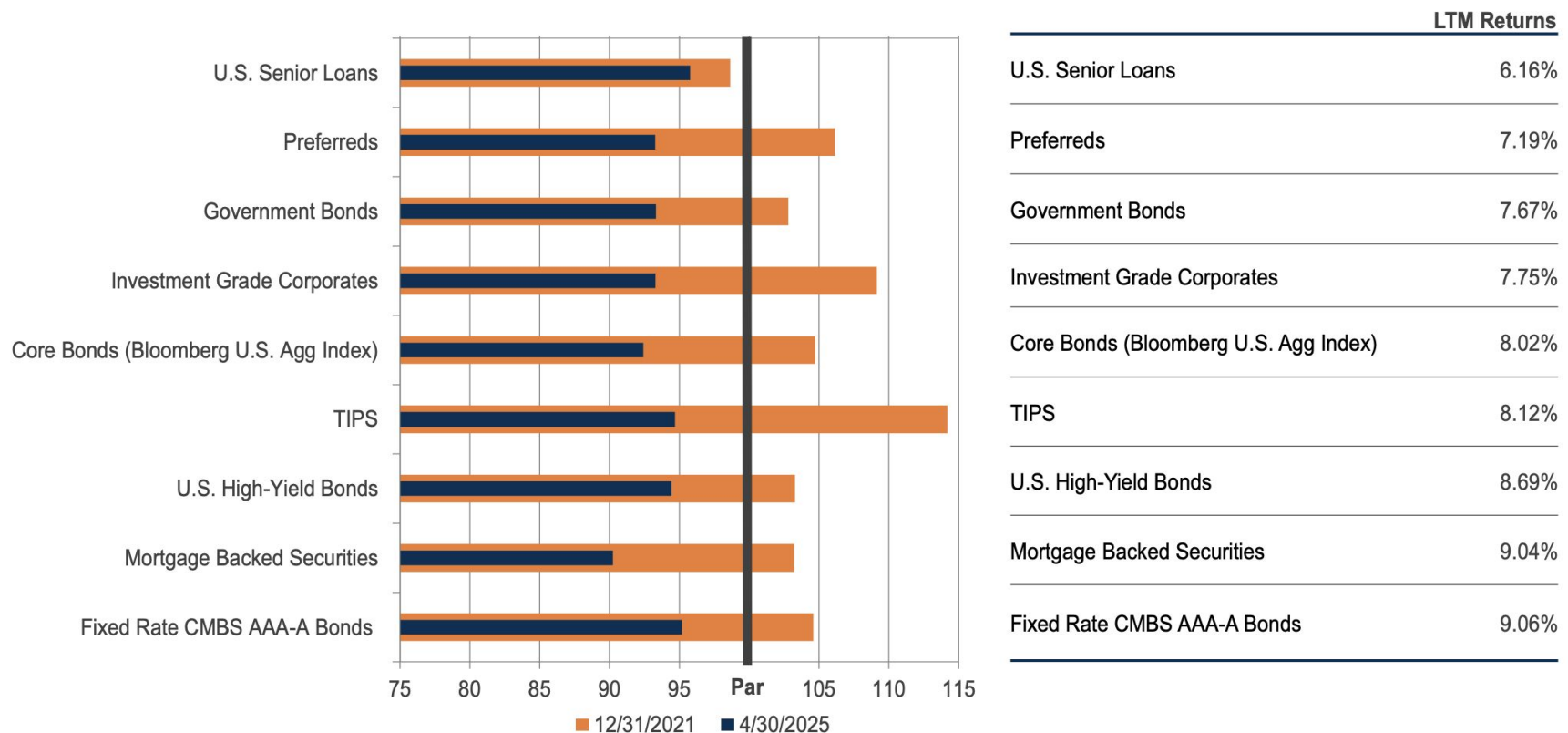
Source: FT, Bloomberg

Note:

Portfolio Management Implications

- **Duration bets (short vs long): In DM, curve positioning is critical: being long duration in 2022 meant equity-like losses; tactical short duration can hedge inflation shocks**
- Inflation protection (TIPS, commodities): Real assets and inflation-linked bonds provide a buffer when nominal bonds fail as hedges
- Flight-to-safety allocation (USTs): Even after the 2022 rout, Treasuries remain the global safe-haven in systemic stress
- **Cross-market relative value (UST vs Bunds vs JGBs vs EM debt)**
- Yield differentials and policy divergence create spread opportunities (ex. UST-Bund trades, EM vs DM rotations)

FIXED INCOME OPPORTUNITY SET



21.17

217.88

MIA

GIES

MACRO INVESTING ASSOCIATION

Lecture-4 Fixed Income(Credit, Corporate and Emerging Markets)



Agenda



Market News	4
Introduction to Corporate and Credit Bonds	5
Corporate Bond Structure	6
Investment Grade vs High Yield	8
Credit Spreads/Ratings	9
Bond Pricing, Yield, and Risks	10
Case Study	12
Portfolio Management Implications	13

Introduction to Corporate and Credit Bonds

- Debt instruments issued by corporations to raise capital (funding expansion, M&A, refinancing)
- Similar to Treasury Bonds as Investors receive coupon payments and principal at maturity
- Risk-free benchmark (sovereign bonds) vs **credit-bearing debt (corporates, EM)**
- **Credit Risk** is the main proponent of Corporate bonds that differs from Government bonds

Economic & Policy Role

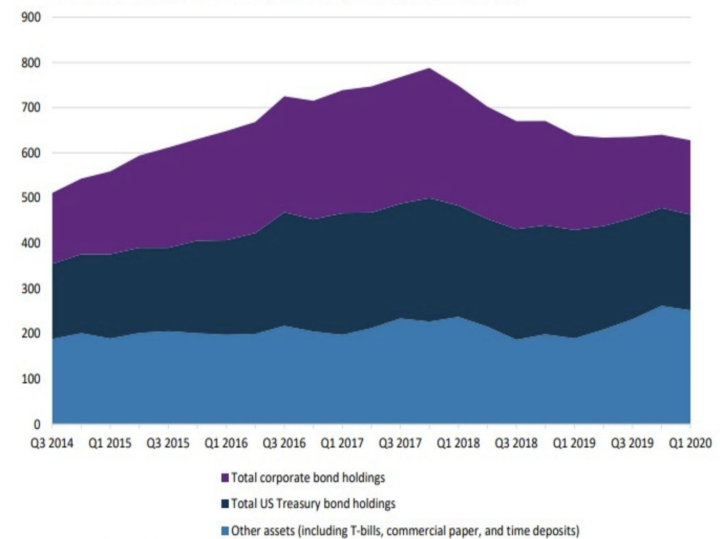
- **Key Economic Indicator in terms of Economic Health**
- Firm stimulator for expansion, especially in times of tight bank lending (Covid, 2008 market crash)
- Used as a stress barometer for consumer confidence
- US bond market ~ 10 trillion, heavily invested by foreign entities -> global consequences

Investor Perspective

- Predictable income stream, more so than equities
- More attractive yield vs government bonds, for the more risk-seeking individual
- Variable Risk – fix your asset risk between safety (investment) or yield (junk)
- **Diversification vs equities: historically negative correlation (though regime-dependent)**
- Hedge in Credit Default Swaps (CDS) to speculate on credit risks



Figure 2: A big pile of fixed income
Investment holdings of 11 of the largest U.S. companies* by selected buckets, \$bn



Credit Bond Structure

Municipal Bonds

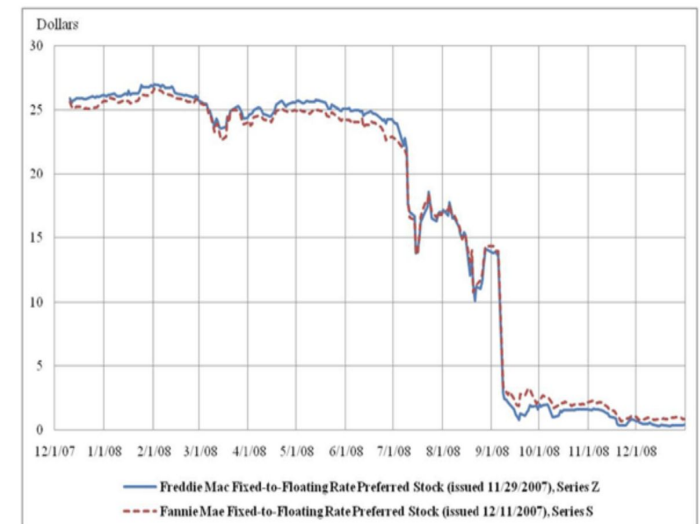
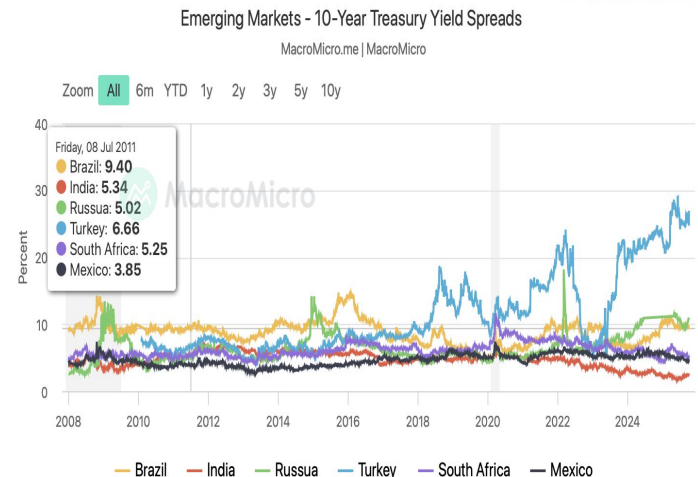
- Issued by States or other localized government for funding of public services
- Widely used for large scale infrastructure projects
- Government Obligation (Tax-backed) vs Revenue bonds (project-backed)
- **Key feature** - Interest that is tax-exempt
- Generally safe but default risk in poorly managed local gov. (Detroit, Puerto Rico)

Agency Bonds

- Issued by Government Sponsored Enterprises (Fannie Mae, Freddie Mac)
- Very high credit rating, typically AA/AAA
- Funding role primarily for mortgage market and policy initiative
- Widely regarded as an extremely safe collateral for repo and banking
- Not without risks – 2008 housing crisis, 200 billion in debt repayment to bond holders
- **Stock plunged by 90%, equity vs bond lesson**

Emerging Market Bonds

- Issued by developing countries (either sovereign or corporate)
- Higher risk bond due to policy, inflation, and FX risk
- Local Currency vs Hard bonds – helps lower exposure for investors
- Certain options have equity-like movements (Argentina peso, Turkish Lira)



The Problem with Argentina

Context

- **Argentina has defaulted 9 times in sovereign bonds**
- Due to large fiscal deficit, which turns into massive devaluations when investor confidence drops
- Reliant on commodities, therefore subject to shocks, makes the government revenue volatile
- Because investors know past information, they demand high yields, which further drive the deficit down

Yield Spike

Yields on Argentina's inflation-linked local bonds have spiked on default fears

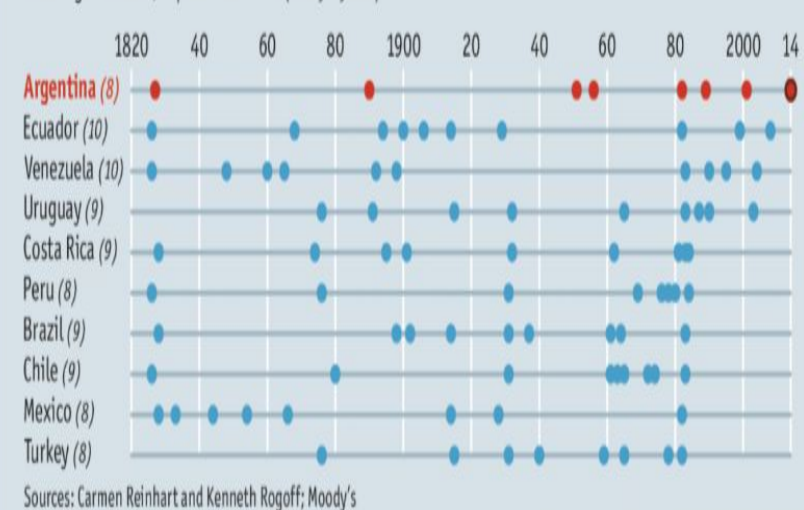


Result

- Creates a cyclical default period that is almost "untouchable" to investors
- **August 2020 - Last default, restructure of \$65 billion in bonds**
- Bonds traded 30 cents on the dollar, maturity extensions, interest rate reductions
- Argentina has since been widely shut out of markets, as well as indirectly affected other sovereigns in Latin America as well

Something of a reputation

Sovereign defaults, top ten countries (no. of defaults)



Corporate Bond Structure

Breakdown of Corporate Bonds

- Investment Grade (IG): Rated BBB- or higher; stable w/ lower yield
- High Yield (HY): Below BBB-: higher yield w/ tighter covenants and risk of default
- Maturities: Short term (<5y), medium (5-12y), long term (12y+)
- **Bullet Maturity** – Principal is repaid in entirety at maturity date

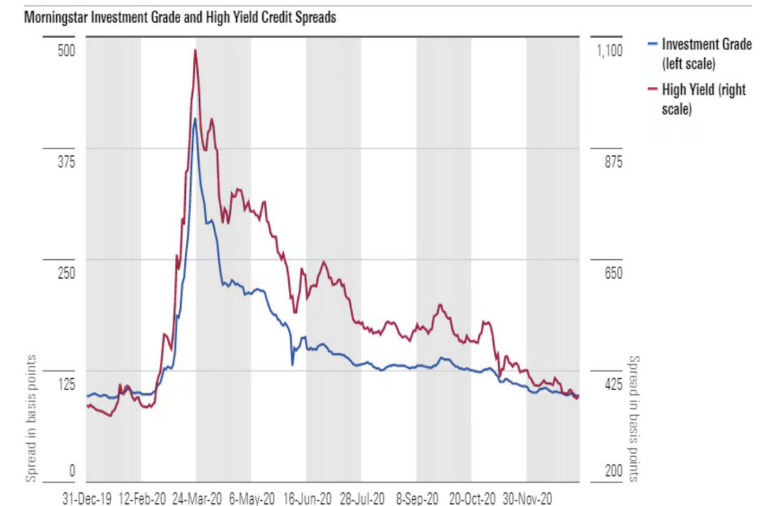
Role in the Financial System

- Major financing channel for companies beyond bank loans and equity issuance
- Pricing establishes **credit spread**, which signals systemic risk and influence corporate borrowing costs
- Core holdings for central banks (reserves), pensions, insurers, and sovereign wealth funds
- Vehicle for monetary policy operations (open market ops, QE/QT)
- Used in repo, collateralized lending, CLOs, linking credit markets with banking system

Global Perspective

- U.S.: ~\$10T market; deepest and most liquid globally
- Europe: Smaller, bank-dominated corporate funding, growing industry
- Asia: Japan/Korea issue IG bonds; rapidly growing credit market in China
- Trend: Rising issuing of sustainable bonds; EM corporate participation growth

S&P	Moody's	
AAA	Aaa	Investment-Grade Bonds
AA	Aa2	
A	A2	
BBB	Baa2	
BB	Ba2	High-Yield Bonds (Junk Bonds)
B	B2	
CCC	Caa	
CC	Ca	
C	C	
D	D	



Source: Morningstar. Data as of January 11, 2021.

Breakdown of Ratings and Investment Grade

Credit Rating

- Independent assessments of an issuer's **ability and willingness to meet debt obligations**
- Allows for investors to compare credit risk across sectors
- Agencies – S&P, Moody, Fitch
- Ratings typically go across AAA -> D

How They are Adjusted

- Micro-Inputs – Cash flow stability, industry outlook
- Macro – Governance, regulatory environment
- **Credit Migrations- typically happen during M&A, sector shocks, or leverage changes**

Investment Grade

- Majority of the Corporate investment spread
- "The Next Step" in terms of risk appetite outside of Treasuries
- Benefit from Credit Ratings because of access to large scale capital (Pensions, Trusts, etc)
- Newer development – Many ETF funds reference credit ratings for sustainable growth

	Rating	Description
Investment-grade	AAA	Lowest level of credit risk
	AA+, AA, AA-	Very low credit risk
	A+, A, A-	Low credit risk
	BBB+, BBB, BBB-	Moderate credit risk
	BB+, BB, BB-	Substantial credit risk
Non-investment-grade	B+, B, B-	High credit risk
	CCC+, CCC, CCC-	Very high credit risk
	CC	Highly speculative
	C	Highest level of credit risk
	D	Currently in default

	U.S. Equities	Investment Grade Bonds	High Yield Bonds	1-3 Year Treasuries	Commodities
U.S. Equities	1.00	-0.17	0.32	-0.24	0.25
Investment Grade Bonds		1.00	0.13	0.80	-0.08
High Yield Bonds			1.00	-0.02	0.24
1-3 Year Treasuries				1.00	-0.09
Commodities					1.00

Source: Bloomberg, Financial Times

Note:

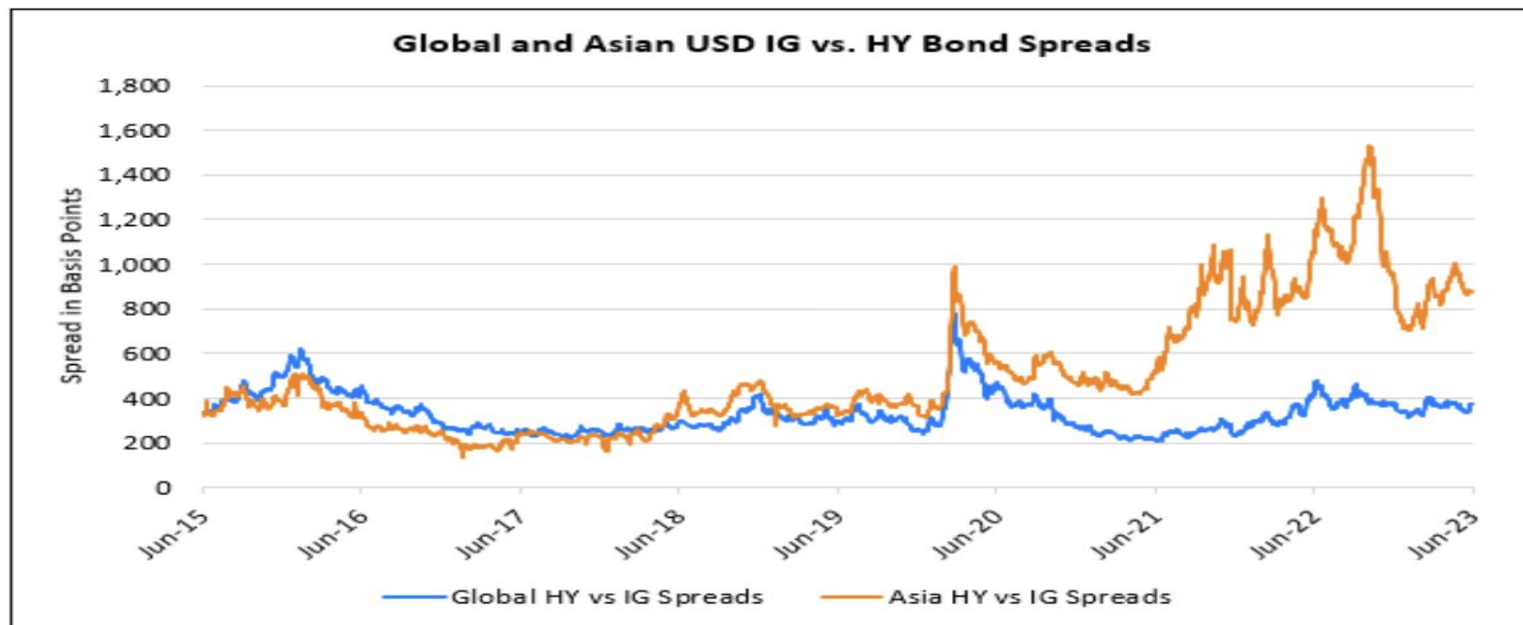
Investment Grade vs High Yield Corporate Bonds

Investment Grade

- Issued by Blue-Chip companies (JPM, APPL, P&G)
- **Tight spread in along Treasuries**
- Risk Profile – very low, <1% default rate
- Used heavily in secondary market for larger scale gains , because of liquidity demand from core base, as well as large benchmark sizes
- Inherently a "sitting" investment but can be played through the secondary market for larger gains.

High Yield

- **Smaller, or More leveraged firms (pre rev)**
- Higher yield, higher spread, especially in stress periods
- Risk profile: 3-4% default rate, jeopardized in crisis periods, spikes up to 15%
- Thinner secondary market because of bond allocation and overall risk exposure to the bonds themselves.
- Popular holdings for Hedge Funds and other Growth Funds



Source: BondValue, Bloomberg | Yield to Worst Differentials Taken

Credit Spread

Definition

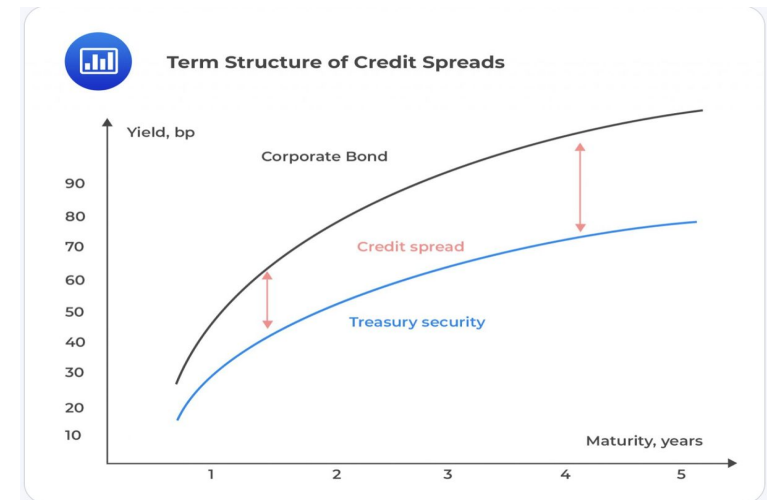
- The yield a credit bond pays above Treasury yield of the same maturity level.
- Compensation for the increased risk exposure in comparison to T-Bills
- **Measured in BPS (Corporate 6% and US Treasury 4% = 200 bps spread)**
- Normal cycle: IG -> 100-150 bps, HY -> 400-600 bps
- Compress in expansionary period, spike in crisis

Role in the Financial System

- Market Stress Indicator
- Macro-Compass – Narrow spreads lead to high consumer confidence, with wide spreads showing consumer wariness of default, poor market health.
- Central Banks use as a gauge in financial conditions, potential intervention (2020 corporate buyback by FED)
- **Announced \$750 billion in potential buyback, only ended up spending \$14 billion**

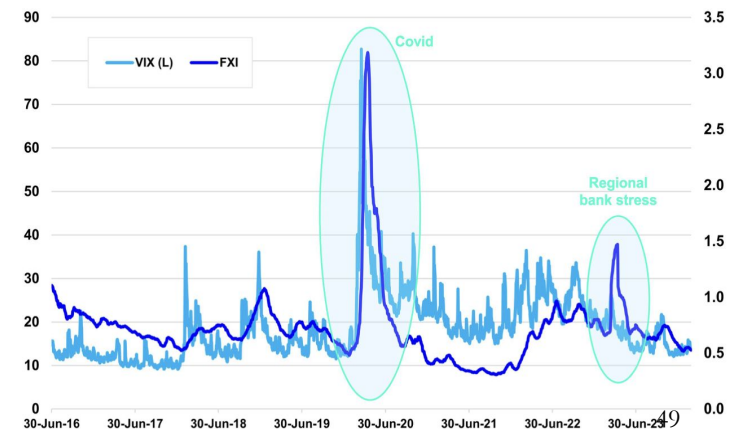
Determinants

- Trend: Rising issuing of sustainable bonds; EM corporate participation growth
- Micro drivers – Leverage and metrics (Debt/EBITDA, Revenue spread)
- Credit Rating – Dependent on Micro drivers more than Macro
- Macro Drivers – Strong GDP + Accommodating FED = tightened spread
- Sector Shocks – affect certain areas dependent on event



Measures of market volatility have moved in tandem with credit spreads in recent months

VIX Index, SOFR Academy's USD Financial Conditions Credit Spread Index (FXI), percent



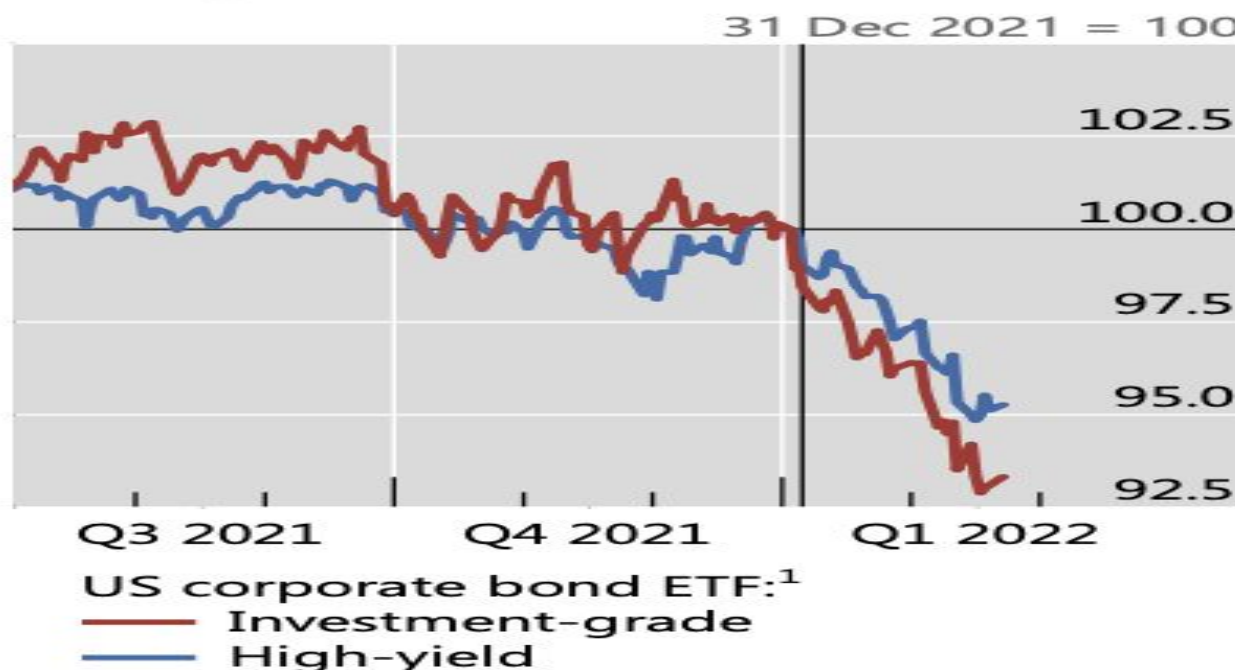
Price–Yield Trade-Off

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Structure of Cash Flows

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- This structural difference explains **why 30-year zeros swing violently while 2-year bills remain stable**

IG bond prices fell more than HY due to longer duration



Risks in Corporate Bonds

Ratings Migration Risk

- **Downgrades from IG -> HY bonds can trigger a forced selling by low-risk funds**
- Spikes spreads for bond and results in large scale losses for smaller investors due to liquidation

Concentration Risk

- Corporate bond issuance is very concentrated within certain markets (Energy, Tech, Finance)
- **Effect – Sector shocks can disproportionately hit the credit market as a result**

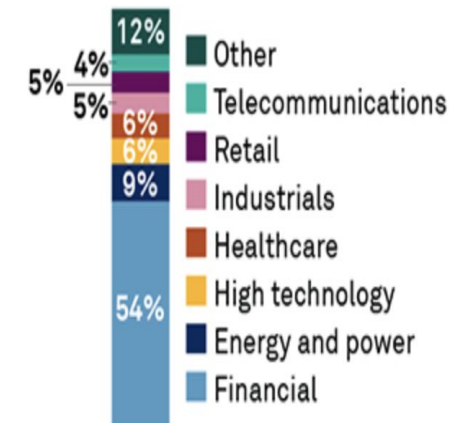
Contagion Risk

- Credit markets share a high correlation with crisis risk
- Investor mentality leads to runs that can affect the whole market even though there is no direct effect on them post event (Lehman 2008)

Average cumulative default rates (%)

Term	1	2	3	4	5 ...	7 ...	10 ...	15
AAA	0.00	0.00	0.07	0.15	0.24 ...	0.66 ...	1.40 ...	1.40
AA	0.00	0.02	0.12	0.25	0.43 ...	0.89 ...	1.29 ...	1.48
A	0.06	0.16	0.27	0.44	0.67 ...	1.12 ...	2.17 ...	3.00
BBB	0.18	0.44	0.72	1.27	1.78 ...	2.99 ...	4.34 ...	4.70
BB	1.06	3.48	6.12	8.68	10.97 ...	14.46 ...	17.73 ...	19.91
B	5.20	11.00	15.95	19.40	21.88 ...	25.14 ...	29.02 ...	30.65
CCC	19.79	26.92	31.63	35.97	40.15 ...	42.64 ...	45.10 ...	45.10

Corporate Bond Issuance by Sector, 2022



Note: May not total 100% due to rounding.
Source: SIFMA

Source: Bloomberg, Financial Times

Note:

Case Study

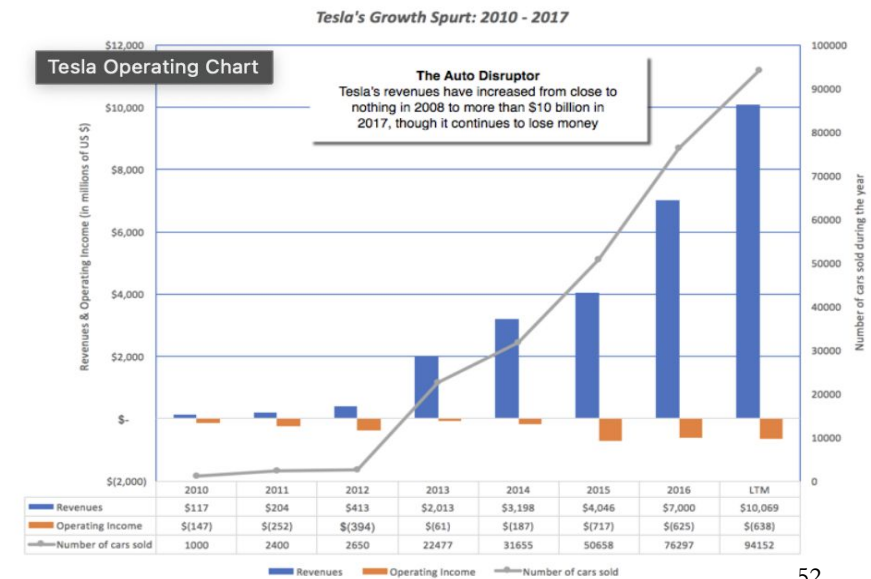
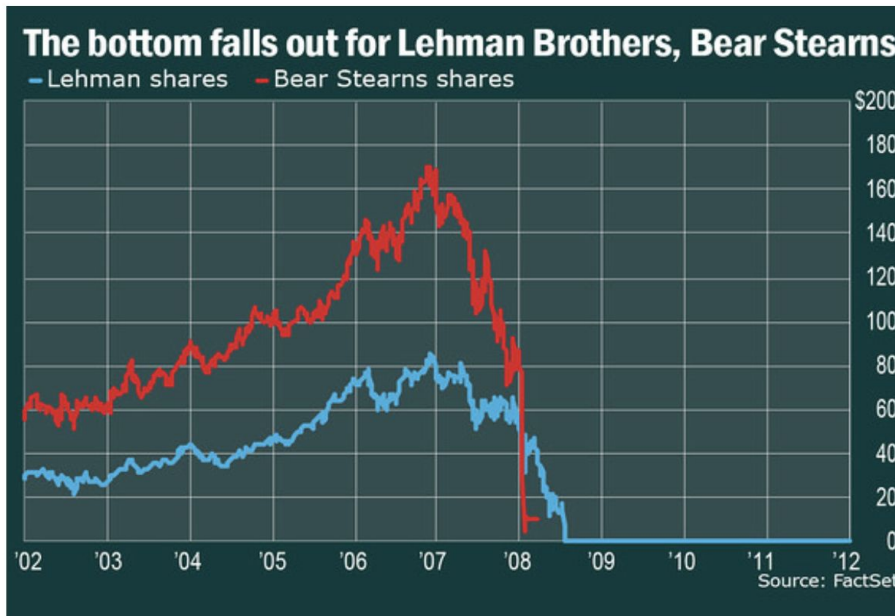


2008 Lehman Brothers Default

- **Lehman collapsed** after bond rating went from 2007 to junk by **Q3 2008**.
- Forced selling – large funds restricted to IG bonds therefore mass hysteria caused liquidation
- **Diversification Fallacy**- Market pooled high-default risk to create a "safe" investment, under the grounds of diversification theory.
- Result: Spread explosion (>1000 bps), Failure of 60/40 portfolio hedge, freezing of corporate bond issuance, **S&P down 38.5%**

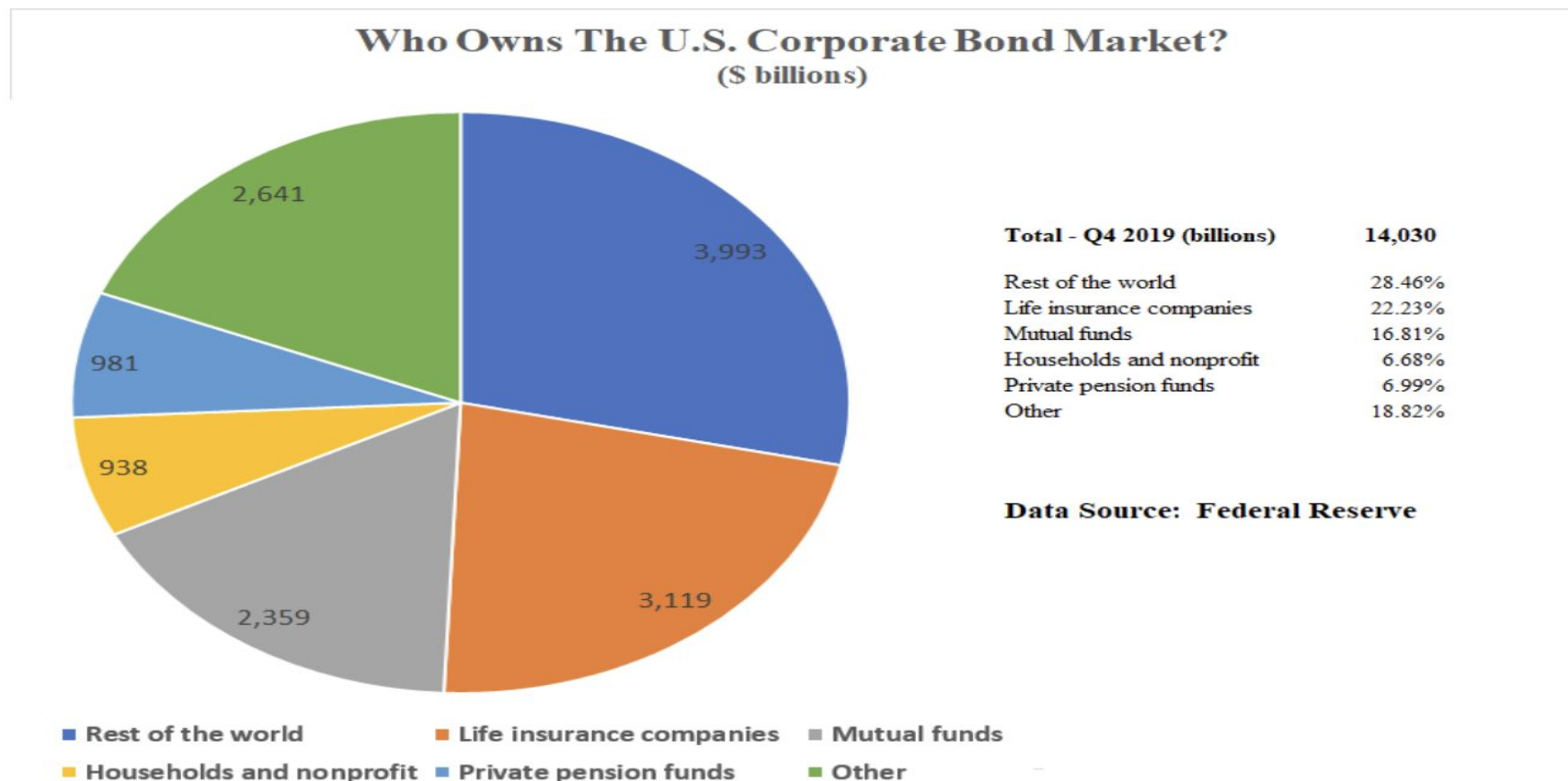
2017 Tesla Credit Jump

- **High Growth Play** - \$1.8B, 8y maturity, 5.3% coupon, B- rated; used to fund MODEL 3
- Regarded as a high-risk investment because of negative cash flow and execution risk
- Demanded by funds looking for a growth story, innovative tech
- As Tesla delivered on their promises and became profitable, credit migrated up sharply to IG standard..
- Yield dropped from 5.3% to 3.5%, **Bonds rallied to \$1.10 on the dollar in secondary market**



Portfolio Management Implications

- **Yield Enhancement-** Relative to treasuries, a key income generator for more aggressive pensions and retirement portfolios
- Can reduce portfolio volatility overall, but subject to equity-like movement in rating migrations (mass sell-offs)
- **Shift allocation based on market.** IG -> downturns, HY -> recoveries
- Spread analysis- Widened margins lead to contrarian entry opportunities, while tight spreads are defensive
- Biggest Risk – Automated portfolio selloff through large funds





Agenda



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Sector Rotation Framework	7
Investor Preferences	8
How Macro Affects Equities	10
Technical Analysis	15
Case Study	16
Portfolio Management Implications	17

Introduction to Equities

Introduction

- Equities represent **ownership** in a company's future earnings and growth
- Stock returns come from **price appreciation** and **dividends**
- **Defensive sectors** hold steady during downturns; **cyclical** sectors outperform in expansions
- Sector and style rotation occur as investors reposition based on changes in economic conditions

Top-Down Analysis

- **Top-down investing** starts with the overall economy and works down to individual stocks
- Focus areas include **GDP growth, inflation, interest rates, and monetary policy**
- The process moves from economy to asset classes to sectors to industries to companies
- The goal is to **align** sector and style exposure with the **current** phase of the business cycle
- **Bottom-up analysis** takes the opposite path, starting with company fundamentals

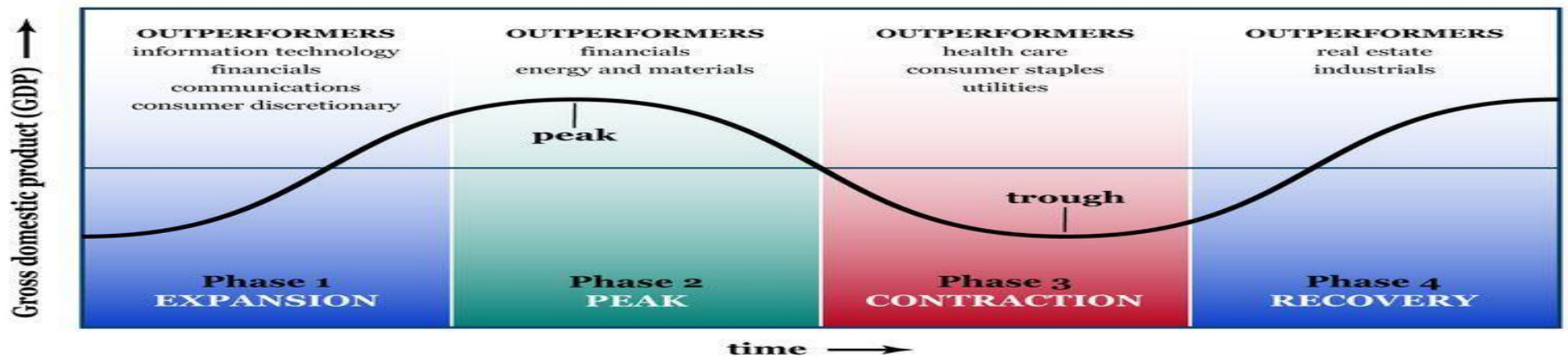


The Business Cycle

Four Phases of Economic Activity

- The economy moves through recurring phases of **expansion, peak, contraction, and recovery**
- **Expansion:** GDP and credit growth strengthen, confidence improves, and cyclical sectors like technology and financials lead
- **Peak:** Growth and inflation reach their highs as central banks tighten, with financials, energy, and materials performing best before slowing
- **Contraction:** Output and spending decline, inflation eases, and defensive sectors such as healthcare, consumer staples, and utilities hold up
- **Recovery:** Activity rebounds, rates stabilize, and early-cycle sectors including industrials and real estate regain momentum

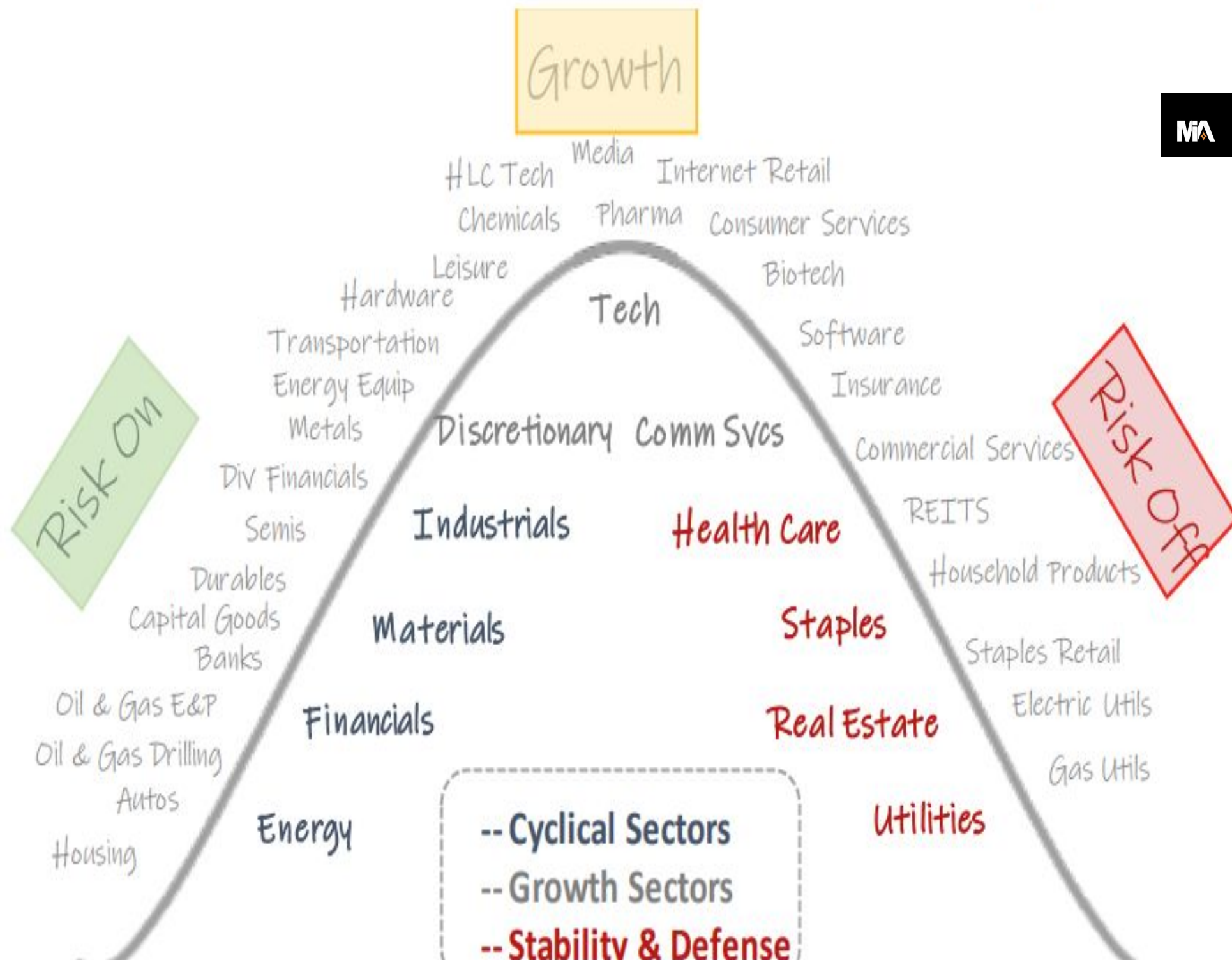
Four phases of an economic cycle

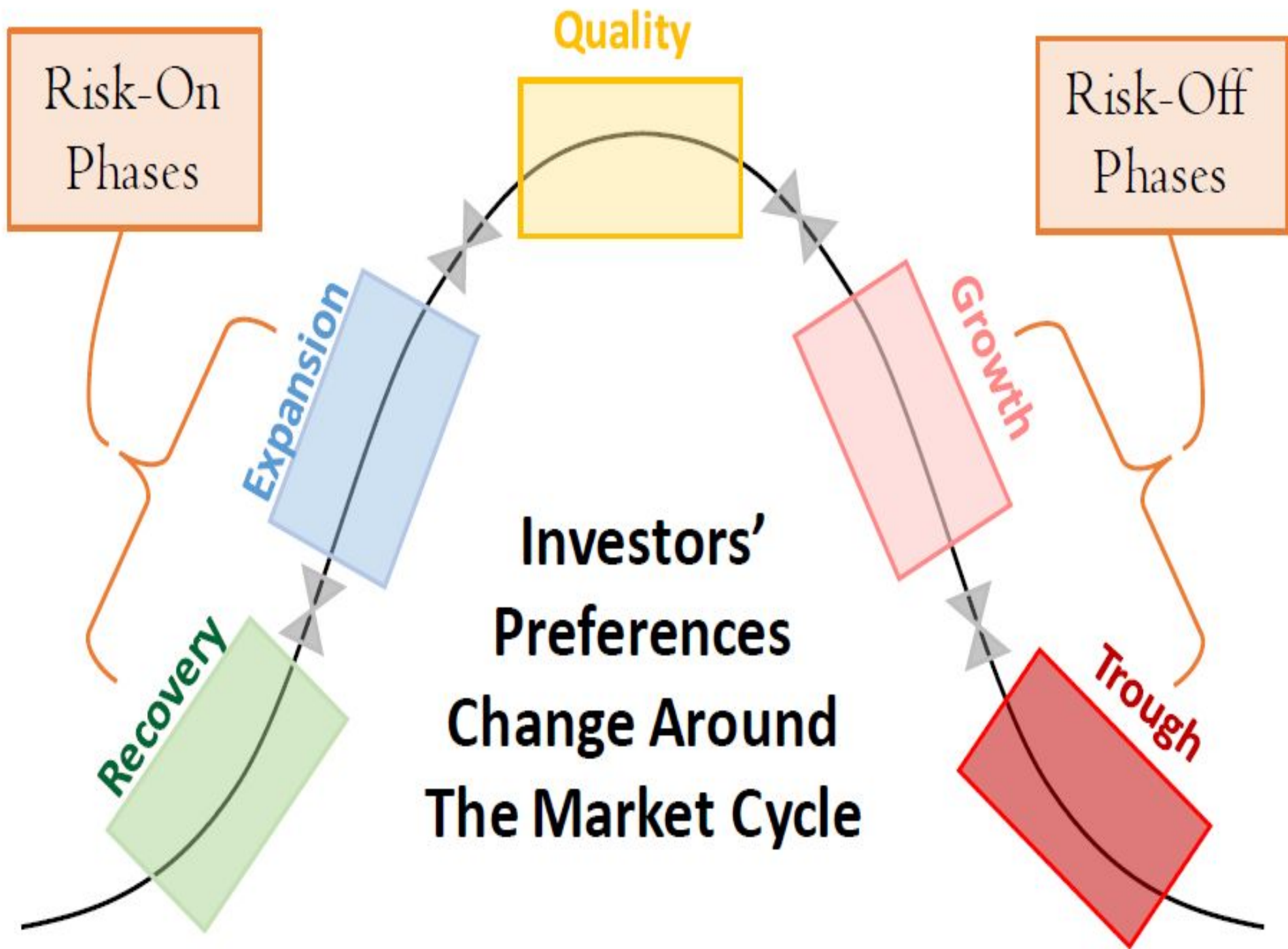


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Source: Bloomberg, Financial Times, WSJ, Reuters

Note:





Investor Preferences



Risk-On Phases

- **Recovery:** Investors shift from cash to cyclical assets (industrials, consumer discretionary, small-caps) as growth returns
- **Expansion:** Confidence builds; capital moves into growth and financial sectors, driven by earnings acceleration and easy liquidity
- **Quality:** As growth peaks, capital shifts toward companies with strong balance sheets and steady cash flow

Risk-Off Phases

- **Growth:** Investors favor quality and large-cap growth as earnings stabilize and monetary easing supports recovery momentum
- **Trough:** Defensive and value sectors (healthcare, staples, utilities) maintain leadership until confidence returns and cyclicals re-emerge



Source: Bloomberg, Financial Times, WSJ, Reuters

Note:

Defensive Sector Overview

Definition: Sectors with consistent earnings and demand regardless of economic cycles

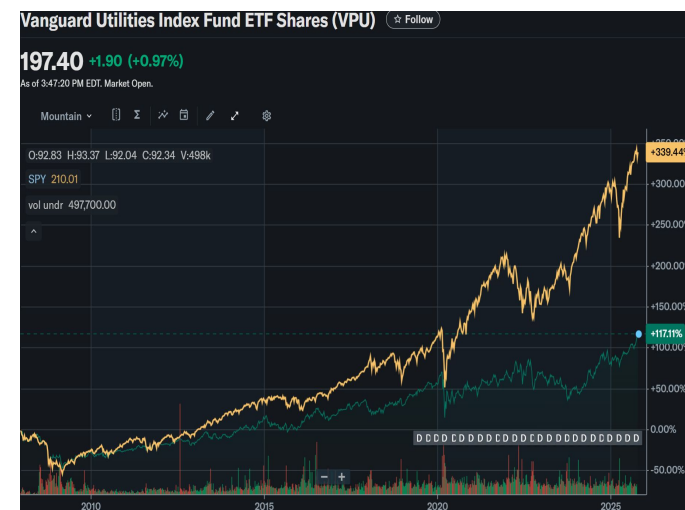
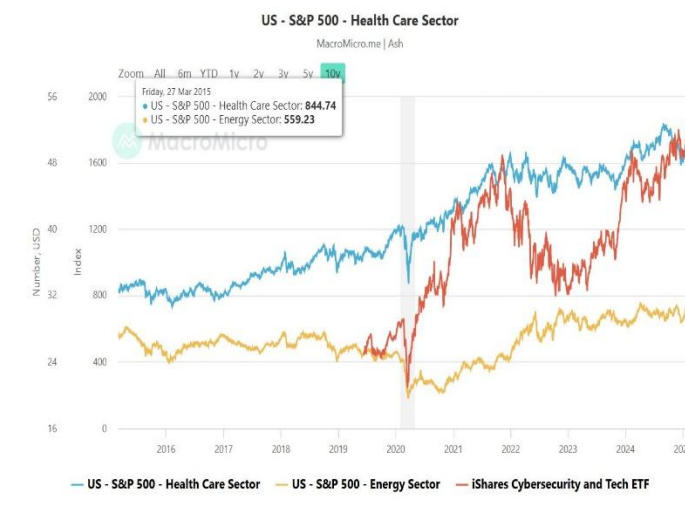
Key Traits: Low volatility, steady dividends, high regulation, essential goods/services

Health Care

- Healthcare is a **defensive** sector with stable demand across all economic conditions
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- Aging populations** drive long-term demand for pharmaceuticals and medical services
- The sector remains **resilient** during recessions as medical needs persist regardless of the economy
- Government spending through **Medicare, Medicaid, and subsidies** impacts provider and pharma revenues
- Policy reforms** and **healthcare regulations** create both risks and investment opportunities

Utilities

- Utilities are a classic **defensive** sector with stable demand and predictable cash flows
- Higher interest** rates increase borrowing costs for capital-intensive infrastructure projects
- Lower rates** make financing cheaper, supporting expansion and dividend stability
- Rising inflation** raises operating and maintenance costs, while **lower inflation** eases input pressures and supports margin stability
- Energy prices and fuel costs** affect profitability and rate-setting decisions
- Government **policy** and **regulation** drive investment toward renewables and grid upgrades
- During recessions, utilities typically **outperform** due to consistent consumer demand



Source: Bloomberg, Financial Times, WSJ, Reuters

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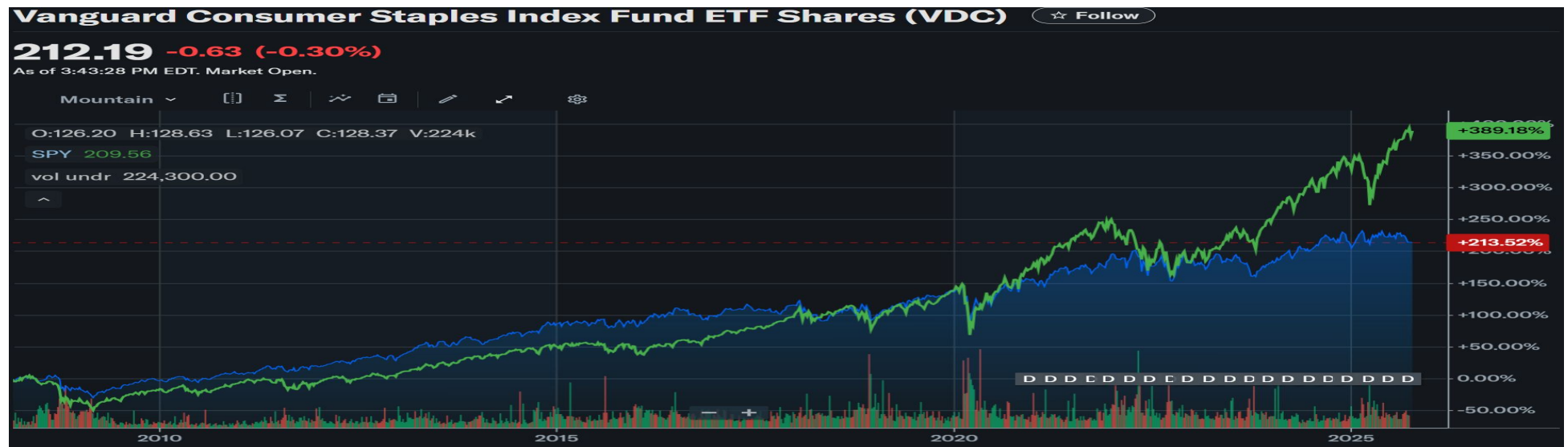
Defensive Sector Overview

Definition: Sectors with consistent earnings and demand regardless of economic cycles

Key Traits: Low volatility, steady dividends, high regulation, essential goods/services

Consumer Staples

- Consumer staples are a **defensive** sector driven by steady demand for essential goods
- Demand remains **stable during recessions** as households prioritize necessities
- **Higher inflation** raises input and packaging costs but allows selective price increases to protect margins
- **Lower inflation** stabilizes costs and supports stronger profitability through improved purchasing power
- **High interest rates** have limited impact given predictable cash flows and strong balance sheets, while **lower rates** enhance dividend appeal and valuation support
- Consistent dividends and reliable earnings make the sector **attractive** in volatile markets



Source: Bloomberg, Financial Times, WSJ, Reuters

Note:

Growth Sector Overview

Definition: Firms with high earnings potential driven by innovation and market expansion

Key Traits: High valuations, reinvestment focus, rate sensitivity, strong in low-rate periods

Technology

- Technology is a **semi-cyclical sector** with moderate correlation to the overall economy
- Lower interest rates** encourage venture capital funding and corporate investment
- Higher rates** make it harder for firms to raise capital and slow innovation spending
- Low borrowing costs** support research and development of new technologies
- Privacy laws, antitrust actions, and cybersecurity regulations impact profitability
- Global trade policies and semiconductor supply chains influence production and costs
- Macro stability supports long-term tech growth, while volatility disrupts demand and funding

Communications

- Communication services are **semi-cyclical**, balancing stable subscriptions with advertising exposure
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- Lower rates and cooling inflation** support digital investment and media expansion
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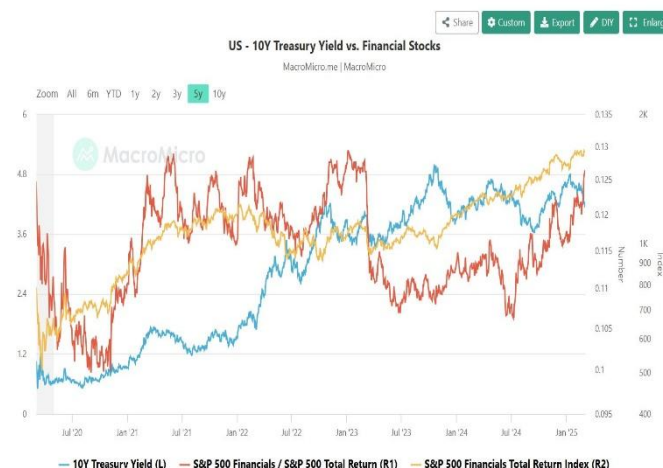
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Financials

- Cyclical sector heavily influenced by **economic growth, credit conditions, and rate policy**
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- When **inflation rises**, banks face higher costs and uncertain loan demand; when **inflation falls**, stability returns, and profit margins improve
- **Market cycles** drive volatility in investment banking, trading, and asset management revenues

Industrials

- Highly cyclical sector tied to **GDP, investment, and trade flows**
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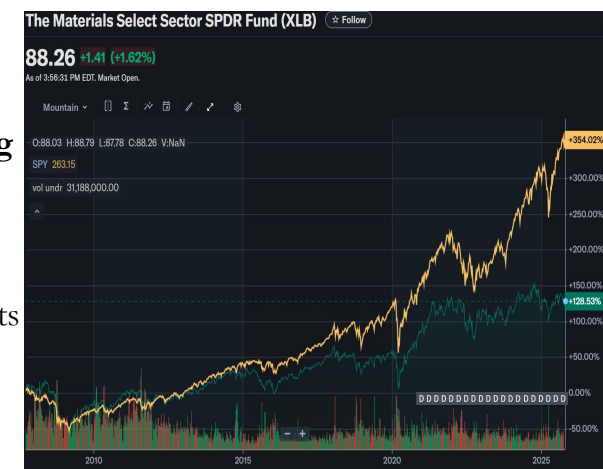
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Energy

- **Highly cyclical**, macro-driven sector tied to global demand and commodity prices
- **Oil and gas prices fluctuate with geopolitical events**, OPEC policy, and regulation
- **Higher inflation** boosts revenues but raises input and exploration costs, while **lower inflation** stabilizes expenses and reduces price swings
- **Higher rates** increase financing costs for capital projects, while **lower rates** support investment and production growth
- **Energy transition policies, carbon taxes, and ESG mandates** shape profitability and long-term capital allocation
- **Geopolitical shocks and supply disruptions** drive volatility

Materials

- Highly cyclical, macro-sensitive sector driven by **industrial production, construction, and global trade**
- Demand for metals, chemicals, and building materials **rises in expansions and contracts during slowdowns**
- **Commodity prices, input costs, and supply chain dynamics** heavily influence margins and profitability
- **Higher inflation** increases prices and supports producers but raises energy and raw material costs
- **Lower inflation** stabilizes input prices and improves cost visibility, supporting steady margins
- **Higher rates** slow construction and capital spending, while **lower rates** stimulate demand for materials and infrastructure investment



Source: Bloomberg, Financial Times, WSJ, Reuters

Note:

Technical Analysis



Why it is useful?

- Improves timing by **identifying entry and exit points** using trends, momentum, and volume signals
- **Confirms direction** by measuring trend strength and spotting early reversals
- **Enhances risk control** with clear stop loss and target levels to manage downside and protect gains
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Indicators

- **Support & Resistance:** Show where price tends to pause or reverse; breakouts indicate new momentum
- **Volume:** Confirms conviction behind price moves as high volume validates trend continuation
- **Trendlines:** Reveal direction and strength of a market trend; breaks suggest potential reversals
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Source: Bloomberg, Financial Times, WSJ, Reuters

Note:

Case Study - Post-COVID Economic Recovery (2020–2022)

After the 2020 market crash, unprecedented fiscal and monetary stimulus triggered one of the fastest recoveries in modern history as investors rotated aggressively across styles and sectors as the macro environment evolved

Phase 1: Early Recovery (Mid-2020 – Early 2021)

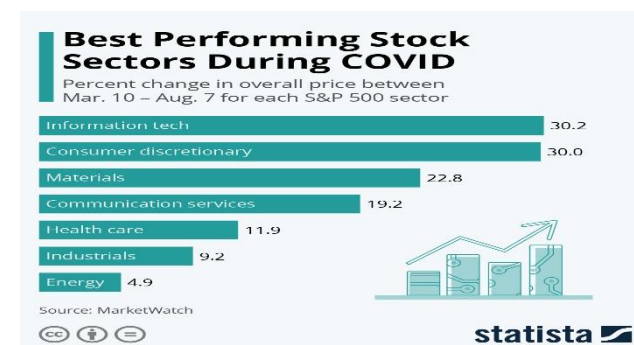
- The economy rebounded quickly after COVID, helped by record stimulus and near-zero interest rates
- Investors moved out of defensive sectors like healthcare and staples into cyclical and growth areas
- Technology, consumer discretionary, industrials, and small-caps led as optimism returned
- Growth and momentum strategies outperformed as earnings recovered

Phase 2: Expansion to Late Cycle (2021 – Mid-2022)

- As growth continued, inflation surged and the Fed began raising rates
- Investors shifted away from expensive growth stocks toward value and quality names
- Energy, financials, and materials benefited most from higher prices and stronger demand
- Value and quality factors led as markets focused on steady earnings and cash flow.

Phase 3: Slowdown & Risk-Off (Late 2022 – 2023)

- Growth slowed as higher interest rates made borrowing harder, tightening credit and cooling market momentum
- Investors turned cautious, favoring defensive and dividend-paying sectors
- Healthcare, utilities, and consumer staples held up best
- Low-volatility and dividend yield strategies regained leadership as safety became the priority



2 Year Treasury Rate (1:2YTCMR)
3.52% for Oct 10 2025



Source: Bloomberg, Financial Times, WSJ, Reuters

Note:

Portfolio Management Implications



Macro Drives Allocation

- Economic outlook, inflation trends, and policy shifts **shape asset allocation** across sectors and styles.
- Investors **overweight** sectors expected to outperform in the current phase (e.g., cyclicals in early expansion) and **underweight** laggards (e.g., defensives during growth peaks).
- Align positioning with **macro regime**: growth, inflation, liquidity, and rate direction

Sector & Style Rotation

- Equity sectors perform differently across the business cycle; rotating exposure boosts returns and limits drawdowns.
- In expansions, favor growth and cyclicals (tech, industrials); in slowdowns, shift to defensives and value (healthcare, staples).
- Style factors (growth, value, quality, momentum) rotate with changes in rates and risk appetite.

Technical Analysis for Execution

- Use **trendlines, support/resistance, and volume confirmation** to refine entry and exit points
- Momentum and relative strength indicators** help identify sector rotation early and confirm trend durability
- Combine macro-driven top-down allocation with technical signals** for precision timing and risk control

Bull Case

Industry	Weight	S&P 600	Post Weight
Energy	EW	4.50%	4.50%
Materials	OW	5.00%	5.50%
Industrials	OW	18.00%	18.75%
Consumer Discretionary	OW	13.00%	13.50%
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Healthcare	UW	12.00%	11.00%
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Information Technology	OW	12.00%	13.00%
Communication Services	UW	3.00%	2.50%
Utilities	UW	2.50%	1.75%
Real Estate	EW	8.00%	8.00%
Total		100.00%	100.00%

Bear Case

Industry	Weight	S&P 600	Post Weight
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Information Technology	EW	13.00%	13.00%
Communication Services	EW	3.00%	3.00%
Utilities	OW	2.00%	2.50%
Real Estate	EW	7.50%	7.50%
Total		100.00%	100.00%

Source: Bloomberg, Financial Times, WSJ, Reuters

Note:

MACRO INVESTING ASSOCIATION

Lecture-5 Intro to Equity Markets





Agenda



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Introduction to Equities

Introduction

- Equities represent **ownership** in a company's future earnings and growth
- Stock returns come from **price appreciation** and **dividends**
- **Defensive sectors** hold steady during downturns; **cyclical** sectors outperform in expansions
- Sector and style rotation occur as investors reposition based on changes in economic conditions

Top-Down Analysis

- **Top-down investing** starts with the overall economy and works down to individual stocks
- Focus areas include **GDP growth, inflation, interest rates, and monetary policy**
- The process moves from economy to asset classes to sectors to industries to companies
- The goal is to **align** sector and style exposure with the **current** phase of the business cycle
- **Bottom-up analysis** takes the opposite path, starting with company fundamentals

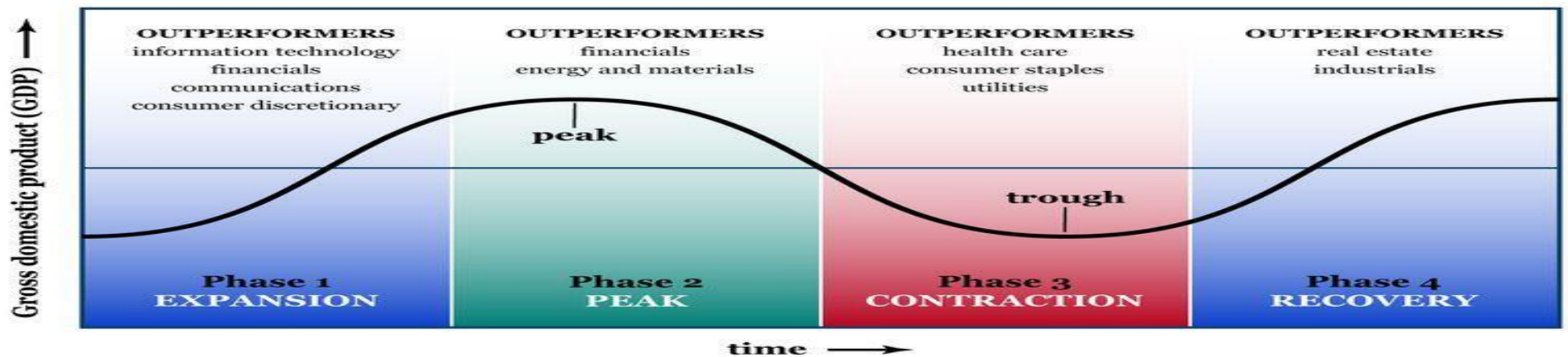


The Business Cycle

Four Phases of Economic Activity

- The economy moves through recurring phases of **expansion, peak, contraction, and recovery**
- **Expansion:** GDP and credit growth strengthen, confidence improves, and cyclical sectors like technology and financials lead
- **Peak:** Growth and inflation reach their highs as central banks tighten, with financials, energy, and materials performing best before slowing
- **Contraction:** Output and spending decline, inflation eases, and defensive sectors such as healthcare, consumer staples, and utilities hold up
- **Recovery:** Activity rebounds, rates stabilize, and early-cycle sectors including industrials and real estate regain momentum

Four phases of an economic cycle



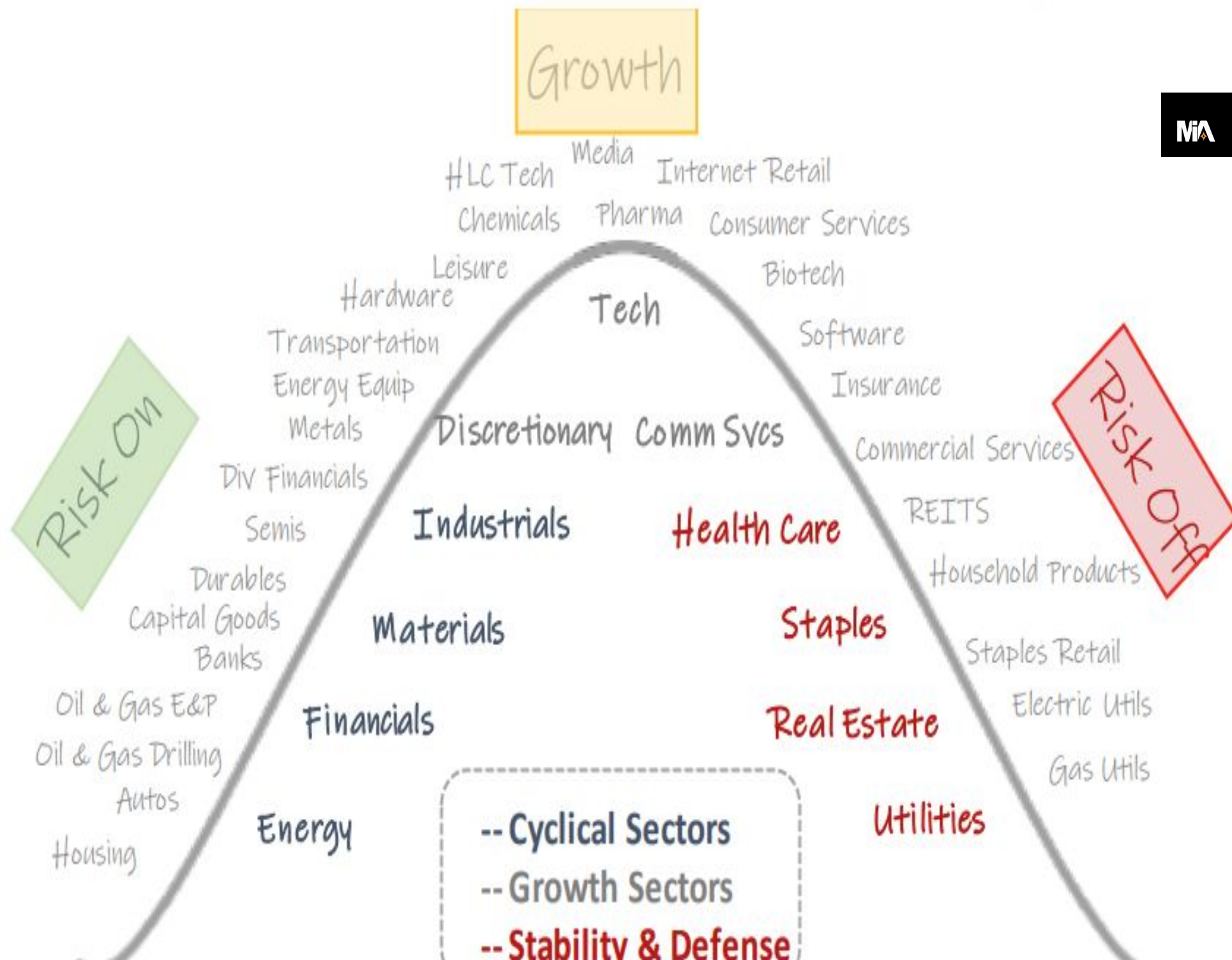
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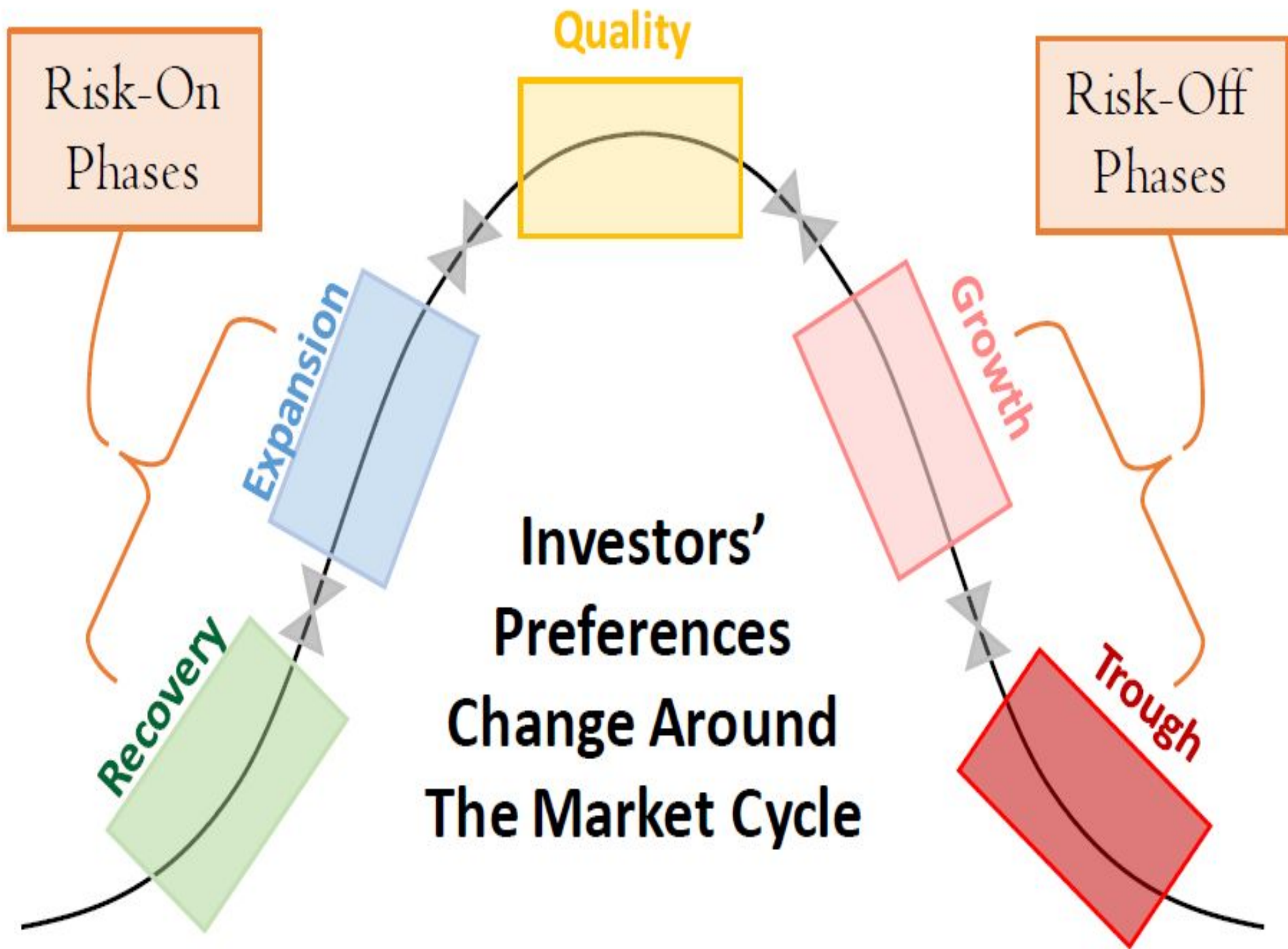
Source: Bloomberg, Financial Times, WSJ, Reuters

Note:

Sector Rotation Guide Around The Business Cycle

NIA





Investor Preferences

Risk-On Phases

- **Recovery:** Investors shift from cash to cyclical assets (industrials, consumer discretionary, small-caps) as growth returns
- **Expansion:** Confidence builds; capital moves into growth and financial sectors, driven by earnings acceleration and easy liquidity
- **Quality:** As growth peaks, capital shifts toward companies with strong balance sheets and steady cash flow

Risk-Off Phases

- **Growth:** Investors favor quality and large-cap growth as earnings stabilize and monetary easing supports recovery momentum
- **Trough:** Defensive and value sectors (healthcare, staples, utilities) maintain leadership until confidence returns and cyclicals re-emerge



Source: Bloomberg, Financial Times, WSJ, Reuters

Note:

Defensive Sector Overview

Definition: Sectors with consistent earnings and demand regardless of economic cycles

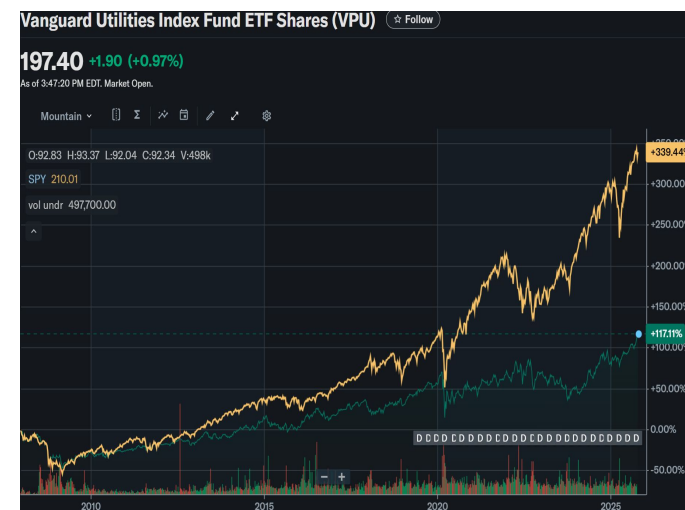
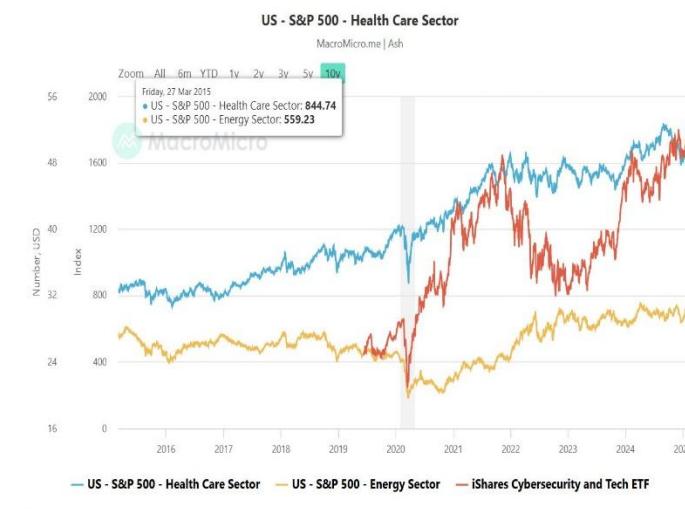
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Health Care

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Utilities

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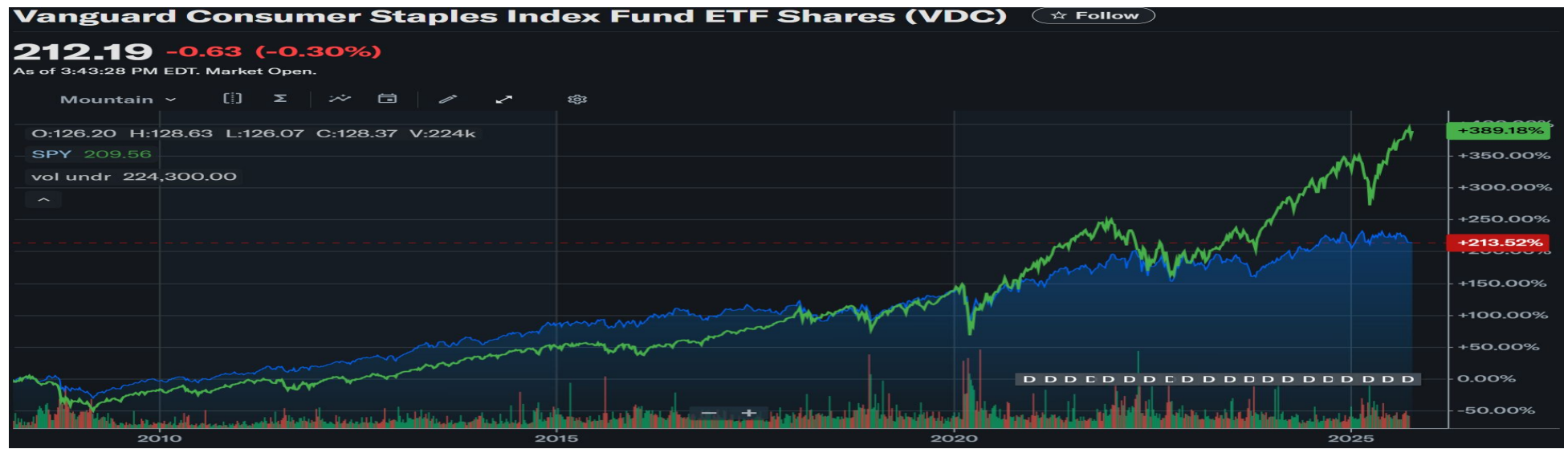
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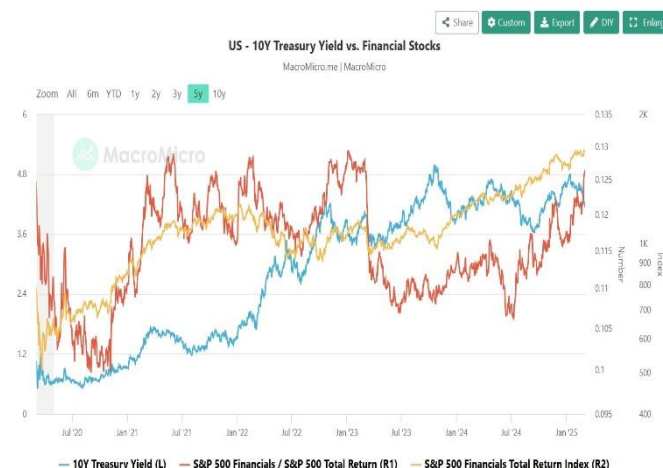
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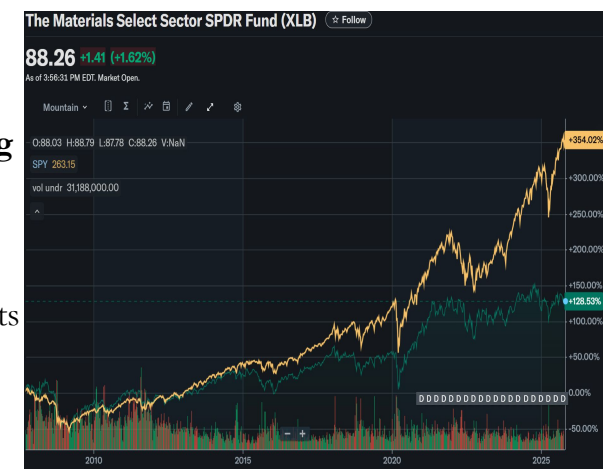
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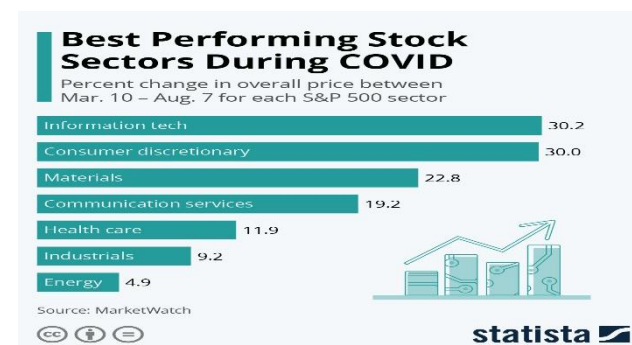
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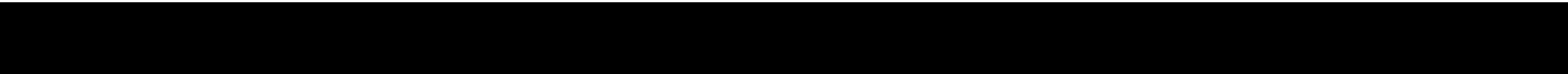


MACRO INVESTING ASSOCIATION

Lecture-7 G7 Market Equities



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Valuation & Performance Trends	9
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INTRO TO G7 MARKETS

Introduction to G7 Markets

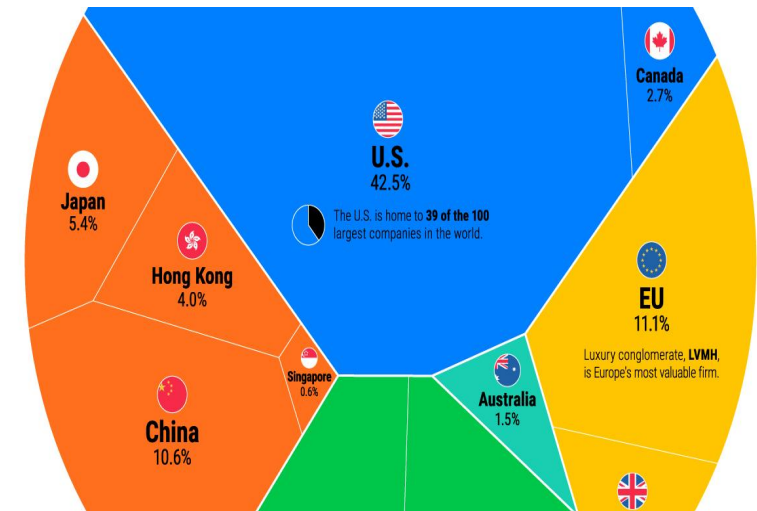
- United States, United Kingdom, Japan, Germany, France, Italy, Canada
- Represent the world's most advanced developed economies, coordinating on fiscal, monetary, and trade policy
- **Collectively account for approximately. 24-50% of global GDP, as well as 60% equity market capitalization**

Role in Financial Markets

- **The core of global equity allocation, base foundation of the MSCI and major ETFs**
- Institutional anchor for capital flows, sets tone for global risk appetite
- Largest and most liquid capital markets, which influence overall market momentum through US, Japan, and the UK
- Safe Haven – In volatile periods, capital flows into G7 equities and bonds, serving as a non-systemic mitigator
- Innovation – G7 exchanges (NYSE, LSE, TSE, Deutsche Borse) serve as jump starters for IPOs and tech growth

Market Characteristics

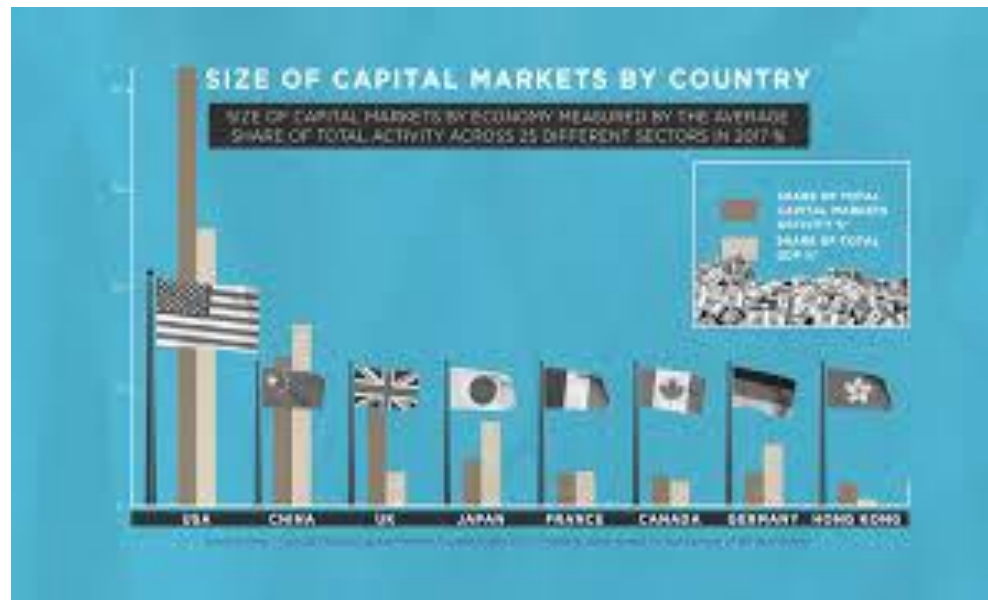
- USA – Innovation-driven, tech-heavy, deepest capital markets
- Europe (UK, Germany, France, Italy): Industrial base, dividend culture, rate sensitivity
- Japan – Export led, yen-sensitive, very strong reformative momentum
- **Canada – Energy and Commodities focused, strong banking sector**



Market Characteristics

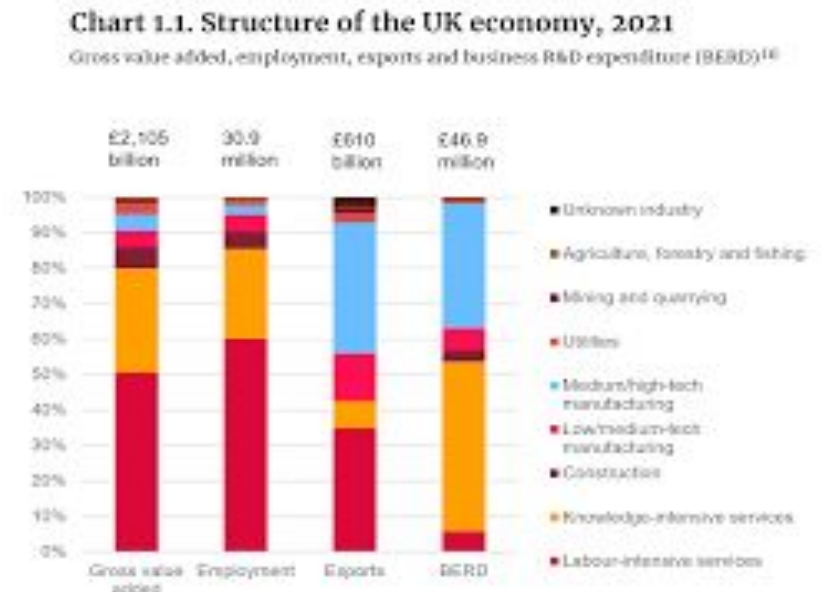
USA

- **Largest equity market globally -> 60% world market cap**
- Dominated by large scale tech, healthcare, and consumer discretionary sectors
- Thrives through innovative cycles and capital market depth
- Valuation – Trades at a premium (S&P 500 P/E 20x) due to growth bias and influence in global tech and AI
- Corporate Giant – High buyback activity, larger EPS growth, share-holder friendly



United Kingdom

- **Dividend-heavy (FTSE 100 yield approximately 4%)**
- Sector comp – Energy, materials, and financials -> value oriented
- High exposure to global commodity cycles and FX fluctuations (GBP/USD)
- Equity behavior – Defensive in global selloffs, big performers when commodities are up
- Shift more towards growth equities post Brexit



Continued



Japan

- Performance is heavily associated with USD/JPY trends, therefore export-driven
- Dominated by Corporate Conservatism, very stable, but slow growth as a nation
- **Beginning shift as an equity driven market "Quiet bull market"**

Eurozone

- Cyclical Markets, reliant on exports and energy input costs
- Germany – Engineering, and Autos. (DAX)
- France – Luxury and Aerospace with energy (CAC 40)
- Italy – Financials and utilities heavy (FTSE MIB)

Canada

- Energy, materials, and financials make up 70% of TSX
- Returns strongly linked to global commodity cycles
- Hence, Outperforms during commodity booms, underperforms in growth rallies)
- Believes in high dividend payout through a strong banking system



Source: Bloomberg, Financial Times

Note:



MACRO DRIVERS

Macro-Drivers for equities

Growth & Earnings Cycles

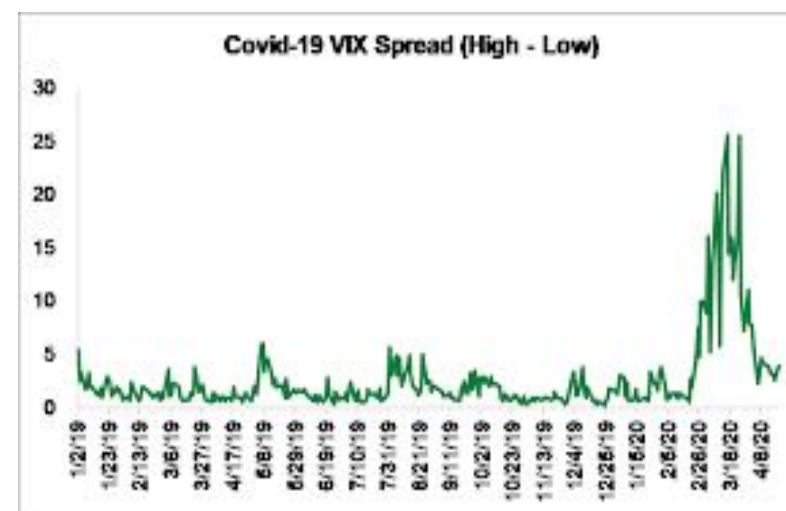
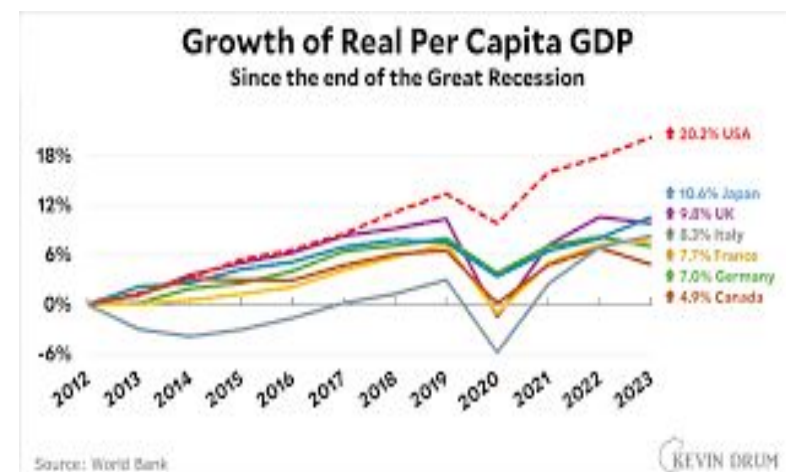
- Equity returns track GDP growth and Corp. Earnings momentum
- USA and Eurozone more aligned with EPS while Japan and Canada more with Export and Commodity cycle
- Slowing Growth -> EPS downgrades and rotation to defensive sectors (Utilities, Healthcare)
- Accelerating growth -> cyclical outperformance (Industrials, Tech)

Inflation Dynamics

- Slight/moderate inflation is equity supportive (through pricing power and growth)
- High inflation compresses P/E as discount rates rise
- UK & Eurozone – Value-oriented, benefit from resilience in energy and materials
- USA – Valuation sensitive, high inflation triggers FED tightening -> equity reprice, typically unfavorable

VIX – Market Volatility

- **VIX (US) and VSTOXX (Europe) measure expected market volatility derived from equity index options (S&P 500, Euro STOXX 50)**
- Called a "fear gauge" for rising volatility because of market stress
- VIX rises -> high volatility -> shift into defensive stocks due to risk aversion
- VIX lowers ->: lower volatility -> rotate into cyclical and return on bullish behavior
- **Different Regional sentiment (VXJ based more on yen sensitivity, VIX global, VSTOXX on ECB uncertainty)**



Valuation & Performance Trends

Relative Valuations Across G7

P/E Valuation is an excellent way to decipher growth ratings and valuations on a broad scale

USA

- Trades at a structural premium (P/E 20-22x) due to tech dominance, profitability potential, and capital efficiency
- Forward earnings multiple remains above the historical median, which establishes as global growth leader

UK and Eurozone

- Discounted valuations (P/E 12-14x) due to cycle exposure, slower earnings growth, and regulation friction
- Bias on Value leads to higher dividend yields (3-4%)

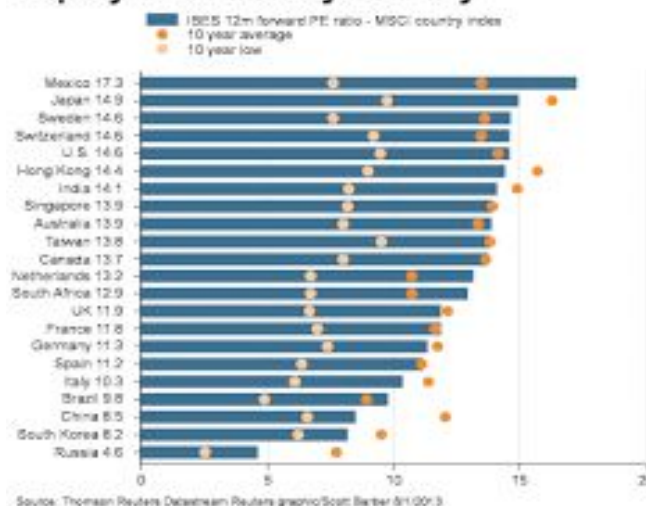
Japan

- Long undervalued at P/E 16x, but reforms and governance push driving a re-rating
- Low return on equity – often sub 6%, in comparison to 10+ % returns on US equities
- Shift in investor agenda has created a boost in Japanese growth (2024 -9 Trillion Yen share buybacks, instead of hoarding)

Canada

- Trades at a moderate multiples (P/E 13x), reflecting commodity-linked market structure
- Offers 3-4% dividend yield, attractive for income investors
- Leveraged widely on oil and metal growth, therefore is supportive of global growth

Equity valuation by country



World - MSCI PE Ratio by Country

MacroMicro.me | MacroMicro



Continued

Sectoral Performance Trends

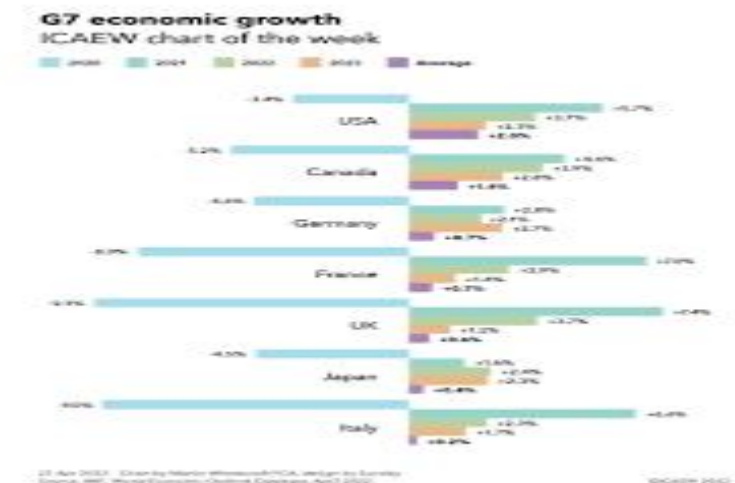
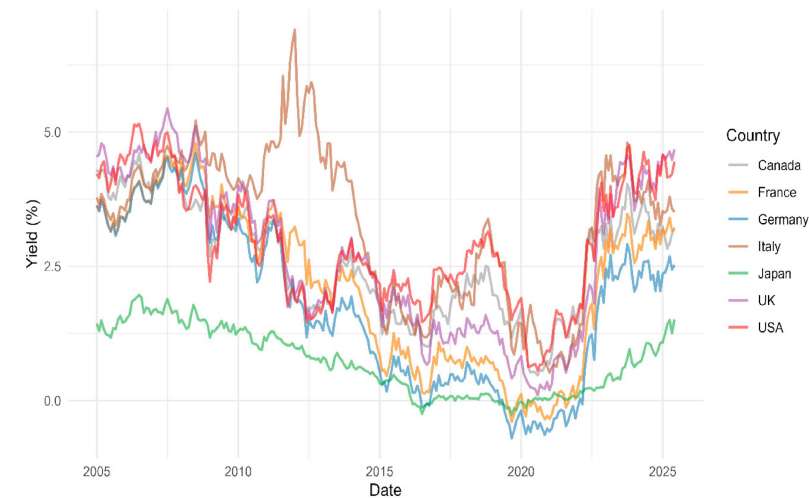
- Post-2020: US mega-cap tech drove global equity gains (AI, digital transformation)
- 2022-2023: Rate hike led to rotation into value – energy, financials, and industrials outperformed
- 2024 – present: Renewed growth leaders in tech in US in Japan has led Europe to lag behind because of policy uncertainty and weak productivity
- Defensive Sectors perform best during inflation or rate uncertainty

Earnings Growth & Return Dispersion

- US – EPS growth >10% CAGR over past decade; driven by innovation and share buybacks
- Europe – Slower EPS trajectory, fragmented markets, lower tech exposure
- Japan – Strong rebound post-reform; ROE rising from 6% to 10% in decade span
- Divergence across G7 now driven more by **sector mix** than geography itself

Market Correlation & Diversification

- **G7 equities remain highly correlated during global shocks (2008,2020), but diverge during recovery cycles**
- Japan and UK offer low beta diversification due to FX and sector structure
- Combining G7 exposures reduces volatility and enhances Sharpe ratios in global portfolios



Policy & Structure

Monetary Policy Divergence

US FED

- Aggressive tightening cycle (2022-2024) raised real yields and compressed growth-stock valuations
- **Rate-cut expectations now driving a rotation back toward tech and duration-sensitive equities**

European Central Bank

- Cautious normalization; equities remain rate-sensitive and cyclical, especially financials and industrials

Bank of Japan

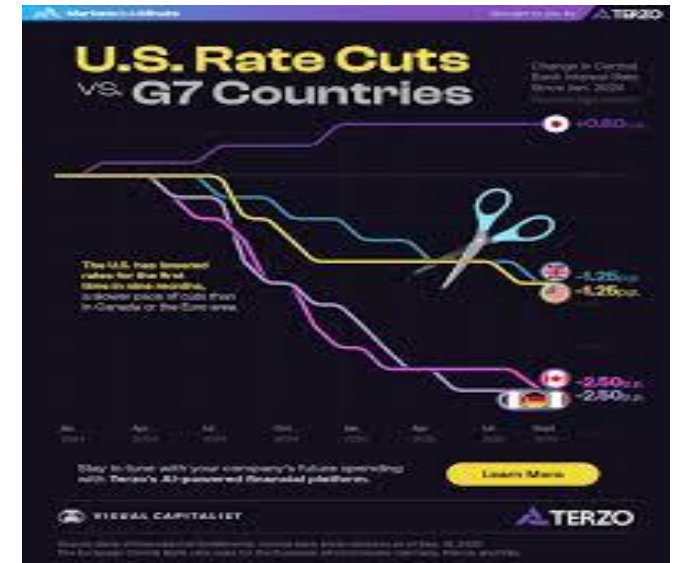
- Long-running yield-curve control and negative-rate policy keeps liquidity abundant
- Weak yen boosts exporters and global investors' returns

Bank of England & Bank of Canada

- Balancing inflation control with fragile growth; policy direction directly impacts dividend-heavy sectors (banks, energy, housing)

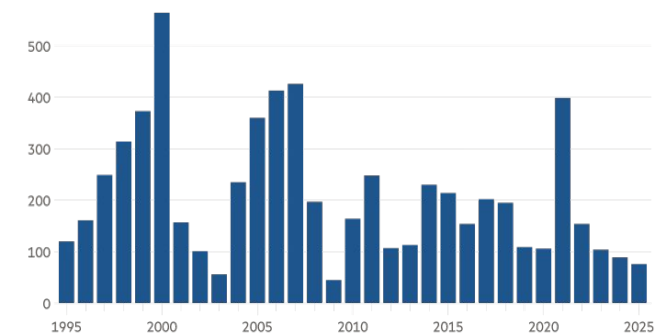
Capital Market and Governance Evolution

- Japan – Ongoing corporate-governance reforms (TSE directives) improving ROE and global investor pull
- UK & Europe – Listing reforms(London,Paris) aimed at reviving IPO activity post-Brexit and amid stricter EU regulation



Europe's IPO market remains downbeat

Number of listings



Year to October 9 2025
Source: Dealogic

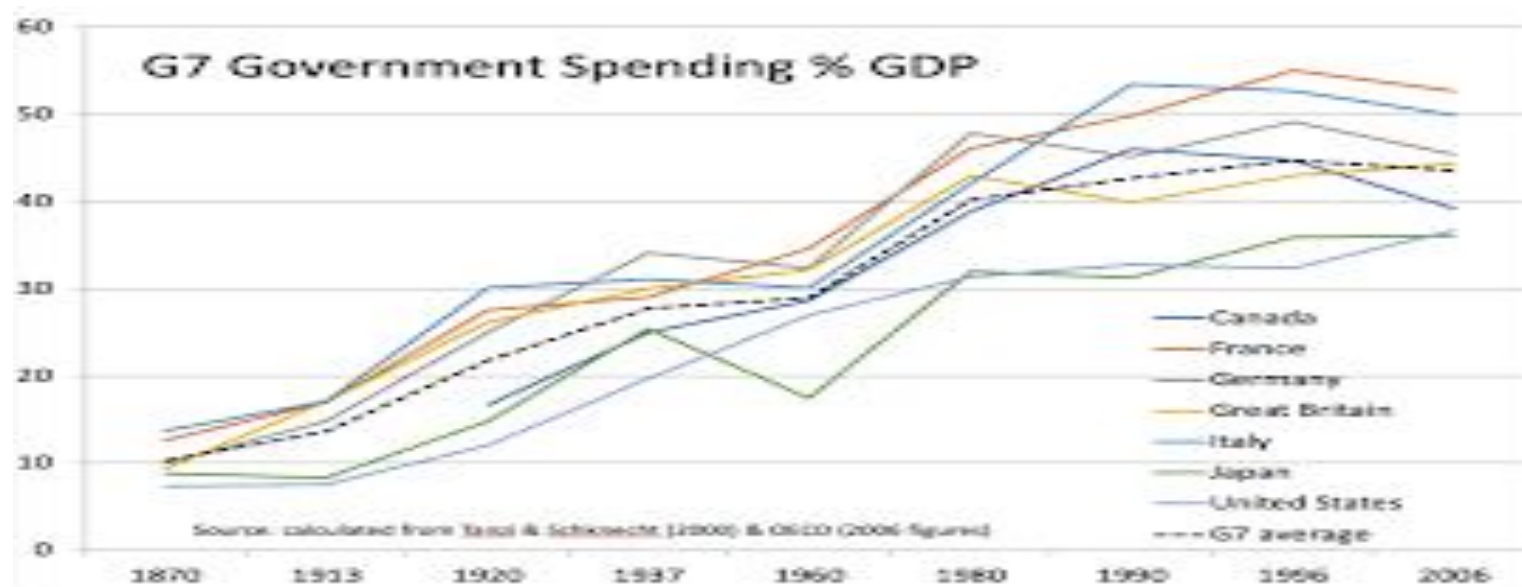
Continued

Fiscal Strategy & Gov Spending

- US – Structural deficits and industrial policy fueling on-shoring and capex-driven equity opportunities
- **Europe – Targeted fiscal support for energy transition and defence; fiscal fragmentation limits scale**
- Japan – continued fiscal stimulus and public investment underpin domestic demand and corporate earnings
- Canada – Balanced fiscal stance; infrastructure and green-energy incentives support commodity and industrial plays
- Fiscal expansion across G7 has created sustained earnings tailwinds for construction, defense, and energy infrastructure

Shifts Transforming Equity Markets

- ESG & Energy Transition – Major capital reallocation toward renewables, electrification, and carbon-reduction technologies
- Digitalization & Productivity Push – AI, automation, and robotics driving secular growth, especially in US and Japanese industrials (key for aging demographics)
- Demographics & Labor Markets – Aging populations in Japan, Europe, and Canada constrain labor supply and long-term growth potential. AI is a solution that is being used to boost equity allocations





CASE STUDY

Case Study

2022 US Equity Drawdown

- FED executed one of the fastest tightening cycles in history (10 year yields surged 1.5% -> 4.5%)
- Valuation Compression – 22x to 16x, which led to a stall in growth which led to NASDAQ –33%
- **Energy and Value outperformed amid inflation and geopolitical shocks**
- QT and rising yields reduced risk appetite across global equities
- Outcome – Sparked equity dispersion into Europe, Japan, with heavy investments into Canada and UK



Japan Equity Renaissance (2023-2025)

- Japan was labeled as value trap, because of low growth **perogatives and cheap valuations**
- Corporate conservatism led to depression of shareholder returns
- Tokyo Stock Exchange pushed firms to take corrective action
- This coincided with ultra-loose BOJ rates that acted as a catalyst for a boom
- Outcome - Nikkei 225 hit 35-year highs in 2024, because of mass foreign capital flow, closed the valuation gap between them and US
- Excellent example of how structural reform can change market perception



PORTFOLIO MANAGEMENT IMPLICATIONS

Portfolio Management Implications

Allocation & Diversification

- G7 equities anchor global portfolios
- Differing sector compositions and policy cycles provide cross-market diversification
- **Balancing these factors enhances return stability and reduces drawdown risk during global shocks**

Positioning

- Rising rates -> favor value (Euro, Canada)
- Falling rates -> favor growth (US, Japan tech)
- Income focused – stick to Canada and UK
- **ESG & Energy Transition – Key alpha sources across G7**

Valuation and Risk

- US equities demand a structural premium, but are not guaranteed returns, Japan and Europe provide potential
- G7 markets are crisis correlated but diverge in recovery, hence tactical rotation periods
- Volatility profile – Canada and UK are defensive, Japan and US cyclical; Eurozone driven by growth sensitivity



Source: Bloomberg, Financial Times

Note:



Market News

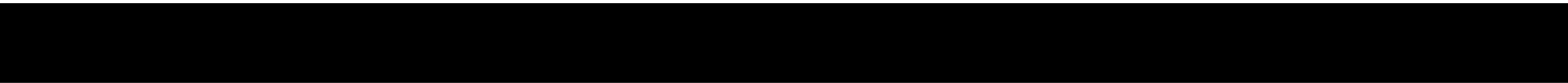


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Market News

Yen & Crypto Liquidation Event



Introduction : Current Market state

- A massive liquidation event has triggered a crypto flash crash today, wiping out over **\$630 million** in leveraged positions and sending Bitcoin tumbling toward the **\$85,000** support level.
- The Bank of Japan (BOJ) Governor Kazuo Ueda signaled a **76% probability** of an interest rate hike at the upcoming December 19 meeting, causing Japanese 2-year bond yields to spike to **1.84%** (the highest since 2008).
- We are witnessing a **perfect inverse correlation**: As the Yen strengthens (and JGB yields rise), cryptocurrency prices plunge. The Yen is currently acting as the "kryptonite" to crypto's leverage.

Mechanics of the Crash

The "Carry Trade" Collapse: For years, investors borrowed Yen at near-zero interest rates to buy high-yield assets like Bitcoin. This is known as the "Carry Trade." Today's signal suggests that this cheap borrowing window is slamming shut.

The Debt Squeeze: As the Yen rises in value against the Dollar (USD/JPY falls), the *real value* of the loans traders took out increases. If you borrowed 1 million Yen when it was "cheap," and the Yen suddenly becomes "expensive," you effectively owe more money than you started with.



Source: Bloomberg, Financial Times, Trading View, Investing.com

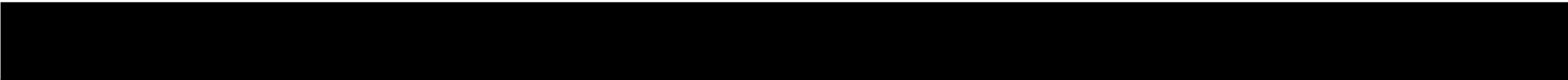
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MACROINVESTING ASSOCIATION

Lecture-9 Commodity Markets



Agenda



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Commodity Markets

Introduction

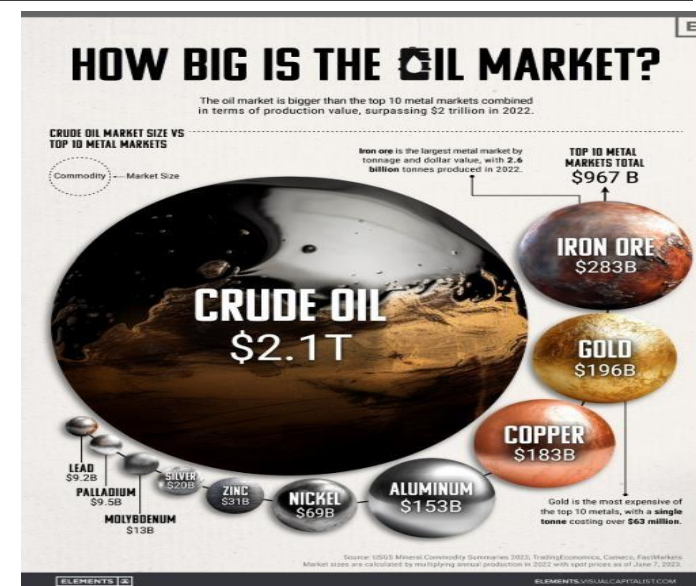
- Commodity markets are global marketplaces where raw materials and primary products are traded
- They are inputs to the global economy – everything from crude oil and wheat to copper and gold
- There are various players involved in the market from producers, consumers and investors

Breakdown of the market

- There are broadly 2 type of commodities which are hard and soft commodities.
- Hard Commodities include energy, metals such as gold, copper and aluminum
- Soft Commodities include agriculture products like wheat, corn, soybeans etc.

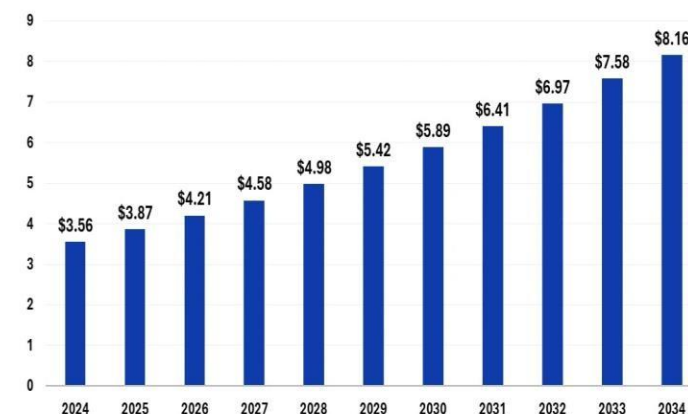
Value of the Market

- Global commodity derivative market is valued at over \$20 billion
- Physical commodity trading exceeds \$10 trillion annually in global trade value
- Energy is the largest in the commodity market, with metals and Agri coming i second and third



Precedence
RESEARCH

Commodity Services Market Size 2024 to 2034 (USD Billion)



Source: <https://www.precedenceresearch.com/commodity-services-market>

Breakdown of Commodity Trading Exchanges around the World

North America

- CME Group is the largest commodity exchange in the world
- ICE Futures US is major for soft commodities and energy benchmarks
- Montreal Exchange which focus on environmental and energy derivatives

Europe

- London Metal Exchange is the world's leading metal market
- ICE Futures Europe is a major energy and emission trading hub
- European Energy Exchange is Europe's main electricity and carbon trading platform

Asia-Pacific

- Shanghai Futures Exchange is China's largest commodity exchange
- Dubai Mercantile Exchange which is the benchmark for Middle Eastern Crude



Ways to Trade Commodities

Spot Market

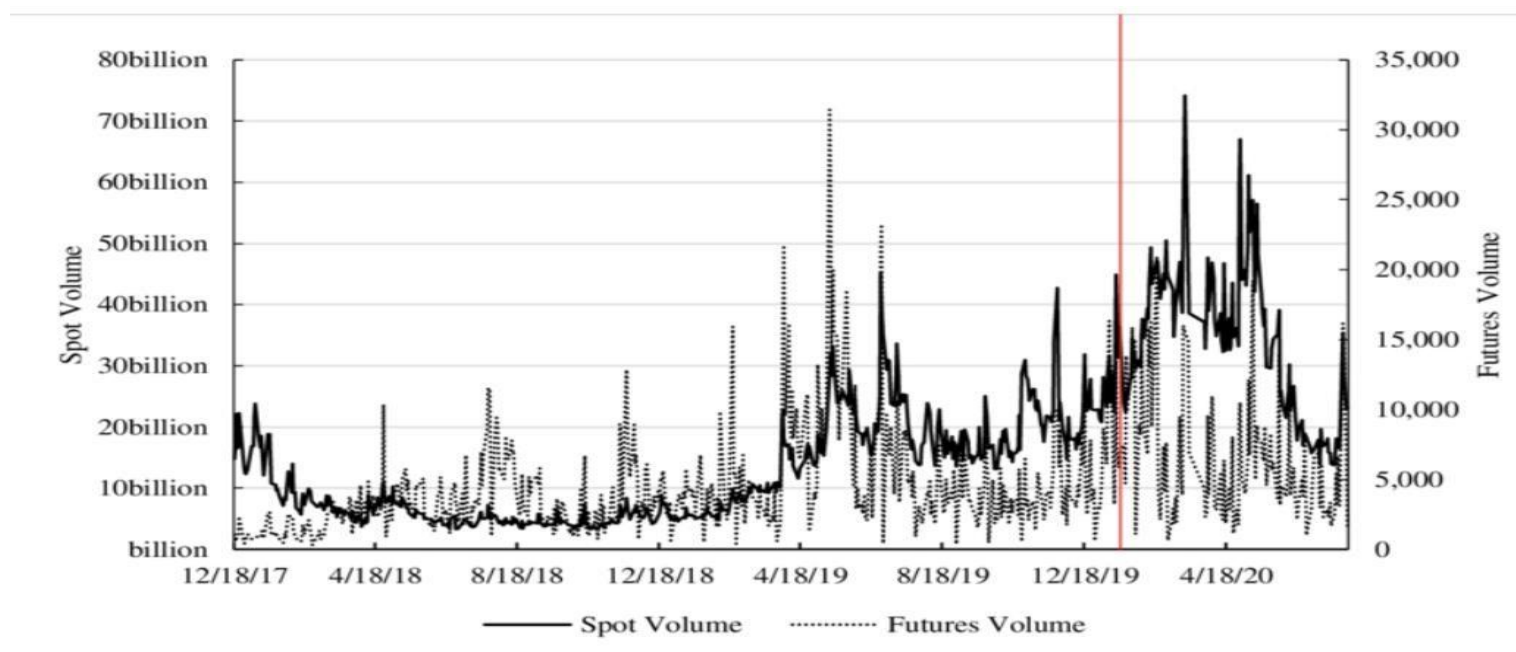
- Immediate purchase/sale of the physical good at today's price
- This is mostly used by producers, refiners, exporters etc.
- **Spot Market trading is also known as physical trading**

Futures Contracts

- Agreement to buy or sell a specific commodity at a fixed price in a future date
- This instrument is widely used by hedge funds, investors, producers and speculators

ETF Index Funds

- Track the price of one or more commodities, they traded like any other stock
- They are massively used by retail investors and passive funds
- Ex SPDR Gold Shares, Invesco DB Commodity Index Tracking Fund



Source: Bloomberg, Financial Times

Note:

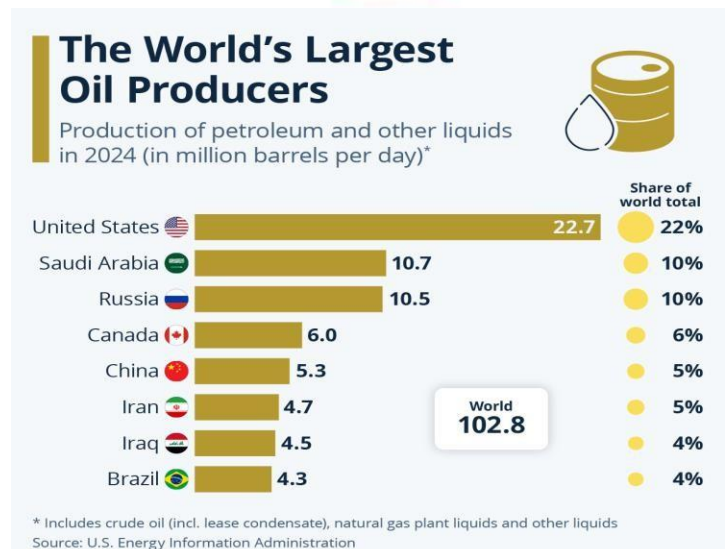
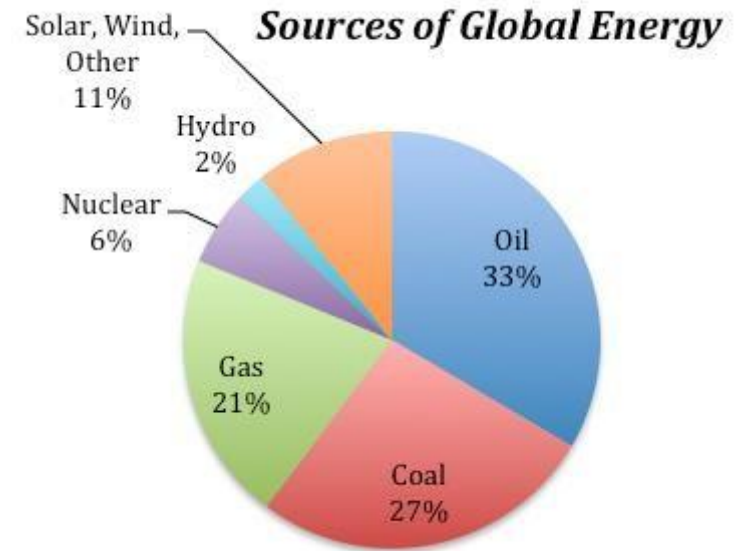
Introduction to Energy Commodities

Energy Commodities Overview

- **Energy Commodities include crude oil, natural gas, coal, and refined products**
- Foundation of transportation, power generation, and manufacturing
- Highly liquid and **VERY** globally interconnected market
- Crude Oil measurements
 - Varies in density – light vs heavy – Directly impacts market value
 - API (American Petroleum Institute) scale
 - A **higher** API gravity means the oil is lighter.
 - More valuable; easier to refine; more desirable products like Gasoline, diesel, and jet fuel; heavier crude is used for Asphalt, and other less profitable products

Role in Energy Markets

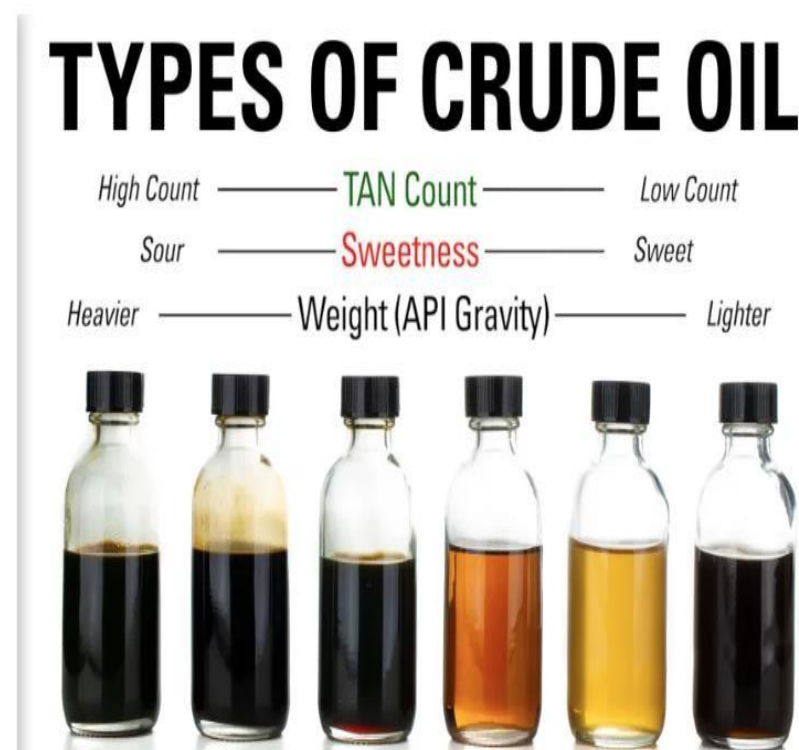
- **Energy prices directly affect inflation, growth, and currency**
- Investors (Pension funds) buy energy to protect against rising prices
- Oil prices shape central bank policy and consumer confidence
 - Central banks use energy prices indirectly to decide interest rates
 - What happens to interest rate policy when oil prices rise?
- Energy sector represents a major portion of global indexes.



Energy Commodities

Market characteristics

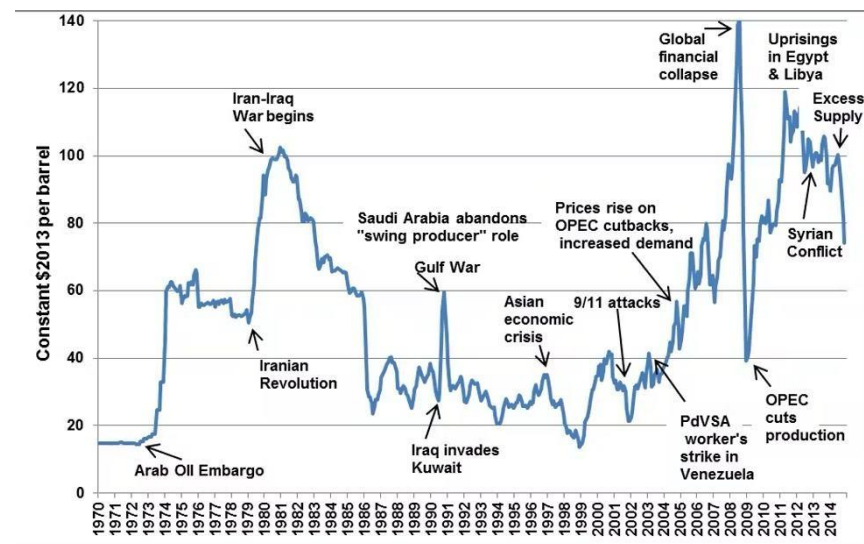
- **Futures contracts**
 - Standardized by **quantity, grade, and delivery location**
- **Quantity**
 - Standard contract size is 1,000 U.S. barrels (42,000 gallons)
- Contract sizes are standardized, allowing uniformity for trading and risk management
- **Grade**
 - Futures commonly specify "Light Sweet Crude Oil" – WTI (West Texas Intermediate) grade
 - The crude oil **must meet quality standards** such as:
 - <.42% sulfur by weight
 - API gravity 37° and 42° (lightness).
 - This specification ensures consistent product quality for delivery or settlement.
- **Delivery**
 - The primary delivery location for crude oil futures is Cushing, Oklahoma.
 - Major storage and pipeline hub; influences physical supply, which influences prices
 - Impacts
 - Shows supply/demand; Quality issues; traders watch inventory for market signals.
 - Contracts specify delivery terms for where physical oil will be delivered if held to expiration.



Energy Commodities

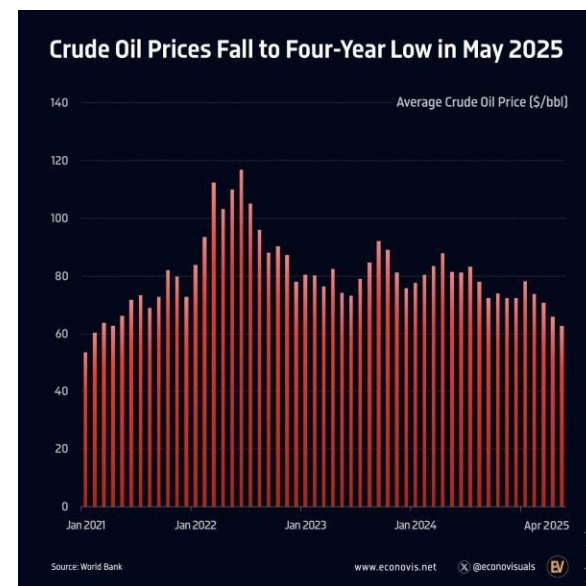
Macro Factors Affecting Energy

- Prices are driven by **global demand, geopolitical events, and currency fluctuations**
- **OPEC+** (Organization of the Petroleum Exporting Countries) decisions
 - Output quotas and significantly influence prices
- U.S. dollar is inversely correlated with oil prices
 - Oil is priced in USD globally, so a stronger dollar makes oil more expensive in other currencies, reducing demand and lowering prices.
- War is a big factor
- When the Russia-Ukraine war began oil prices surged to levels that were never seen before. Since Russia is a big exporter of oil, people feared for supply chain disruptions in the future and heavily increased their demand for oil



Current Trends in Energy Markets

- Volatility is still present due to Geopolitical tensions
- **OPEC+ supply management is still an ongoing issue**
 - Adjusts production to match supply and demand
- Energy **trends towards renewables** and electricity are also emerging
- ESG (Environmental, Social, and Governance) goals are putting



Introduction to Industrial Commodities

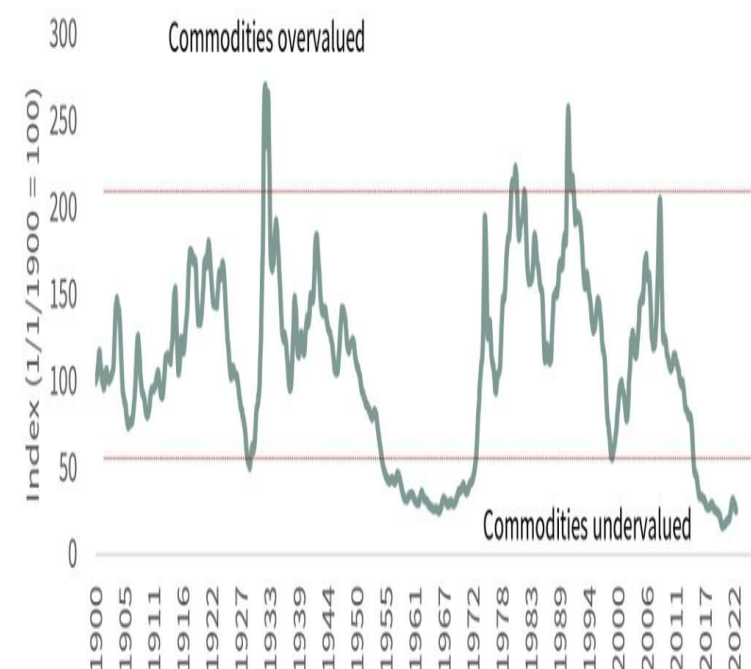
Industrial Commodities Overview

- Industrial commodities include **metals, construction materials, and energy inputs**. All of which are essential for manufacturing and infrastructure
- Demand is closely tied to global economic growth and continued industrial production
 - During times of **strong GDP growth, demand and price for industrial commodities go up**
- Industrial commodities serve as leading indicators of global manufacturing growth and even the momentum of trade
 - If copper and steel prices rise, it suggests that factories are ramping up production and overall economic growth usually follows

Role in Financial Markets

- Industrial metals are heavily used in futures and derivative markets to help with both price estimates and are commonly used for hedging risk by locking in current prices for future sales
- Many institutions and portfolios include industrial commodities in diversified portfolios as **inflation hedges**
- Price trends influence **equity indices, producer margins, and capital investment cycles**
- Rising metal prices boost revenues and margins for mining companies but raise costs for manufacturers. These price changes can influence decisions made by mining companies.

Chart 8 Commodities vs. Dow Jones Industrial Average



Source: Bloomberg and G&R Models.

Industrial Commodities

Market Characteristics

- **Global Economic Sensitivity:**
 - Industrial Commodity prices track global industrial activity such as manufacturing and construction demand
 - Loosely follow the global levels of global growth/recession
- **Supply Chain Dynamics:**
 - Supply disruptions, geopolitical tensions, and trade policies create volatility
- **Energy and Raw Materials:**
 - Fluctuations in energy prices and input costs like mining iron ore and coal influence production and prices
- **Price Volatility and Market Cycles:**
 - Industrials face very significant price swings through factors such as supply/demand balance, inventory, speculative activity
- **Sustainability and Regulation:**
 - Increasing environmental regulations impact production, which influence costs of mining
- The most influential aspects as of late have been **trade policies and tariffs** which can disrupt supply chains and increase costs

Industrial Commodities

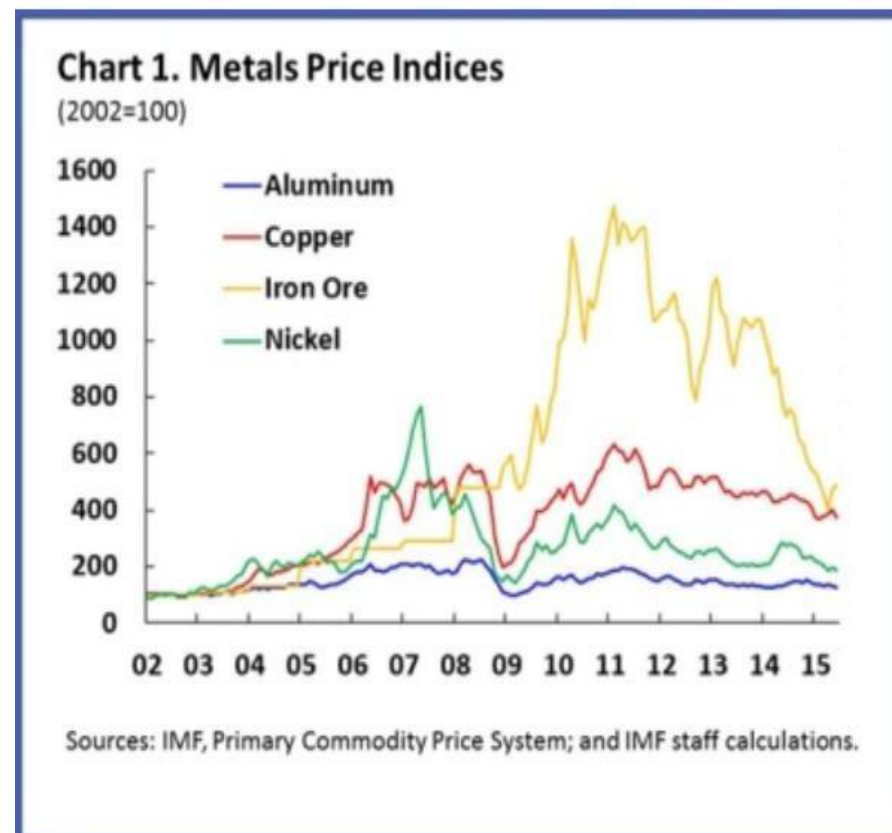
Market Characteristics (Specific commodities)

Copper

- Key uses include electrical wiring and equipment (power generation and transmission), plumbing and piping (water, gases, heating), industrial machinery, transportation (cars, planes, trains, EV batteries, inverters, charging), and renewable energy systems (solar panels and wind turbines)
- Prices are mainly driven by manufacturing demand and supply
- Most of the copper used in the market is from Chile and Peru

Steel

- Alloy primarily composed of iron and carbon and accounts for ~90% of total metal production
- Steel is used in almost everything, like buildings, bridges, cars, trains, ships, appliances, pipelines, wind turbines, tools, and medical equipment.
- Prices are mainly driven by raw material costs and energy costs
- Most of the steel used in the market is from China and India



Industrial Commodities

Macro Factors Affecting Industrials

- Global Economic growth; expansion boosts demand and price while recessions reduce demand and price
- Interest rates & USD strength; strong dollar and higher interest rates both harm commodity price
- Trade policies & Tariffs
 - Import/export restrictions and tariffs directly affect pricing by disrupting global supply chains
- Change in energy
 - Changing to electric car, green energy and other methods of energy can increase demand for industrial commodities

Current Trends in Industrial Markets

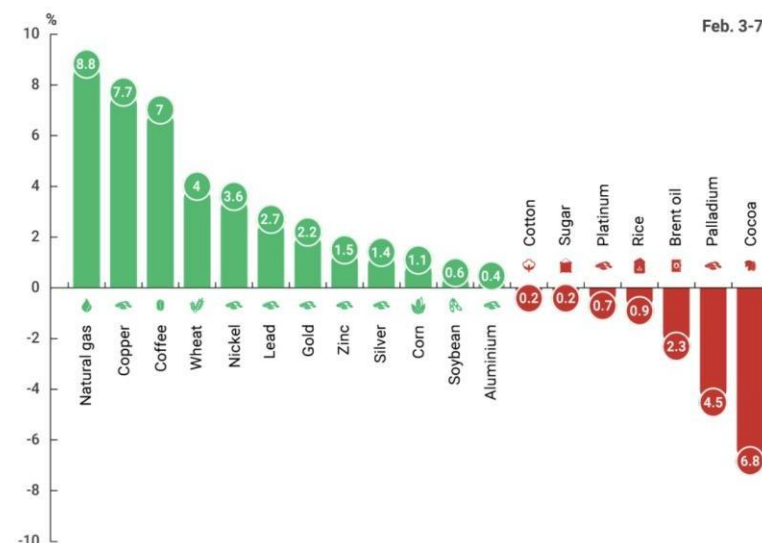
- Growing demand for metal; driven by global energy transition and infrastructure investments
- Supply chain disruptions continue due to a variety of geopolitical tensions, causing price volatility and higher lead times
- Rising focus on sustainable and ethical sourcing; changing operational processes and production
- Inflationary pressures and fluctuating energy costs impact production expenses across all industrial commodities
- Technological innovations (automation and AI) improve efficiency but also change the demand pattern for certain commodities

Sourced: Federal Reserve, Macro Micro

Note: Policy shifts, trade negotiations, and tariffs are all shaping

Trump's tariff statements continue to influence commodity markets

Commodity markets saw mixed movements last week amid US President Donald Trump's tariff decisions and released macroeconomic data



Intro to Agriculture Markets

Introduction

- The agriculture (Agri-commodity) market involves the trading of farm-produced raw materials — crops and livestock — that are essential for food, fuel, and industrial production.
- The global agriculture market is valued at over \$3 trillion annually which 10-12% of total commodity market
- Future and derivatives trading volume exceed \$20 trillion notion value per year

Grains & Oilseeds

Commodity	Key Producers	Key Consumers	Exchange/Benchmark
Wheat	Russia, US, Canada, EU	China, Egypt, India	CBOT, Euronext, SAFEX
Corn (Maize)	US, Brazil, China	Livestock, ethanol	CBOT, DCE
Soybeans	Brazil, US, Argentina	China	CBOT, DCE
Rice	China, India, Thailand, Vietnam	Asia (90% of consumption)	TOCOM, ZCE

Soft Commodities (Tropical Crops)

Commodity	Key Producers	Exchange
Coffee	Brazil, Vietnam, Colombia	ICE US
Cocoa	Côte d'Ivoire, Ghana, Nigeria	ICE Europe
Sugar	Brazil, India, Thailand	ICE US, NCDEX
Cotton	India, US, China	ICE US, MCX



Source: Federal Reserve, Macro Micro

Note:

Coffee Markets

Important Factors

Weather & Climate

- Coffee Trees are sensitive to temperature and rainfall
- **Droughts in Brazil** which is the largest producer can cause sharp price spikes

Supply Side Dynamics

- Takes 3-4 years to mature yields
- Directly correlated to fertilizer prices

Policy & Sustainability Factors

- Export restrictions, tariffs, or government subsidies can distort supply and pricing.
- For example, a government limiting exports to control domestic prices can create global shortages

Demand Side Factors

- Rising demand in **emerging markets** (China, India, South Korea) and continued strong consumption in the

U.S. and Europe sustain higher price levels.

- Conversely, economic downturns or shifts toward

Source: Bloomberg, like tea or energy drinks) can depress

Note: demand.



Soybean Market

Important Factors

Weather & Crop Yields

- Extremely sensitive to precipitation and temperature patterns
- **Brazil, United States and Argentina are the largest producers**

Global Supply & Harvest Cycles

- Soybeans have 2 major growing seasons which is April – October in the US and September – April in Brazil
- These cycles stagger global supply, so markets price in the expectations months in advance

Demand from China

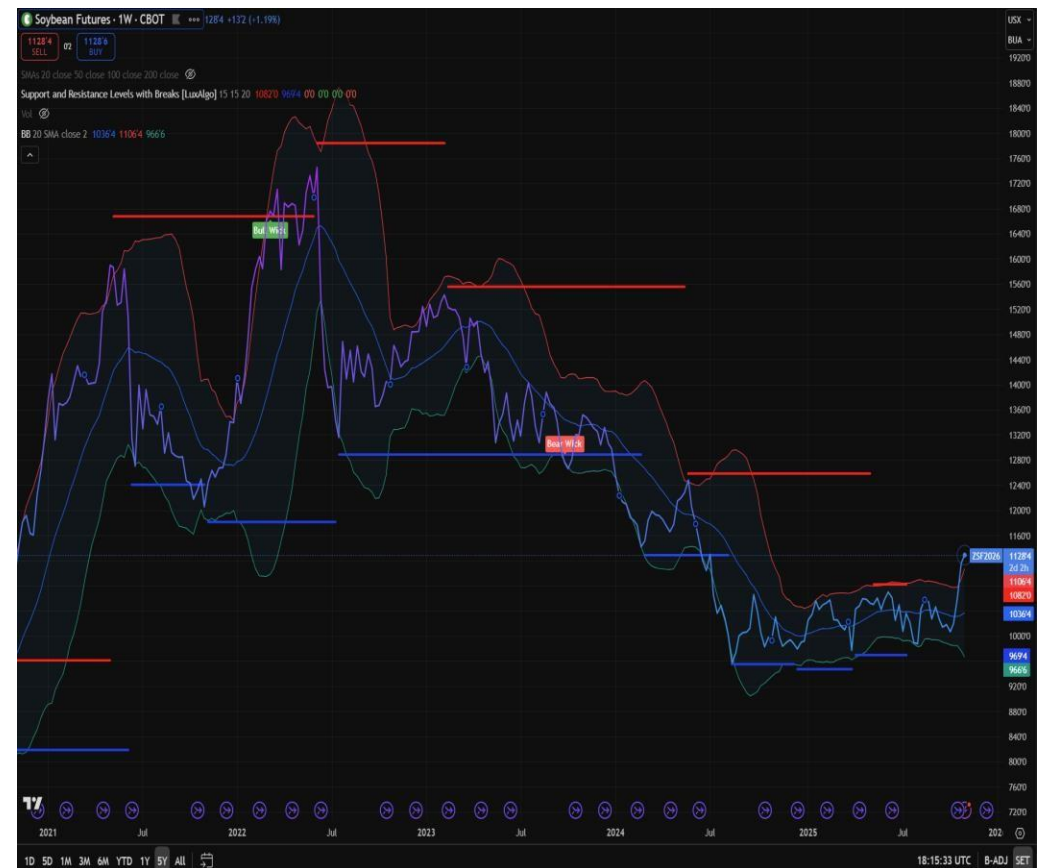
- China has 60% of the global soybean imports
- Economic health of China's livestock sector drives massive demand swings

Trade Policies & Geopolitics

- Tariffs, export bans and trade wars have huge effects
- Ex : Export restrictions, Trade Agreement, and diversification of supply

Currency Movements

- A weaker Brazilian real or argentic peso encourages exports which puts a downward pressure on prices
- A strong US dollar makes US soybeans less competitive compared to global prices



Wheat Markets

Important Factors

Weather & Crop Conditions

- Highly sensitive to temperature, rainfall and drought
- **China, India and Russia are the largest producers**

Geopolitics & Trade Tensions

- One of the most geopolitically exposed grains because supply is concentrated in a few regions
- Export ban and tariffs can instantly squeeze global supply

Global Supply and Demand Balance

- Global wheat consumption rises 2% annually, driven by population growth
- If stock-to-use ratios fall, prices rise sharply

Currency Movements

- Strong dollar makes wheat more expensive globally
- Weak Russian ruble can lead to more exports

Energy & Fertilizer Costs

- Wheat farming is energy-intensive – diesel for tractors, natural gas for fertilizer
- When oil price rise, production costs soar which lead to higher wheat prices



Intro to Gold

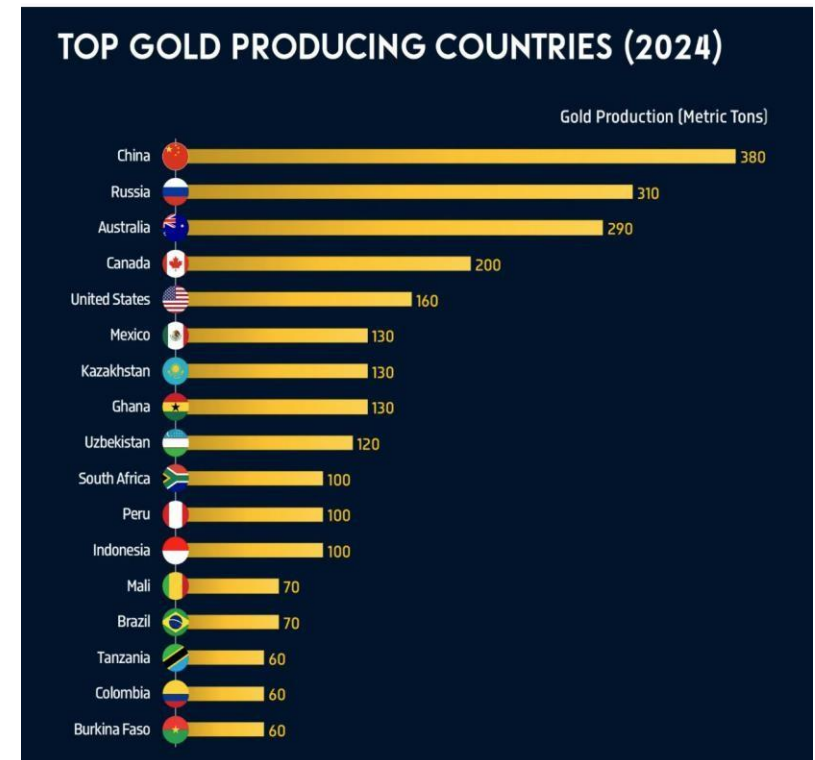
Introduction

- Gold is a precious metal used both as **a store of value and as an industrial input**
- It plays 3 major roles in global finance
 - Monetary Asset – Central Banks hold it as reserves
 - Investment Asset – Safe haven & inflation hedge
 - Industrial/consumer good – jewelry, electronics, technology

Size of Gold Market

There is about 210,000 metric tons of gold with annual production of around 3,000 – 3200 tons per year

The daily trading volume is around \$120-150 billion/day which makes it the second largest commodity after oil in terms of financial volume



Why is Gold considered the store hold of value?

- There is a limited supply and scarcity of the commodity which gives it intrinsic value and is a good hedge against inflation
- For over 5000 years, gold has been recognized across civilizations as a medium of exchange of wealth

• Not dependent on any institution's solvency

Source: Federal Reserve, Macro Micro

Note:

Factors affecting Gold Prices

Important Factors

US Dollar

- The dollar is inversely correlated with gold
- As the dollar gets stronger -> Gold is cheaper abroad -> prices fall

Interest Rates

- Gold is inversely correlated with interest rates

Inflation Expectations

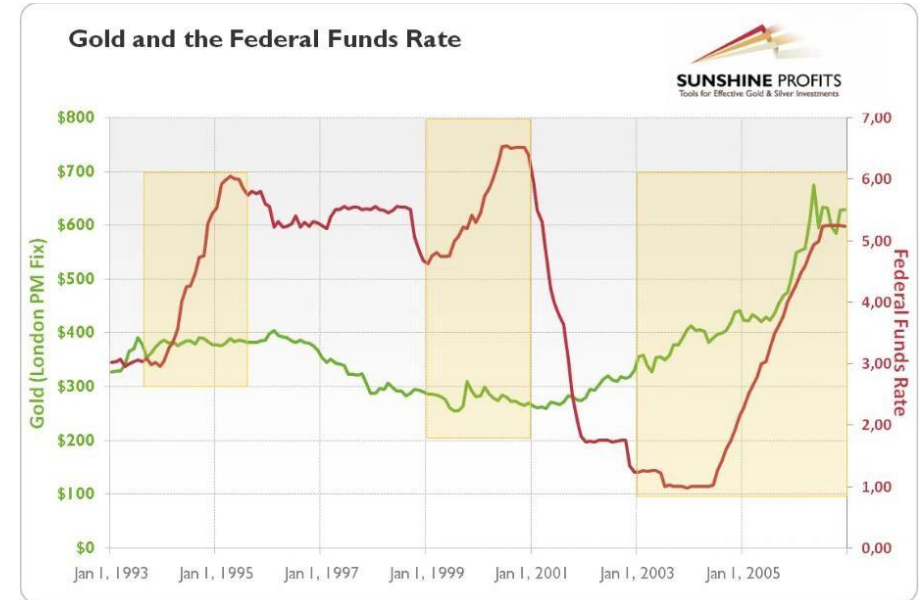
- Inflation expectations is directly correlated with the value of gold
- Weak yen boosts exporters and global investors' returns

Geopolitical Risk

- Political risk is directly correlated with gold
- During bad or uncertain economic times, investors are looking to park their capital into safe assets

Central Bank Policy

- The central bank has strong influence on the value of gold
- Higher Rates -> Strong dollar -> Gold prices fall
- Lower Rates -> Weak dollar -> Gold prices rise



1973 Oil Crisis



https://www.youtube.com/watch?v=YMb_hbt1NLw

1973 Oil Crisis Explained

1. Background: Pre-1973 Oil Market

Before 1973, global oil prices were stable and low — around \$3 per barrel.

Western oil companies (the “Seven Sisters”) dominated global production and pricing.

The U.S. was the world’s largest producer but had started to depend on imports.

Middle Eastern nations — especially Saudi Arabia, Iran, Iraq, Kuwait, and the UAE — began pushing for greater control and revenue.

2. Trigger Event: The Arab-Israeli (Yom Kippur) War

In October 1973, Egypt and Syria launched a surprise attack on Israel during Yom Kippur.

The U.S. and several Western countries provided military support to Israel.

In response, the Arab members of OPEC (Organization of Petroleum Exporting Countries) retaliated with an oil embargo against the U.S. and its allies.

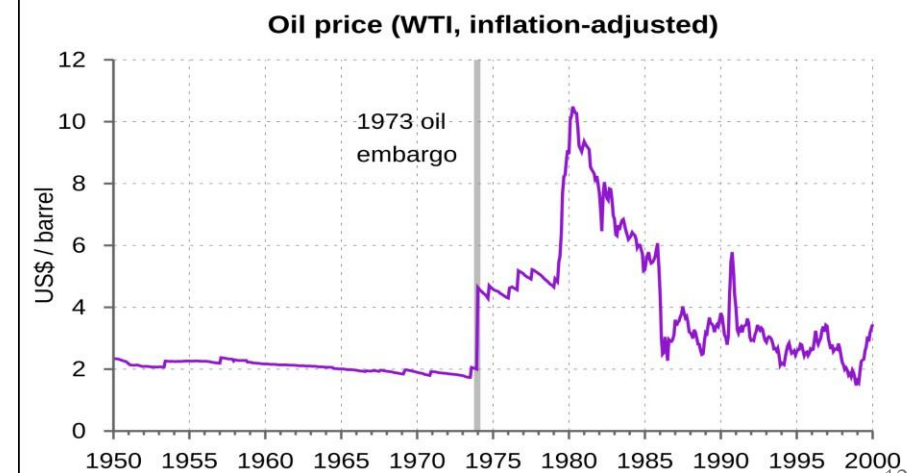
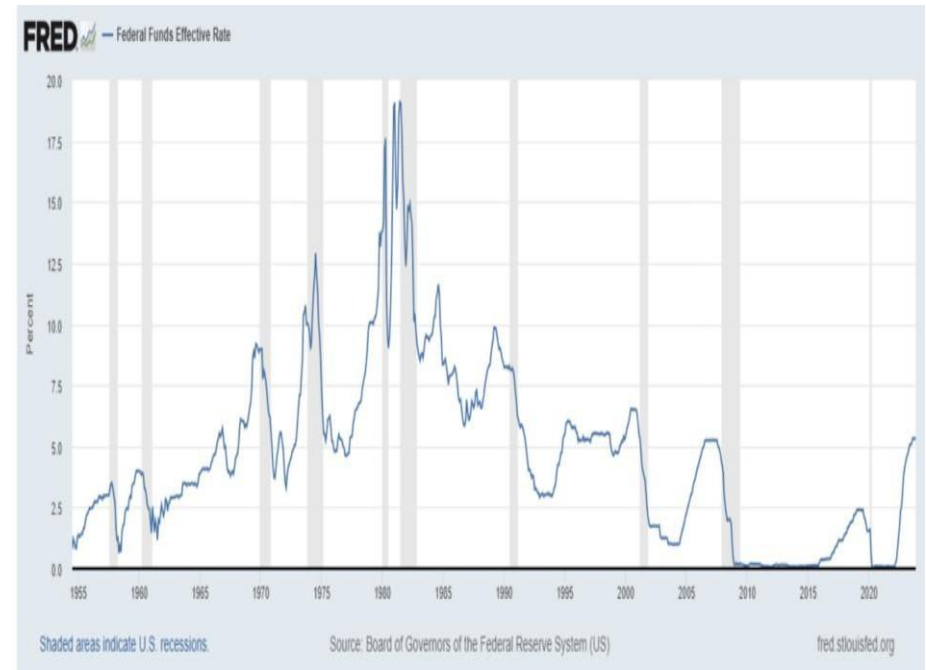
3. The Embargo & Production Cuts

The Arab Oil Embargo targeted the U.S., the Netherlands, Japan, and other pro-Israel nations.

OPEC nations cut oil production by 5% per month until demands were met.

Oil shipments to embargoed countries were suspended completely.

Note: The embargo lasted from October 1973 to March 1974.



Hunt Brothers and the Silver Squeeze

1) Massive Silver Accumulation:

- Texas billionaires Nelson and William Hunt tried to corner the silver market by buying up to one-third of the world's silver supply, both physically and through futures contracts.
- Their belief: inflation would erode the dollar's value, so silver was a safe hedge.

2) Price Explosion and Market Panic:

- Silver prices soared from around \$6 to nearly \$50 per ounce by January 1980 — a more than 700% increase.
 - This triggered a speculative frenzy and fears of manipulation, with margin requirements raised by exchanges.
- The Commodity Exchange (COMEX) and Chicago Board of Trade (CBOT) introduced “Silver Rule 7” in January 1980: Traders could only liquidate existing positions — not open new long positions. Margin requirements were raised drastically.

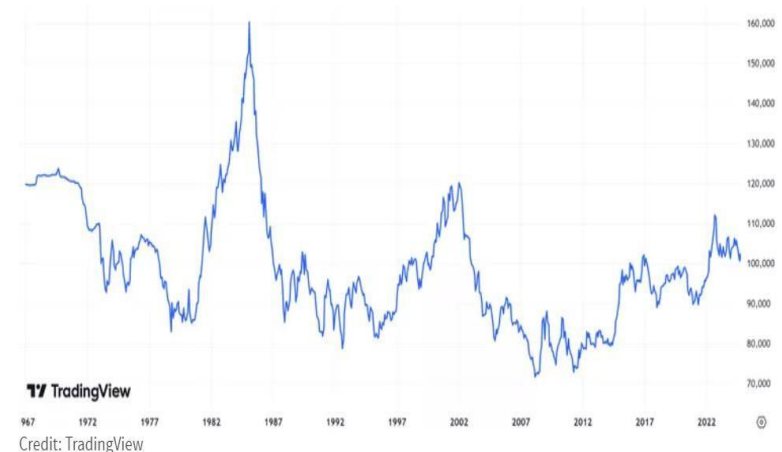
3) Collapse and Aftermath:

- When regulations tightened and margins increased, the Hunts couldn't meet margin calls — causing silver to crash back to ~\$10.
- The brothers faced bankruptcy, lawsuits, and tighter CFTC oversight

to prevent future commodity manipulation.

Source: FT, Bloomberg

Note:



PORTFOLIO MANAGEMENT IMPLICATIONS

Portfolio Management Implications

Allocation & Diversification

- Commodity have low or negative correlation with assets like equities or bonds
- During high inflation environment commodity prices tend to rise as a hedge against inflation
- In 2022, while equities fell, energy and agricultural commodities outperformed, cushioning portfolio drawdowns

Tactical Allocation

- Commodity prices move in **cycles tied to global growth, supply disruptions, and policy shifts**.
- Active managers can exploit these cycles through **tactical overweight or underweight positions** in specific sectors (e.g., metals during industrial recovery).
- Futures and ETFs allow investors to **express short-term macro views efficiently**

Valuation and

Risk

- Commodities are **highly volatile** and sensitive to unpredictable factors like weather, OPEC decisions, or geopolitical events.
- Futures trading introduces **leverage and margin risk**.
- Effective portfolio management requires **position sizing, stop-loss rules**, and possibly **hedging through options** or cross-asset exposure



Source: Bloomberg, Financial Times

Note:



Intro to FX, Spot, Futures and Forwards

Introduction to FX

Introduction

- Foreign Exchange (FX) is the **global market** where currencies are bought and sold
- Currencies **trade in pairs**, showing the value of one currency relative to another (EUR/USD)
- The FX market operates **24 hours a day**, 5 days a week, moving with global financial centers (Asia → Europe → U.S.)
- FX is the **most liquid market** in the world, with roughly \$7 trillion traded daily
- Major participants include **central banks, commercial banks, hedge funds, corporations, asset managers, and retail traders**

Why FX Matters

- FX reflects the **economic strength** and **political stability** of countries
- Every **international investment, trade, and payment flows** through FX
- PMs use FX to **hedge** global portfolios, **reduce** foreign currency risk, and **manage** volatility
- FX provides macro signals, revealing shifts in **capital flows, risk sentiment, and global policy trends**
- Because of its **high liquidity**, FX is one of the cleanest, most technically traded markets, making trends and price levels highly relevant



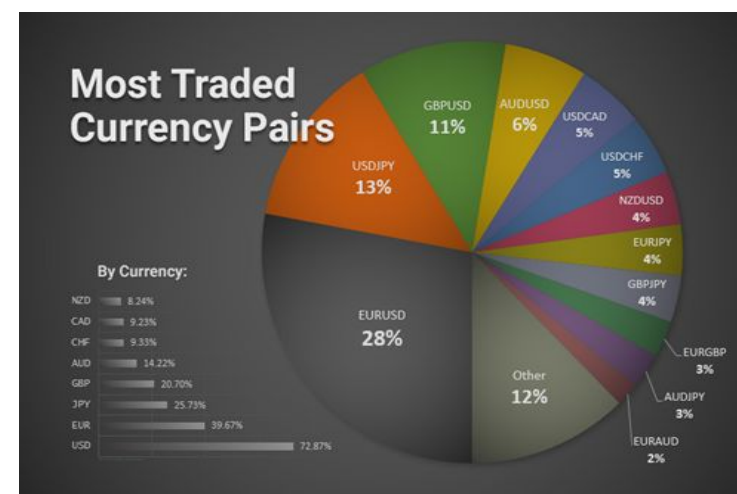
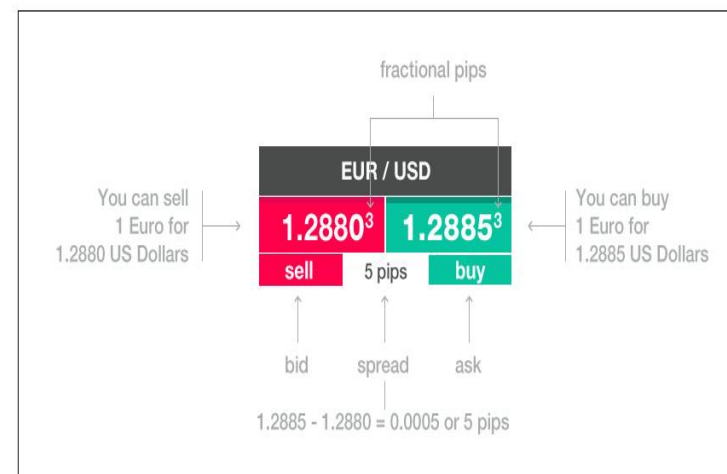
Currency Pairs, Quotes, and Market Mechanics

How Currency Pairs Work

- Currencies **always trade in pairs**, showing how much of one currency you need to buy another (EUR/USD = price of 1 EUR in USD)
- Base currency:** The first currency in the pair (EUR in EUR/USD)
- Quote currency:** The second currency (USD in EUR/USD)
- When the pair rises, the **base currency strengthens** relative to the quote currency
- When the pair falls, the **base currency weakens**
- Major pairs include **EUR/USD, USD/JPY, GBP/USD, USD/CHF** as these are the most liquid and lowest spread

How FX Prices Move

- Pips** measure the **smallest price move** (usually 0.0001), similar to ticks in futures
- Bid/ask spread represents transaction cost; **tighter spreads = high liquidity**
- FX trades 24 hours**, rotating through Asian, European, and U.S. sessions
- Liquidity peaks when **London and New York overlap** as most price action happens here
- High liquidity means cleaner trends and more reliable technical behavior, which is why **FX is heavily traded by technical and systematic funds**



Source: Bloomberg, Financial Times, WSJ, Reuters

Note:

FX Spot, Forwards, and Futures

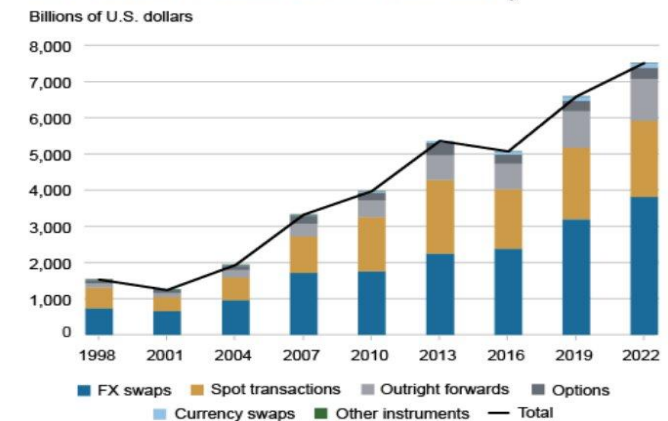
Spot FX

- Spot FX is the **standard way currencies are traded**, involving the near-immediate exchange of one currency for another
- Spot trades settle in about **T+2**, but execution is instant and extremely liquid
- Most FX price movement, trends, and technical signals **come from the spot market**
- **Banks, hedge funds, and asset managers** dominate spot trading, taking directional views based on macro news and market sentiment
- Because FX trades nearly 24 hours a day, spot markets **smoothly transition** from Asia → Europe → U.S., creating continuous price flow

Forwards & Futures

- FX Forwards **are private agreements** to lock in an exchange rate today for a currency transaction that will happen in the future
- They are widely used by corporations and portfolio managers to **hedge foreign revenues, expenses, or international positions**
- Forwards **do not trade on an exchange** they are negotiated with banks (OTC), flexible in size and date
- FX Futures are **standardized contracts** that trade on exchanges (like CME) and behave similarly to forwards but with fixed contract sizes and set expirations
- Asset managers use FX futures to quickly **adjust currency exposure, hedge international positions**, or express broad macro themes with leverage

The FX Market Continues to Grow Substantially



Source: Bloomberg, Financial Times, WSJ, Reuters

Note:



Fundamental & Macro Drivers

FX Drivers



Economic Strength

- Strong economies attract investment → **currency strengthens**
- Weak economies lose investment → **currency weakens**
- Markets watch **growth, employment, inflation**, and business activity as signals
- FX often moves **before economic data** because traders price expectations early

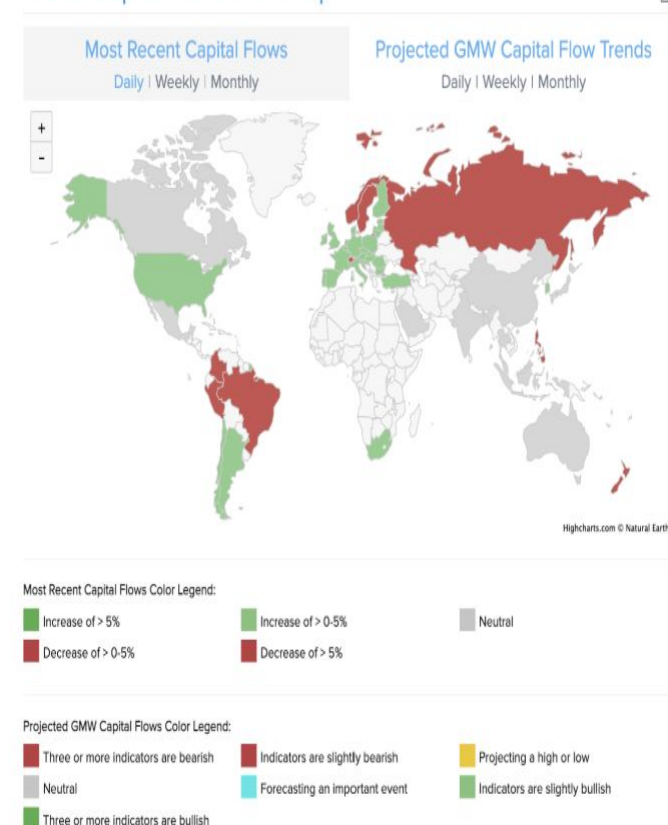
Capital Flows

- When global investors buy a country's stocks, bonds, or real estate → **currency rises**
- When capital leaves due to fear, recession, or uncertainty → **currency falls**
- **Foreign direct investment (FDI)** and **sovereign wealth flows** can move currencies fast

Politics, Policy, and Market Sentiment

- Stable governments and credible central banks **attract global capital**
- Instability, conflict, or unclear policy **pushes money out of a country**
- FX reacts instantly to political events (**elections, policy shifts, geopolitical risk**)
- Because FX is extremely liquid, technical levels and momentum often **amplify these fundamental moves**

Global Capital Flows Heat Map



Source: Bloomberg, Financial Times, WSJ, Reuters

Note:

Central Banks & Interest Rates

Why Interest Rates Matter

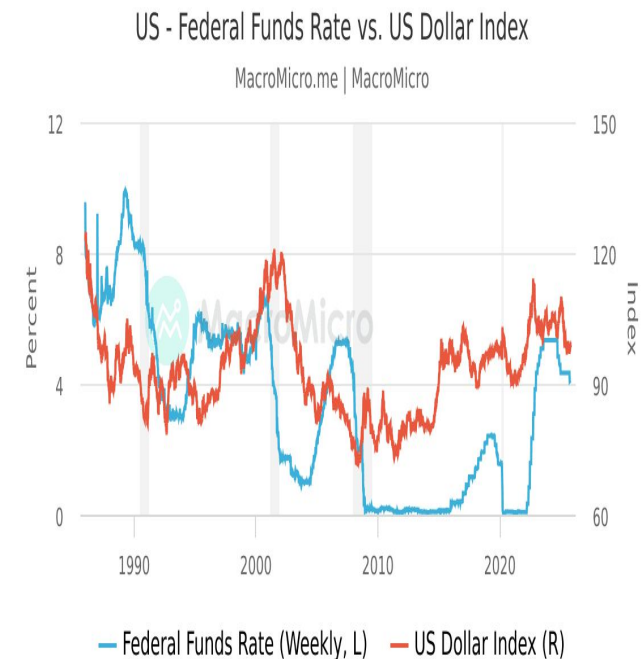
- When a country raises interest rates, investors earn more on cash and bonds → money flows in → **currency strengthens**
- When a country cuts rates, returns fall → money flows out → **currency weakens**
- FX traders focus strongly on **rate expectations**, not just actual changes

What Central Banks Do

- **Policy direction:** Are rates trending up or down?
- **Credibility:** Does the central bank stick to its guidance?
- **Surprises:** Unexpected decisions create some of the largest FX moves
- FX traders combine **macro tone + expectations + technical levels** to position ahead of central bank events

What Markets Watch

- Central banks (**like the Fed, ECB, BOJ, BOE**) set monetary policy that directly affects currency values
- Hawkish tone (concern about inflation) → markets expect higher rates → **currency often rises**
- Dovish tone (concern about growth) → markets expect lower rates → **currency often falls**
- **Press conferences and speeches** can move FX instantly



Source: Bloomberg, Financial Times, WSJ, Reuters, MacroMicro

Note:

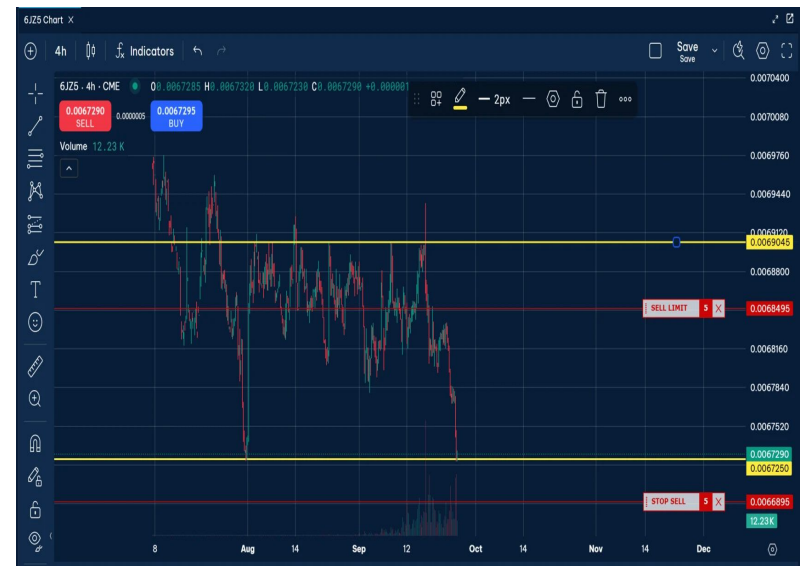
Technical Analysis in FX

Why Technicals Matter More in FX

- FX is the **most liquid market in the world**, so price reacts instantly and cleanly
- Because millions of traders and algorithms watch the same levels, **technical zones create real order flow**
- **Breakouts, momentum, and trend systems** work well because FX markets run 24 hours with no gaps
- Major price levels (like USD/JPY 150 or EUR/USD 1.10) act as **psychological turning points**
- **Central banks** themselves sometimes react to key price levels (especially BOJ)

Key Technical Indicators Used in FX

- **Trendlines & Moving Averages:** FX trends are long and smooth, making moving averages extremely effective for identifying direction
- **Support/Resistance:** FX respects horizontal levels because of concentrated global order flow
- **Breakouts & Range Trades:** FX frequently ranges during low volatility and explodes on macro catalysts; traders target breakouts.
- **Macro + Technical Alignment:** Strongest FX trades happen when fundamentals and technical levels point the same way



Source: Bloomberg, Financial Times, WSJ, Reuters

Note:



Carry Trade

The Carry Trade



What the Carry Trade Is

- The carry trade means buying a currency with a **higher interest rate** and selling one with a **lower interest rate**
- Traders **“earn the difference”** in yields while holding the position
- It is one of the oldest and most widely used FX strategies

Why It Works

- **High-yield currencies** attract more investment during stable markets
- **Low-yield currencies** are often used for funding (historically JPY and CHF)
- When global risk sentiment is strong, capital flows into high-yield countries, **strengthening those currencies**
- Carry tends to work best when **volatility is low** and markets are in **“risk-on”** mode.

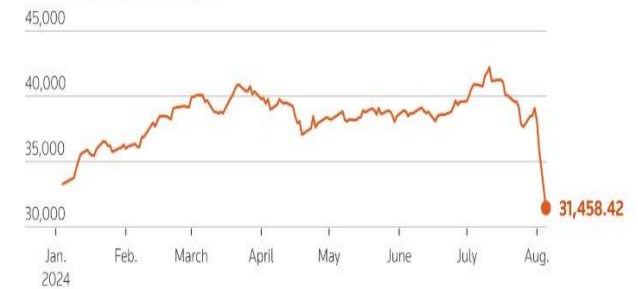
Risk of the Carry Trade

- Carry breaks down during **risk-off events** when investors flee into safe assets
- **High-yield currencies** can fall sharply, wiping out months of positive carry
- Political shocks, central bank surprises, and economic stress **can quickly reverse carry flows**

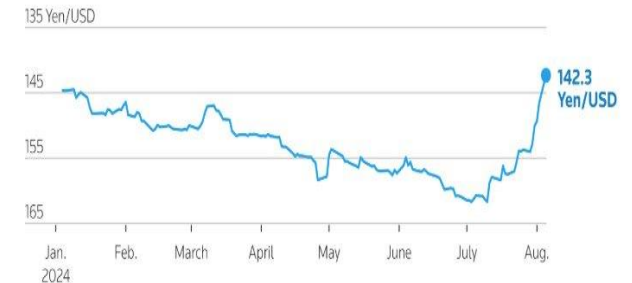
Nikkei vs yen

Japanese stocks collapsed on Aug. 5 in their biggest single day rout since the 1987 Black Monday sell-offs, driven by last week's plunge in global stock markets, economic concerns and worries investments funded by a cheap yen were being unwound.

NIKKEI 225 INDEX CLOSE



YEN



Source: Bloomberg, Financial Times, WSJ, Reuters

Note:



Case Study

Case Study - FX Crises & Soros' Black Wednesday Case Study (1992)

The Soros trade is the most famous example of how macro imbalances, weak policy, and capital flows can break a currency peg and create a massive asymmetric opportunity

Why the UK's Currency Peg Was Vulnerable

- The UK joined the Exchange Rate Mechanism (ERM), **pegging the British pound to the Deutsche Mark**
- But the **UK economy was too weak** to sustain the high interest rates required to defend the peg
- Germany had **raised rates** to fight post-reunification inflation, but the UK was in a **recession**, a policy mismatch that made the peg unstable
- An **overvalued pound** and weak fundamentals created the conditions for failure

How the Crisis Unfolded

- The Bank of England **spent billions** in reserves trying to keep the pound at its target level
- They even **raised interest rates to 15%** in a desperate attempt to attract capital and defend the currency
- But Soros and other macro hedge funds kept selling pounds, **betting the peg would break**
- Eventually, the UK exited the ERM, **the pound collapsed**, and the central bank lost the battle

The Soros Trade

- Soros' Quantum Fund found the pound was **overvalued relative to the economy**
- Also noticed **UK fundamentals were weak**, could not sustain high rates and ERM membership was politically unsustainable within the UK
- Soros took a **massive leveraged short position** in GBP using spot and forwards
- His downside was limited to a controlled loss if the peg held, but the **upside was enormous if it broke**
- When the peg collapsed, Soros reportedly made **\$1–2 billion from the trade**



Source: Bloomberg, Financial Times, WSJ, Reuters

Note:

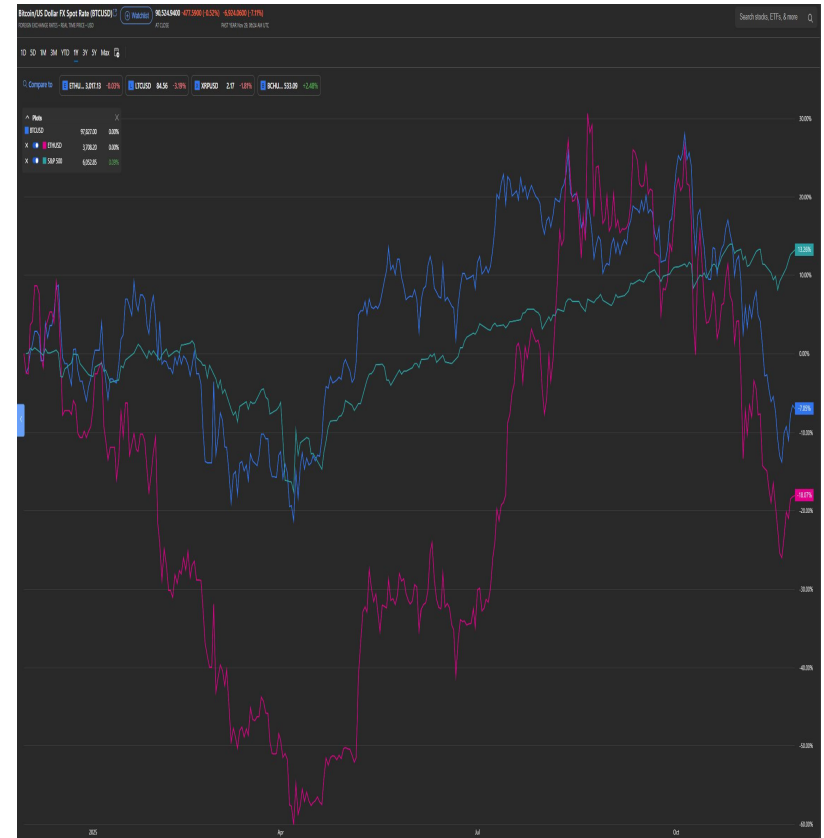
Introduction to Crypto

Introduction

- Crypto is a **24/7 global** market where digital assets like **Bitcoin (BTC)** and **Ethereum (ETH)** trade across thousands of venues
- Prices move based on **supply, demand, liquidity, sentiment, and macro conditions**, similar to FX pairs like BTC/USD or ETH/BTC
- Bitcoin acts like a **macro-driven asset**, reacting to inflation expectations, interest rates, and global liquidity trends
- Ethereum acts more like a **technology network**, where value comes from usage, transaction fees, apps, and scaling activity on the Ethereum ecosystem.

Why Crypto Matters

- Bitcoin has become a **major institutional macro asset**, used to express views on liquidity cycles, digital store-of-value themes, and overall risk appetite
- Ethereum represents platform adoption, with **smart contracts, DeFi activity, and Layer-2 ecosystems driving long-term value**
- Both assets now have regulated ETFs, futures, and institutional custody, making them part of **mainstream financial markets**
- Because crypto trades nonstop, BTC and ETH often provide **early signals of shifts in global sentiment and capital flows**



Source: Bloomberg, Financial Times, WSJ, Reuters

Note:

Global Liquidity & Macro Trends

- Crypto **reacts strongly** to changes in global financial conditions.
- When liquidity **increases** or **real interest rates fall**, investors take more risk, which usually pushes BTC and ETH higher
- When **liquidity tightens** or **real rates rise**, crypto tends to drop quickly because it relies on risk-seeking capital
- Since crypto trades 24/7, it adjusts to macro news **faster than most markets**

Flows, ETFs & Trading Activity

- **Spot Bitcoin ETFs** now play a major role by showing clear, trackable flows
- **ETF inflows** add steady buying, helping lift prices
- **ETF outflows** mean less demand, which can accelerate sell-offs
- **Stablecoin creation/redemption** and **exchange balances** show whether money is entering or leaving the ecosystem
- **High trading activity** usually leads to stronger short-term price moves

Regulation, Narratives, and Major Catalysts

- **Regulation** can **shift crypto sentiment** quickly by shaping who can participate in the market
- **The Genius Act** is a U.S. proposal that aims to clarify how digital assets are defined, how exchanges should operate, and how investors are protected, giving institutions more confidence to participate
- Events like **Bitcoin halving cycles**, **Ethereum software upgrades**, and major crypto narratives (scaling, tokenization, AI themes) influence long-term trends
- Because crypto runs nonstop, it **reacts immediately** to major headlines, policy changes, and global events



Case Study - Case Study: Bitcoin's Drop to 90K

Bitcoin's sudden drop to 90K shows how liquidity shocks, leverage, and reflexive selling can cascade through the entire crypto market.

What Triggered the Drop

- BTC began falling as markets priced in **tighter financial conditions** and expectations of a **stronger U.S. dollar**
- ETF data **showed net outflows**, signaling that institutions were **reducing exposure**
- Large numbers of **leveraged positions** were built up in BTC futures and perpetual swaps, leaving the market vulnerable
- Once selling started, these leveraged long positions began **getting squeezed**

How The Crash Played Out

- As Bitcoin **broke key support levels**, forced liquidations occurred exchanges automatically closed over-leveraged trades
- These liquidations created a chain reaction, adding **additional sell pressure** on each leg down
- Market makers **widened spreads** during volatility spikes, which amplified downward moves
- **ETF outflows, lower liquidity, and negative funding rates** all combined to accelerate the decline

Lessons for Portfolio Managers

- **Crypto is driven by liquidity first:** When the macro environment tightens, BTC is one of the first assets to drop
- **Leverage intensifies selloffs:** Forced liquidations and deleveraging are major drivers of sharp declines
- **Flows matter as much as fundamentals:** ETF inflows/outflows and stablecoin supply often dictate short-term direction
- **Technical levels influence speed:** Once major price levels break in a 24/7 market, selling often accelerates into cascading moves



Source: Bloomberg, Financial Times, WSJ, Reuters

Note:

Portfolio Management Implications

Capital Flows Drives Markets

- FX and crypto both react strongly to where **money is moving globally**
- When investment flows into a country (or into BTC/ETH ETFs), **prices usually rise**
- When capital leaves because of fear, recession risk, or tighter financial conditions, **currencies and crypto tend to fall**
- **Understanding inflows/outflows** helps PMs anticipate big macro trends early

Political Stability & Policy Shape Long-Term Trend

- Stable governments, credible central banks, and clear policy typically **support them**
- Uncertainty, elections, or policy mistakes usually **weaken them**
- Crypto reacts similarly: regulation like the Genius Act improves confidence and encourages institutional adoption
- PMs should **monitor political cycles and policy** because major policy shifts often lead to large FX or crypto moves.

Technicals Matter in Both Markets Because Liquidity Is High

- FX is one of the **most technically traded markets** due to its size and global participation
- Crypto **behaves the same way**: with 24/7 trading and high liquidity, price levels break cleanly and trends often extend
- PMs **combine fundamentals, flow data, and key technical levels** to time entries and manage risk
- When fundamentals and technicals align, **both FX and crypto** offer clear, high-conviction setups.

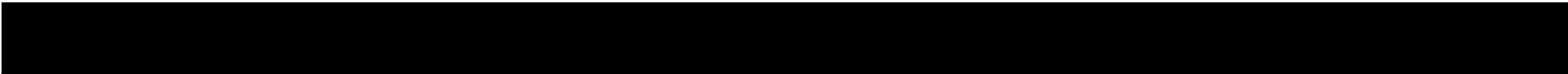


MACRO INVESTING ASSOCIATION

Futures



Agenda



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Introduction to Futures

Introduction

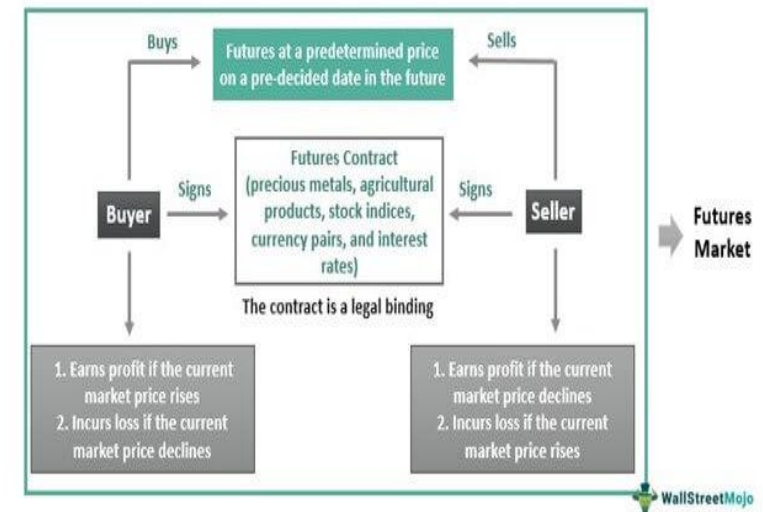
- Futures are **standardized** contracts that lock in the price of an asset for a future date
- They give exposure to **equities, rates, FX, and commodities** without owning the underlying asset
- Returns come from price movements in the underlying, amplified through margin-based leverage
- Investors use futures to **hedge risk, speculate on direction**, and express macro views as market conditions change.

Futures Contract Overview

- Futures trading begins with standardized contracts listed on regulated exchanges like **CME, ICE, and Eurex**
- Each contract **defines the underlying asset**, contract size, tick size, tick value, and delivery or cash-settlement terms
- Markets offer **full-size, mini, and micro contracts** to give traders scalable exposure with different notional values
- Margin requirements **vary by contract**, allowing investors to control large notional amounts with relatively small capital
- Contracts are **marked-to-market daily**, and most traders roll positions forward to avoid expiration and physical delivery



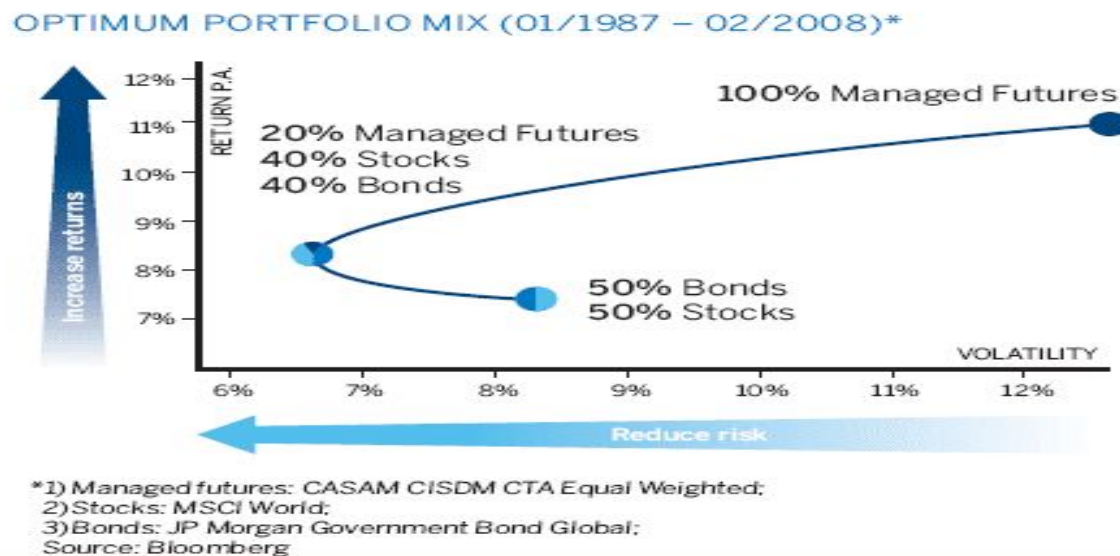
How Does A Futures Market Work?



Why Futures Matters

Future Applications Overview

- Futures give PMs **fast and efficient access** to broad market exposure without trading underlying securities
- They allow managers to **adjust portfolio beta** instantly in response to macro events or risk changes
- Futures serve as a **cost-effective hedging tool** to protect portfolios from adverse market movements
- Their liquidity and tight spreads make them ideal for implementing **tactical or defensive positioning**
- Futures preserve capital by **requiring only margin**, allowing PMs to deploy cash into higher-conviction ideas
- They enable cross-asset strategies and overlays that **enhance diversification and risk management**



Future Contract Specifics



Contract Specifications

- Underlying Asset:** The asset the contract represents, such as equity indices, Treasury bonds, commodities, or currencies
- Contract Size (Multiplier):** The standardized amount of the underlying controlled by one contract (e.g., ES = 50× index level)
- Tick Size:** The smallest allowable price movement for the contract (0.25 points for ES)
- Tick Value:** The dollar value of a single tick movement (e.g., ES tick = \$12.50)
- Expiration Months:** The standardized listing cycle determined by the exchange (often March, June, September, December)
- Settlement Type:** Whether the contract settles in cash (ES, NQ, VIX) or by physical delivery (CL, GC, ZB)

Exposure & Mechanics

- Notional Value:** Total exposure created by the contract, calculated as futures price × contract size
- Mark-to-Market:** Daily settlement where gains are credited and losses debited to the trader's margin account
- Initial Margin:** The required capital to open a position, typically a small percentage of notional value
- Maintenance Margin:** The minimum balance needed to keep the position open before triggering a margin call
- Liquidity:** Depth of market orders and volume that allows PMs to enter and exit positions efficiently
- Rolling Contracts:** The process of closing a near-expiration contract and opening a later one to maintain exposure

Preliminary Contract Specifications

	Micro E-mini S&P 500	Micro E-mini Nasdaq-100	Micro E-mini Russell 2000	Micro E-mini Dow
CONTRACT SIZE	\$5 x S&P 500 Index	\$2 x Nasdaq-100 Index	\$5 x Russell 2000 Index	\$0.50 x D/JIA Index
TRADING HOURS AND VENUE	CME Globex: Sun-Fri: 5:00 p.m. to 4:00 p.m. Mon-Fri: Daily trading halt from 3:15 p.m. to 3:30 p.m.			
MINIMUM TICK/PRICE FLUCTUATION	0.25 Index points	0.25 Index points	0.10 Index points	1.00 Index points
DOLLAR VALUE OF ONE TICK	\$1.25 per contract	\$0.50 per contract	\$0.50 per contract	\$0.50 per contract
PRODUCT CODE	CME Globex: MES CME ClearPort: MES Clearing: MES	CME Globex: MNQ CME ClearPort: MNQ Clearing: MNQ	CME Globex: MZK CME ClearPort: MZK Clearing: MZK	CME Globex: MYM CME ClearPort: MYM Clearing: MYM
CONTRACT MONTHS	Five months in the March Quarterly Cycle (March, June, September, December, March)			Four months in the March Quarterly Cycle (March, June, September, December)
DELIVERY	Cash settlement to Final Settlement Price			
TERMINATION OF TRADING	8:30 a.m. CT on 3 rd Friday of contract delivery month Trading in expiring futures terminates at 8:30 a.m. on Last Day of Trading			



Margin & Leverage

Margin Structure

- **Initial Margin:** The capital required to open a futures position, typically 3–12% of the contract's notional value
- **Maintenance Margin:** The minimum account balance needed to keep the position open before a margin call is triggered
- **Variation Margin:** Daily adjustments to the account as profits are added and losses are deducted through mark-to-market
- **Margin Call:** A requirement to deposit additional funds when the account falls below maintenance margin
- **Exchange vs. Broker Margin:** Exchanges set minimum margins, but brokers may require higher levels depending on risk

Leverage & Exposure

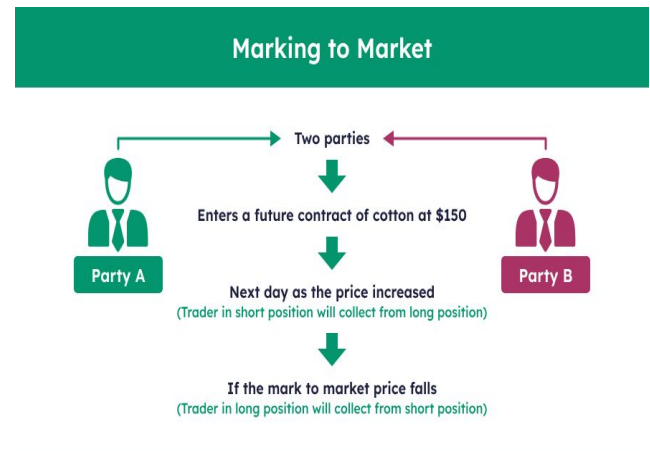
- **Embedded Leverage:** Futures naturally provide high leverage because traders control large notional amounts with limited capital
- **Notional vs. Margin:** A contract's notional value determines exposure, while margin determines required capital these are separate
- **Capital Efficiency:** PMs use futures to adjust exposure without tying up cash needed for other investments
- **Risk Scaling:** Managers adjust position sizes using full, mini, and micro contracts to target specific exposure levels
- **Volatility Impact:** Higher-volatility contracts require higher margin, affecting leverage and position sizing.



Margin Call
[mār-jən 'kôl]

A broker's demand that an investor who has borrowed money to buy securities deposit additional funds when the equity in their account drops below a minimum value, known as the maintenance margin.

Investopedia



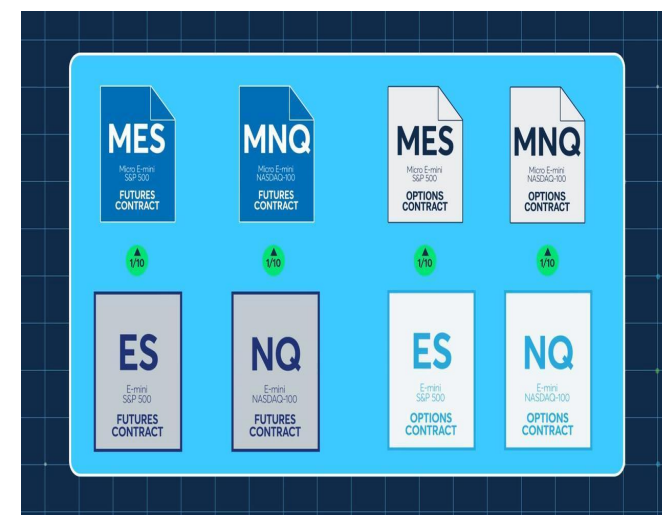
Full, Mini, and Micro Future Contracts

Contract Size Structure

- **Full-Size Contracts:** Standard institutional contracts that provide large notional exposure (e.g., ES, NQ, CL, ZN)
- **Mini Contracts:** Smaller versions of the full contract, typically 1/5 to 1/10 the size, offering more flexible exposure (e.g., E-mini S&P = 1/5 ES)
- **Micro Contracts:** The smallest standardized contracts, usually 1/10 the mini or 1/50 the full-size, ideal for precise sizing (e.g., Micro ES = 1/10 E-mini)
- **Notional Scaling:** Each contract type scales exposure proportionally, allowing PMs to fine-tune position size
- **Margin Scaling:** Margin requirements decrease in proportion to contract size, keeping leverage structure consistent.

Portfolio Applications

- **Precise Position Sizing:** Managers use minis and micros to match target exposures when full-size contracts are too large
- **Risk Management:** Smaller contracts help control volatility and drawdowns by aligning exposure with risk limits
- **Incremental Adjustments:** Micros allow PMs to adjust exposures in small increments rather than large, jumpy moves
- **Multi-Asset Overlays:** Combining full, mini, and micro contracts enables custom beta overlays across indices and asset classes.
- **Accessibility:** Smaller contracts make futures markets usable for funds with smaller mandates or tighter risk budgets.



Notional Value Examples

Equity Index Futures

- **S&P 500 E-mini (ES):** Notional = Index Level $\times$ 50 (e.g., $5,000 \times 50 = \$250,000$ exposure)
- **S&P 500 Micro (MES):** Notional = Index Level $\times$ 5 (1/10th of ES, e.g., $5,000 \times 5 = \$25,000$ exposure)
- **Nasdaq E-mini (NQ):** Notional = Index Level $\times$ 20 (e.g., $18,000 \times 20 = \$360,000$ exposure)
- **Nasdaq Micro (MNQ):** Notional = Index Level $\times$ 2 (e.g., $18,000 \times 2 = \$36,000$ exposure)

Rates & Commodities

- **10-Year Treasury Note (ZN):** Notional = \$100,000 per contract, priced in 1/32 increments
- **Crude Oil (CL):** Notional = 1,000 barrels $\times$ futures price (e.g., \$80 = \$80,000 exposure)
- **Gold (GC):** Notional = 100 troy ounces $\times$ futures price (e.g., \$2,000 = \$200,000 exposure)
- **Micro Gold (MGC):** Notional = 10 ounces $\times$ price (e.g., $10 \times \$2,000 = \$20,000$ exposure)

Calculating Notional Value



$$\text{Notional Value} = \text{Total Units in the Contract} \times \text{Spot Price}$$

How to Size a Futures Position

Determine the Required Exposure

- **Portfolio Value:** Identify the dollar amount of the portfolio you want to hedge or adjust
- **Target Exposure Change:** Define how much beta or notional exposure you want to add or remove
- **Contract Notional:** Use each futures contract's notional value to understand how much exposure one contract provides
- **Hedge or Overlay Ratio:** Calculate required exposure $\div$ contract notional to find how many contracts are needed
- **Choose Contract Size Tier:** Use full, mini, or micro contracts to match the exact exposure target

Practical Position Sizing Steps

Step 1: Decide if you are adding exposure (long futures) or reducing exposure (short futures)

Step 2: Compute the dollar exposure needed to reach your target (e.g., hedge 30% of a \$10M portfolio = \$3M exposure)

Step 3: Divide that exposure by the notional value of the contract (e.g., \$3M $\div$ \$250,000 per ES = 12 contracts)

Step 4: Adjust with minis/micros to fine-tune the position and avoid overshooting

Step 5: Verify margin requirements fit within the portfolio's risk and liquidity constraints

Hedging, Term Structure, and Rolling Contracts

Hedging Market Exposure

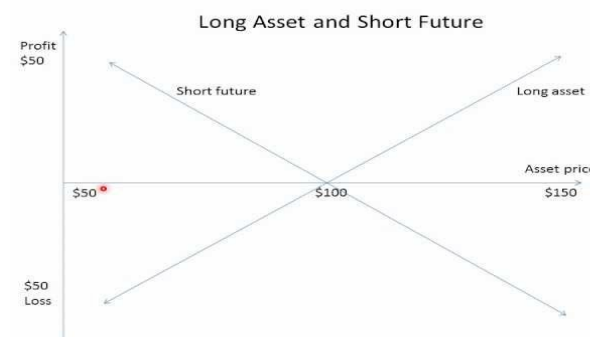
- Futures allow PMs to **reduce portfolio risk instantly** by shorting index, rate, or commodity contracts without selling any positions
- When markets look overextended or macro risk rises, a PM can short a small number of contracts to cut exposure quickly (**e.g., short ES to hedge equity risk or short ZN to hedge duration**)
- This **preserves** long-term holdings while neutralizing unwanted short-term volatility

Understanding Term Structure (Contango & Backwardation)

- Futures are priced for different expiration months, **creating a curve** that reveals the cost of holding exposure over time
- Contango** (upward-sloping curve) means future months are **more expensive**, increasing the cost of rolling a position forward and reducing expected return
- Backwardation** (downward-sloping curve) means front months are **more expensive**, creating favorable roll yield for long positions
- PMs use this curve to decide whether **holding futures is efficient** or if ETFs or cash products are better

Rolling Contracts to Maintain Exposure

- Because futures expire, PMs must **“roll”** by selling the current contract and buying the next month before expiry
- Rolling **avoids physical delivery** in commodities and avoids being locked out of expiring contracts in financial futures
- The timing of the roll affects cost:** rolling early can avoid liquidity drops, while rolling late may benefit from tighter spreads
- Understanding rolls is essential for portfolios using long-term futures exposure (**equity overlays, cash equitization, commodity exposure**)



Applied Futures Strategies: Hedging vs. Speculation

Hedging Strategies (Risk Reduction)

- **Portfolio Beta Hedge:** Short equity index futures to reduce market exposure without selling underlying holdings
- **Duration Hedge:** Short Treasury futures to protect bond portfolios from rising rates
- **Commodity Price Hedge:** Use long or short commodity futures to lock in future prices and stabilize input costs
- **FX Hedge:** Use currency futures to eliminate FX risk in international portfolios
- **Event Hedge:** Short futures ahead of high-risk macro events (CPI, FOMC, earnings) to limit downside volatility

Understanding Term Structure (Contango & Backwardation)

- **Directional Long/Short:** Take outright long or short futures positions to express a macro or market view
- **Event-Driven Speculation:** Use futures to position ahead of economic data, central bank decisions, or geopolitical catalysts
- **Momentum/Trend Following:** Go long strong-trending assets or short weak ones using highly liquid futures markets
- **High-Conviction Tilt:** Increase exposure to favored asset classes using futures without altering core holdings

Spread & Relative-Value Strategies

- **Intermarket Spread:** Long one asset and short another to trade relative performance (e.g., long Nasdaq, short S&P 500)
- **Calendar Spread:** Long one expiration and short another to trade changes in futures curve
- **Cross-Asset Spread:** Combine futures across asset classes (e.g., long equities, short bonds) to express macro regime views
- **Pair Trade Futures:** Long one index and short another to trade style or factor rotations with reduced market risk

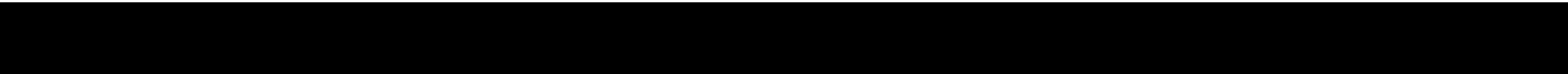


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Options



Agenda



Introduction to Options	3
Options Contract Structure	4
Option Payoff, Profit, and Moneyness	5
The Option Greeks	6
Put-Call Parity & Synthetics	7
Options Matrix	8-9
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Introduction to Options

What are options?

- **Call Option:** Gives the holder the right, but not the obligation, to buy the underlying asset at a specific strike price before expiration
- **Put Option:** Gives the holder the right, but not the obligation, to sell the underlying asset at a specific strike price before expiration
- **Option Buyer:** Pays a premium for the right; risk is limited to the premium paid while upside can be significant
- **Option Seller (Writer):** Receives the premium but takes on an obligation; risk can be substantial depending on direction and strategy
- **Standardized Contracts:** Listed options specify the underlying asset, contract multiplier, strike price, expiration date, and settlement method
- **European vs. American Style:** American options can be exercised anytime before expiration; European options only at expiration

Why are they useful?

- Options allow PMs to express **directional, volatility, and risk-adjusted views** with far less capital than owning the underlying asset
- They provide targeted hedging tools that protect portfolios from downside risk while preserving upside potential
- Options **enhance income generation** through premium collection strategies such as covered calls and cash-secured puts
- They offer **asymmetric payoff structures** that help PMs fine-tune risk exposures without fully altering core holdings
- Options enable **precise tactical positioning** around events, volatility shifts, and market dislocations

Option Contract Structure

Contract Specifications

- **Underlying Asset:** The stock, ETF, index, or futures contract the option is based on
- **Strike Price:** The fixed price at which the option holder can buy (call) or sell (put) the underlying
- **Expiration Date:** The date on which the option expires; can be daily, weekly, monthly, or long-dated (LEAPS)
- **Premium:** The price paid by the buyer, consisting of intrinsic value and time value
- **Contract Multiplier:** Most equity options control 100 shares per contract; index and futures multipliers vary
- **Exercise Style:** American options can be exercised anytime before expiration; European options only at expiration.

How Options Derive Value

- **Intrinsic Value:** The immediate exercise value— $\max(0, \text{Underlying} - \text{Strike})$ for calls; $\max(0, \text{Strike} - \text{Underlying})$ for puts
- **Extrinsic Value:** The time value portion of the premium driven by expiration length and implied volatility
- **Moneyness:** Classification based on price relationship: in-the-money, at-the-money, or out-of-the-money
- **Volatility Impact:** Option premiums rise as expected volatility increases, especially near-the-money options
- **Time Decay:** Options lose time value as expiration approaches, with decay accelerating near the final weeks

Option Payoff, Profit, and Moneyness

Payoff & Profit Fundamentals

- **Call Payoff:** $\max(0, \text{Underlying} - \text{Strike})$
- **Put Payoff:** $\max(0, \text{Strike} - \text{Underlying})$
- **Buyer Profit:** Payoff – Premium Paid
- **Seller Profit:** Premium Received – Payoff
- **Breakeven (Call Buyer):** Strike + Premium
- **Breakeven (Put Buyer):** Strike – Premium
- **Expiration Rule:** At expiration, value = intrinsic value only (time value = 0).

Max Gain / Max Loss for the 4 Basic Positions

- **Long Call max gain:** Unlimited, **Long Call max loss:** Premium paid
- **Short Call max gain:** Premium received, **Short Call max loss:** Unlimited if underlying rises
- **Long Put max gain:** Strike – 0 – Premium (if underlying goes to zero), **Long Put max loss:** Premium paid
- **Short Put max gain:** Premium received, **Short Put max loss:** Strike – 0 – Premium (if underlying goes to zero)

ITM, ATM, and OTM Definitions

ITM (In-the-Money):

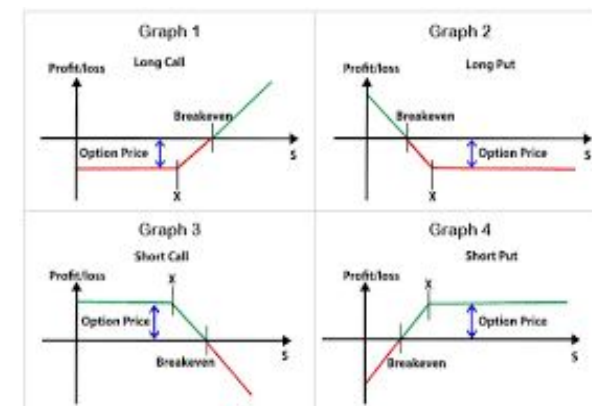
- Call ITM: price > strike
- Put ITM: price < strike
- Has intrinsic value at expiration

ATM (At-the-Money):

- Price $\approx$ strike
- Highest gamma and vega sensitivity

OTM (Out-of-the-Money):

- Call OTM: price < strike
- Put OTM: price > strike
- Zero intrinsic value; expires worthless if still OTM



The Option Greeks

Primary Greeks

- **Delta:** Measures how much the option price changes for a \$1 move in the underlying; also represents directional exposure
- **Gamma:** Measures how quickly delta changes as the underlying moves; highest for near-the-money, near-expiration options
- **Theta:** Measures the rate at which the option loses time value each day; always negative for long options and positive for short options
- **Vega:** Measures the option's sensitivity to changes in implied volatility; higher vega means larger reactions to volatility shifts

How PM's Use the Greeks

- **Delta Exposure:** PMs use delta to understand how options change portfolio risk and adjust beta without trading the underlying
- **Gamma Risk:** High gamma positions require active management because deltas shift rapidly in volatile markets
- **Theta Income:** PMs sell options (covered calls, cash-secured puts) to harvest theta and generate steady premium income
- **Volatility Trading (Vega):** Options allow PMs to express views on volatility buying options in low-vol, selling options in high-vol environments.

The Five Main Greeks				
				
Delta (Δ)	Theta (Θ)	Gamma (Γ)	Vega (V)	Rho (ρ)
Represents the sensitivity of an option's price to changes in the value of the underlying security.	Represents the rate of time decay of an option.	Represents the rate of change of Delta relative to the change of the price of the underlying security.	Represents an option's sensitivity to volatility.	Represents how sensitive the price of an option is relative to interest rates.

Put-Call Parity & Synthetics

Put-Call Parity

- **Put-Call Parity:** $C - P = S - PV(\text{Strike})$, showing the no-arbitrage relationship between calls, puts, and stock
- **No-Arbitrage Principle:** If parity doesn't hold, traders can create risk-free profits through arbitrage
- **Value Components:** Parity links option premiums to stock price, strike price, interest rates, and time to expiration
- **Use Case:** PMs use parity to understand whether options are mispriced relative to stock
- **Foundation:** This formula is the basis for constructing synthetic long/short exposures.

Synthetics

- **Synthetic Long Stock:** Long call + short put at the same strike and expiration
- **Synthetic Short Stock:** Short call + long put at the same strike and expiration
- **Synthetic Long Call:** Long stock + long put
- **Synthetic Long Put:** Long call + short stock
- **Application:** Synthetics let PMs replicate stock exposure, hedge, or restructure risk without buying the underlying asset.

Option strategies: direction and volatility

		Direction		
		Bearish	Neutral/No Bias	Bullish
Volatility	High	Buy puts	Buy straddle	Buy calls
	Average	Write calls and buy puts	Spreads	Buy calls and write puts
	Low	Write calls	Write straddle	Write puts

Directional & Volatility Option Strategies

Directional Strategies (Bullish & Bearish)

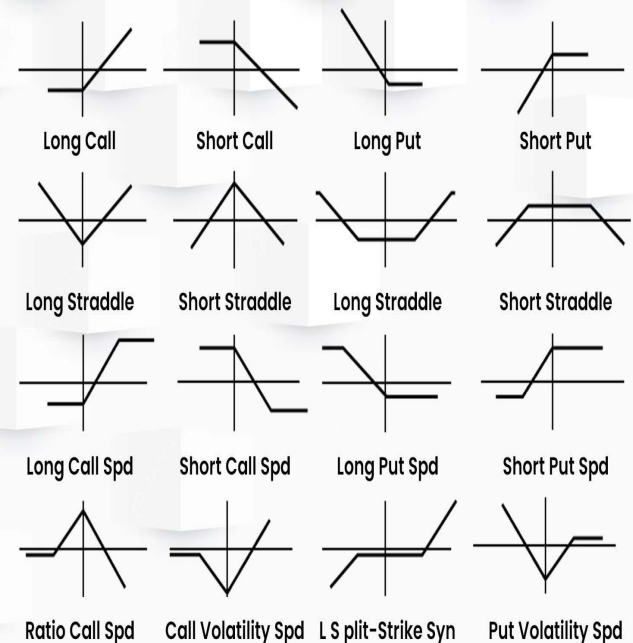
- **Long Call:** Buy a call at a specific strike and expiration to gain upside exposure with limited downside risk
- **Long Put:** Buy a put at a defined strike and expiration to profit from downward moves or hedge single-position risk
- **Bull Call Spread:** Buy a call and sell a higher-strike call with the same expiration to lower cost while capping upside
- **Bear Put Spread:** Buy a put and sell a lower-strike put with the same expiration to reduce premium cost while defining maximum profit
- **Call Replacement:** Buy ATM or slightly OTM calls instead of stock to gain bullish exposure with less capital at risk

Volatility Strategies (Betting on High or Low Vol)

- **Long Straddle:** Buy a call and a put at the same strike and expiration to profit from large moves in either direction
- **Long Strangle:** Buy an OTM call and OTM put with the same expiration to bet on a volatility increase with lower upfront cost than a straddle
- **Short Straddle:** Sell a call and put at the same strike and expiration to profit from low volatility and time decay (high risk)
- **Short Strangle:** Sell OTM call and OTM put with the same expiration to collect premium when expecting stable markets
- **Calendar Spread:** Buy a longer-dated option and sell a shorter-dated option at the same strike to trade time decay and volatility term structure

Different Option Strategies That Investors Should Know

Some Options Strategy Payoffs



Strike

Applied PM Option Strategies

Income & Yield Enhancement

- **Covered Call:** Sell a call at a chosen strike and expiration against an existing stock position to generate premium income while limiting upside beyond the strike
- **Cash-Secured Put:** Sell a put at a target strike and expiration, earning income and potentially acquiring shares at a lower effective price
- **Overwriting Program:** Systematically sell calls with consistent expirations (weekly or monthly) on a portion of the portfolio to harvest steady premium
- **Buy-Write Strategy:** Buy the stock and sell a call at the same expiration to enhance yield during range-bound markets

Hedging & Risk Management

- **Protective Put:** Buy a put with a defined strike and expiration to establish a downside floor while keeping full upside exposure
- **Collar Strategy:** Combine a long put and short call with the same expiration to cap downside and upside while reducing net hedging cost
- **Stock Replacement:** Purchase calls (or call spreads) at specific strikes and expirations to mimic stock exposure with reduced capital and lower downside risk
- **Index Hedging:** Buy index puts or put spreads at chosen strikes and expirations to protect an entire equity portfolio from broad market declines



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Technical Analysis



Trend Signals

Simple Moving Averages

Introduction

- The average price over n periods.
- This is considered to be a smooth and stable indicator and is considered good for identifying long-term trend direction

Types of SMAs

- There are various types of moving averages, the main one being the 20 day, 50 day and 200 day moving average
- **20 Day Moving Average**
 - Average price over the past 20 candles
 - Identify short-term momentum
 - Spot fast trend changes
- **50 Day Moving Average**
 - Average price over the past 50 candles
 - Identify medium-trend direction
- **200 Day Moving Average**
 - The average price over the past 200 candles
 - This indicator is used for long-term trend

How Traders Combine Them

•Golden Cross

- 50 MA crosses above the 200 indicating long-term bull trend starting

•Death Cross

- 50 MA crosses below the 200 indicating long-term bear trend starting



Moving Averages Convergence Divergence(MACD)

Introduction

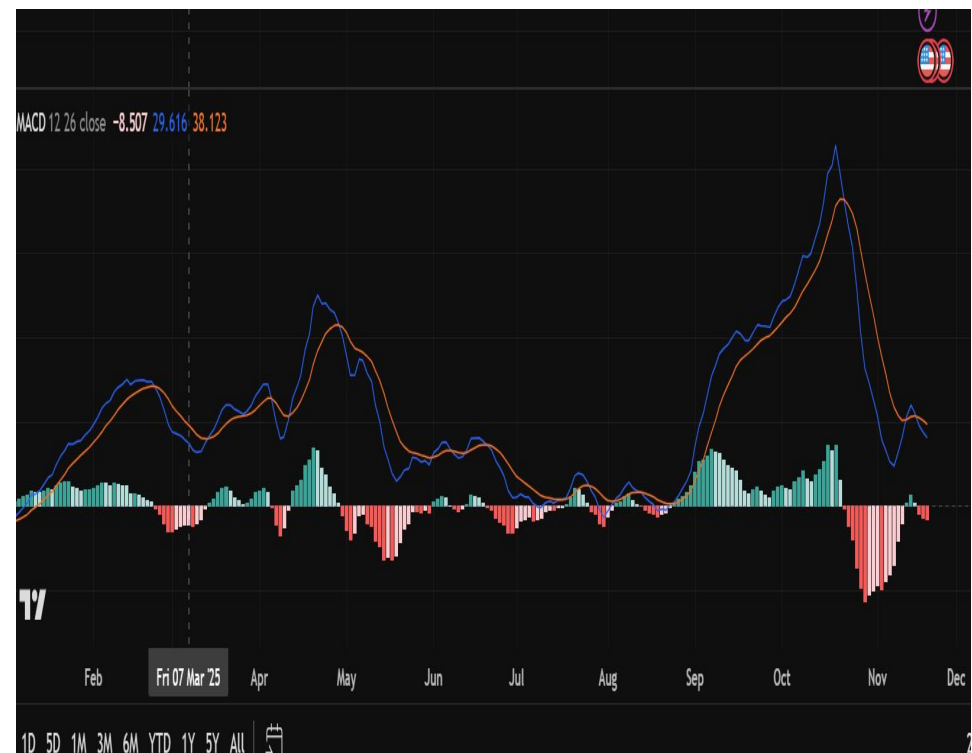
- It is a momentum indicator that measures the relationship between two EMAs
- It helps you see when momentum is increasing or decreasing

How MACD is Calculated?

- The MACD has three components
 - MACD Line**
Trend following momentum indicator
 - Signal Line**
This is the 9-period EMA of the MACD line
 - Histogram**
This is the difference between MACD line and signal line

Main Trading Signals

- If MACD Line Crosses Above Signal Line -> Bullish
- Indicates upward momentum increasing
- Often interpreted as a buy signal
- If MACD Line Crosses Below Signal Line -> Bearish
- Indicates downward momentum increasing
- Often interpreted as a sell signal



Average Directional Index

Introduction

- Technical indicator is used to measure how strong a trend is regardless of it going up or down
- It does not tell the direction only the strength of the index

How ADX is Calculated?

- **ADX is calculated from two other lines**
 - +DI(Positive Directional Indicator)
Trend following momentum indicator
 - DI(Negative Directional Indicator)
This is the 9-period EMA of the MACD line
- Histogram**
This is the difference between MACD line and signal line

Main Trading Signals

- 1)Trend Confirmation
 - Indicates upward momentum increasing
 - Often interpreted as a buy signal
- 2)Avoiding Chop
 - If $ADX < 20$, the market is sideways
- 3)Spotting Trend Reversals
 - When ADX dips from high levels, it signal trend weakning



Support & Resistance Levels

Introduction

Support and resistance are **price levels where market reactions repeatedly occur** because of **liquidity, order flow, and market psychology**.

• **Support** = a price level where **buying interest is strong enough** to stop price from falling.

• **Resistance** = a price level where **selling interest is strong enough** to stop price from rising.

How Price Typically Behave at these Levels

At Support:

- Wick rejections
- Long tails
- Bullish divergence
- Volume drying up
- Market defending the level

At Resistance:

- Rejection wicks
- Lower highs
- Bearish divergence
- Aggressive selling volume



Source: Bloomberg, Financial Times

Note:



Momentum Signals

Relative Strength Index(RSI)

Introduction

- Momentum Oscillator that measures the speed and magnitude of an asset to measure whether
- The RSI answers one question – Are buyers or sellers dominating too aggressively in the short-term

Key Levels

- **RSI < 30 → Oversold**
 - Sellers have been in control aggressively
 - Could signal exhaustion or a potential bounce
- **RSI > 70 → Overbought**
 - Buyers have pushed too aggressively
 - Could signal a potential pullback
- **RSI 45–55 → Neutral / trend transition zone**
 - Useful for trend identification



Indicators to pair RSI with

- Many prof traders cannot use just RSI to predict the direction of an asset. So they use a combination.
- **RSI + ADX** – Provides directional momentum backed by trend strength
- **RSI + Moving Averages** – Shows whether the momentum is aligned with the trend
- **RSI + ATR** – Tells whether the momentum is meaningful or noise from the volatility

Stochastic Oscillator

Introduction

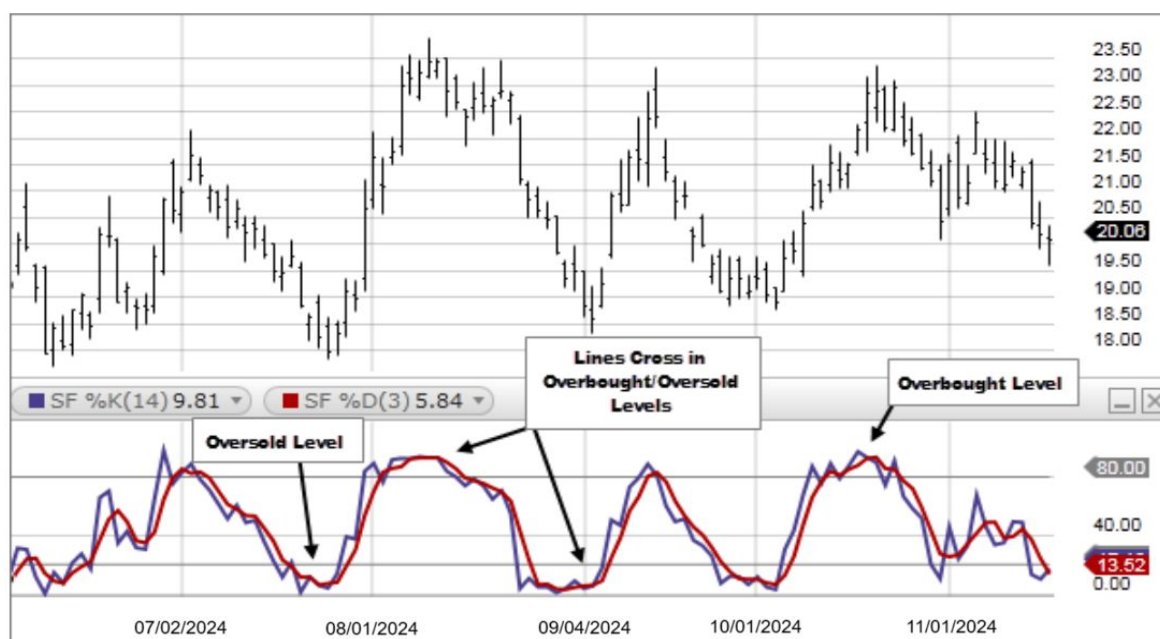
It is a momentum indicator which is used to show where the current price sits relative to its recent trading range

It measures how strong or weak momentum is by comparing current trading prices to the range of highs and lows over a chose time period.

How traders interpret it?

Above 80 = Overbought territory
Strong momentum but possibly topping

Below 20 = Oversold
->Weak Momentum, possibly bottoming





Other Resources



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Tariff Implications



Introduction and Club Format	3-5
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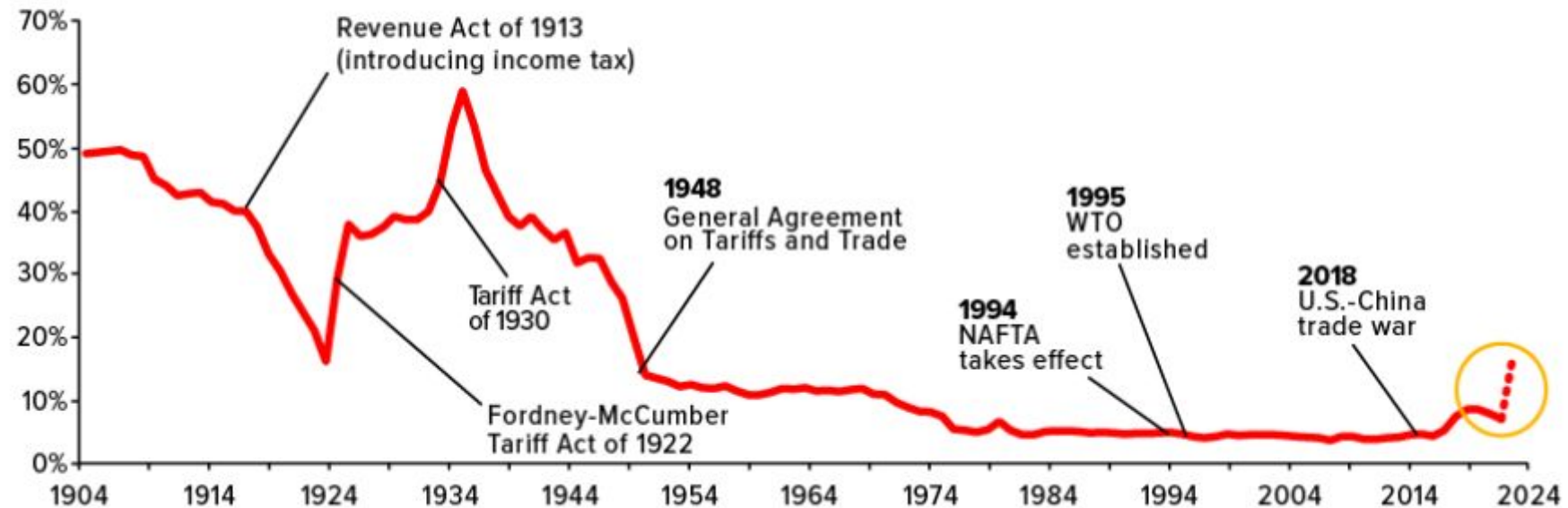


Tariff

Tariffs: A Major Source of Government Revenue before 1900

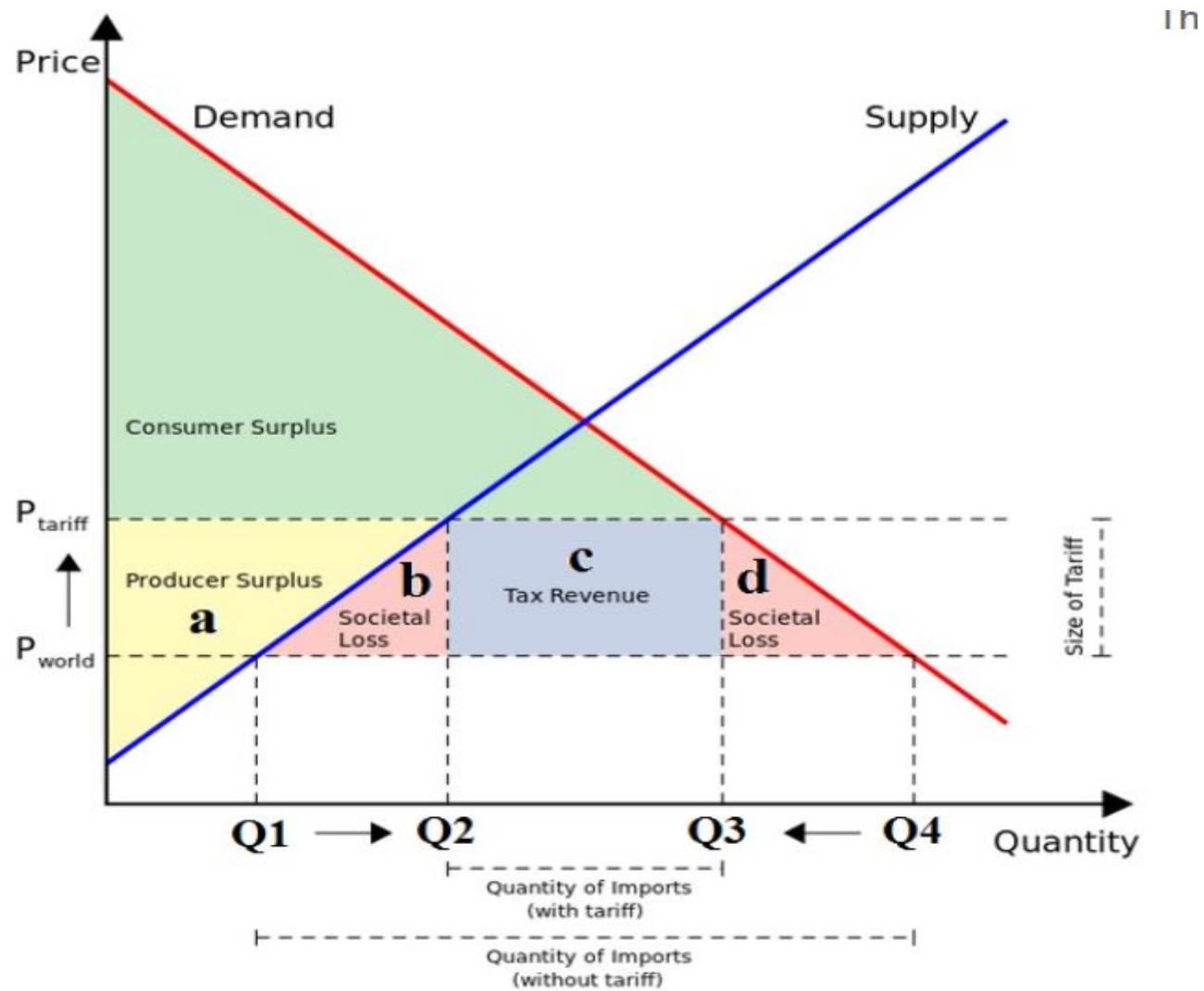
Trump's Tariff Increases Would Be the Largest Since World War II

Average Tariffs on Dutiable Imports | 1900 – 2024



Source: PIIE, U.S. Global Investors

Tariffs: Deadweight Loss



Tariffs: Pros and Cons



Pros

- Increase Government Revenue
- Domestic Producer Benefit
- Protect Infant/Specialize Industry
- Gain Support from Voters/People
- Damage Trading Partners

Cons

- Consumer Pay More
- Downstream Company Pay More
- Retaliation from Trading Partners
- Limit Product Diversification and Competition

TABLE 1: Payoff-matrix for strategies

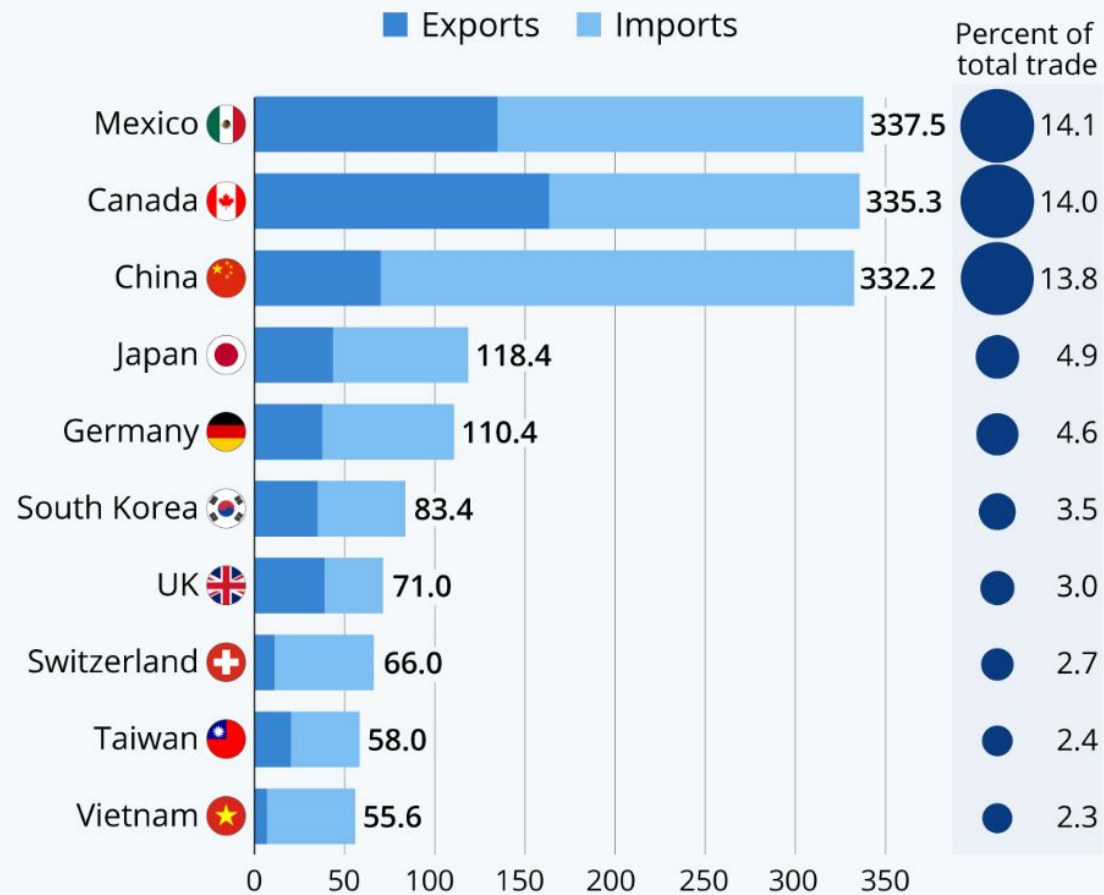
		EU	
		Liberalization	Protectionism
US	Liberalization	Highest, highest	Lowest, high
	Protectionism	High, lowest	Low, low

Source: author's own illustration.

| BertelsmannStiftung

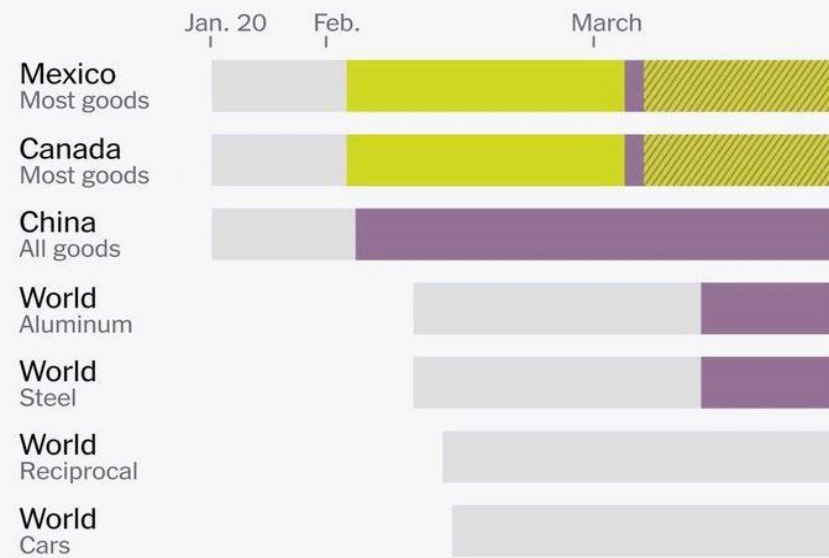
Trading Partners

Top U.S. trading partners for goods only
(in billion of U.S. dollars) *



Tracking Trump's Tariffs

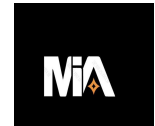
Announced In effect Partly in effect Paused



Note: Goods imported from Mexico and Canada covered under the USMCA trade pact received a temporary reprieve from new tariffs.



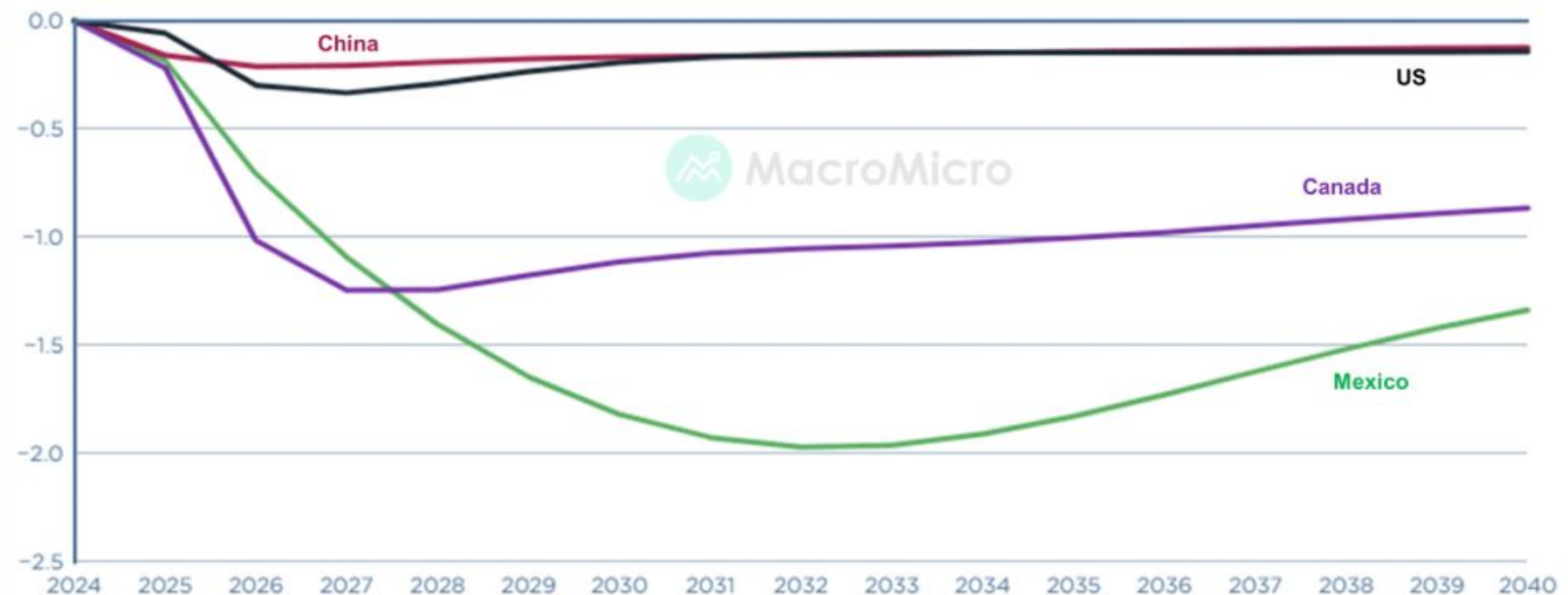
Impact of Tariffs on China, Mexico, and Canada



Canada and Mexico to Face Greater Economic Drag Under New Tariffs

Projected Percent Change in Real GDP from Baseline Forecast (%)

Scenario: US imposes 25% tariff on Canada and Mexico and 10% tariff on China, with retaliation from China

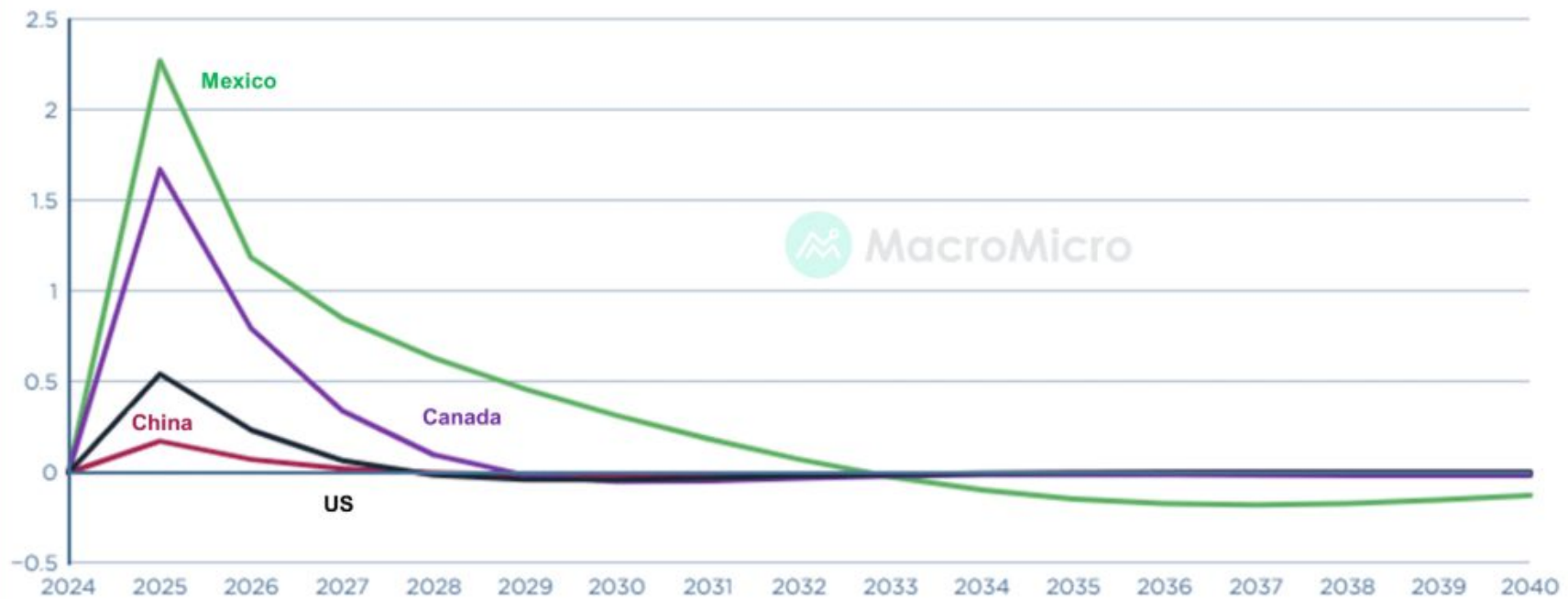


Source: PIIE

Impact of Tariffs on China, Mexico, and Canada

Canada and Mexico to Face Greater Inflation Impact Under New Tariffs

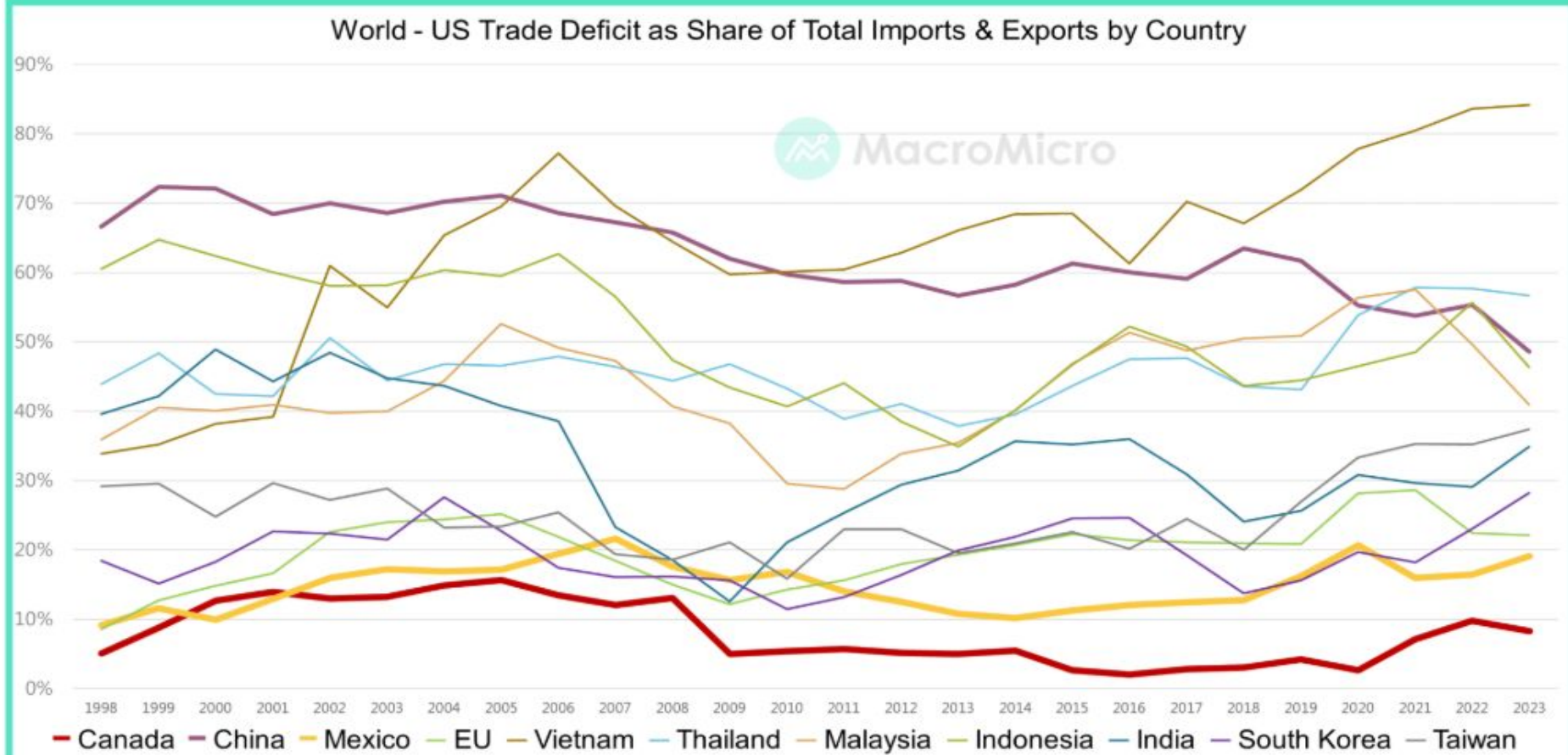
Projected Percentage Point Change in Inflation from Baseline Forecast (ppt)
Scenario: US imposes 25% tariff on Canada and Mexico and 10% tariff on China, with retaliation from China



Source: PIIE

Impact of Tariffs on China, Mexico, and Canada

Canada & Mexico Account for a Small Share of US Trade Deficit



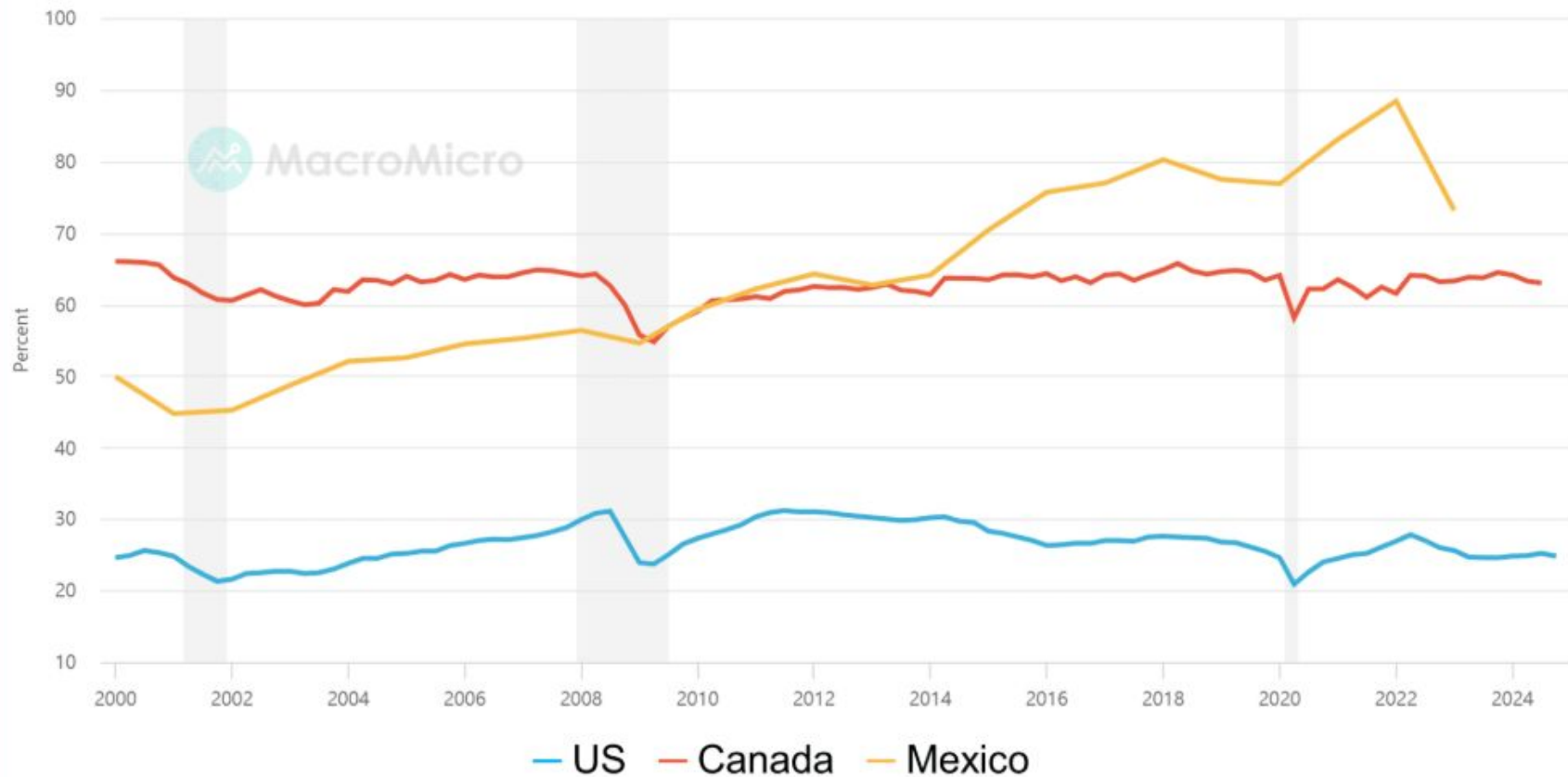
Source: US Census Bureau. Chart compiled by MacroMicro

Impact of Tariffs on China, Mexico, and Canada

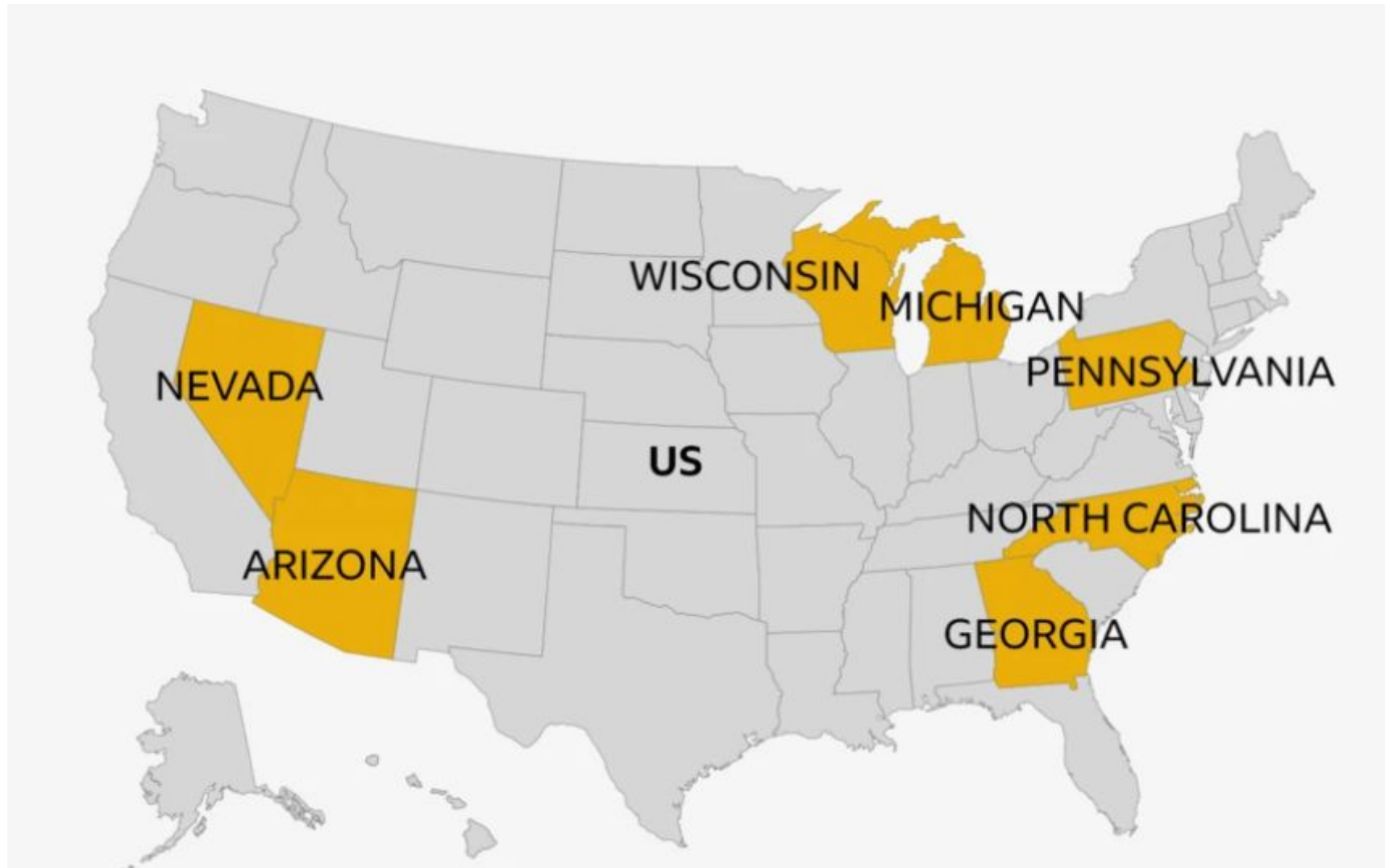
Compared to the US, Canada and Mexico Are More Reliant on Trade

Trade as Share of GDP by Country

MacroMicro.me | MacroMicro



Swing States

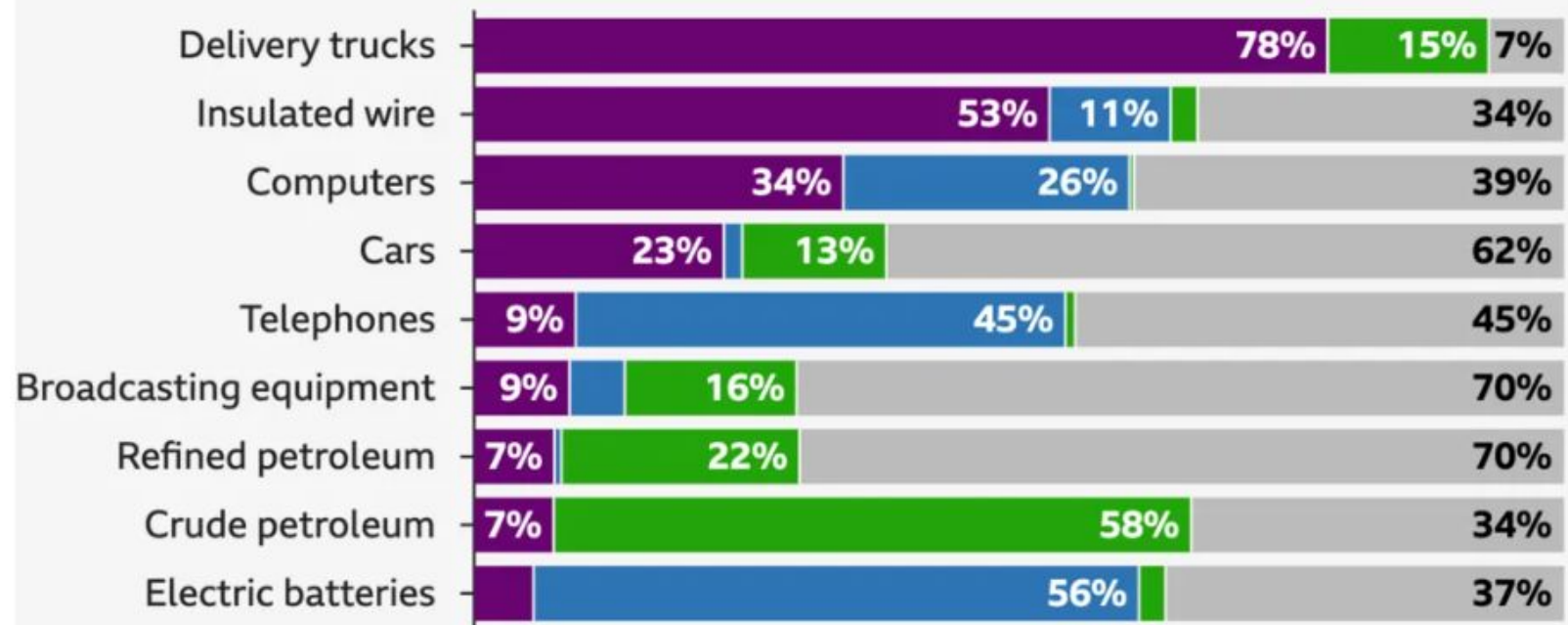


Which products will be affected?

Trump's tariffs: Which products will be affected?

Key items imported to the US from Mexico, China and Canada in 2024

■ Mexico ■ China ■ Canada ■ Others

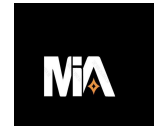


Source: US Census Bureau • BBC calculations based on item's customs value in US\$
2024 data through November



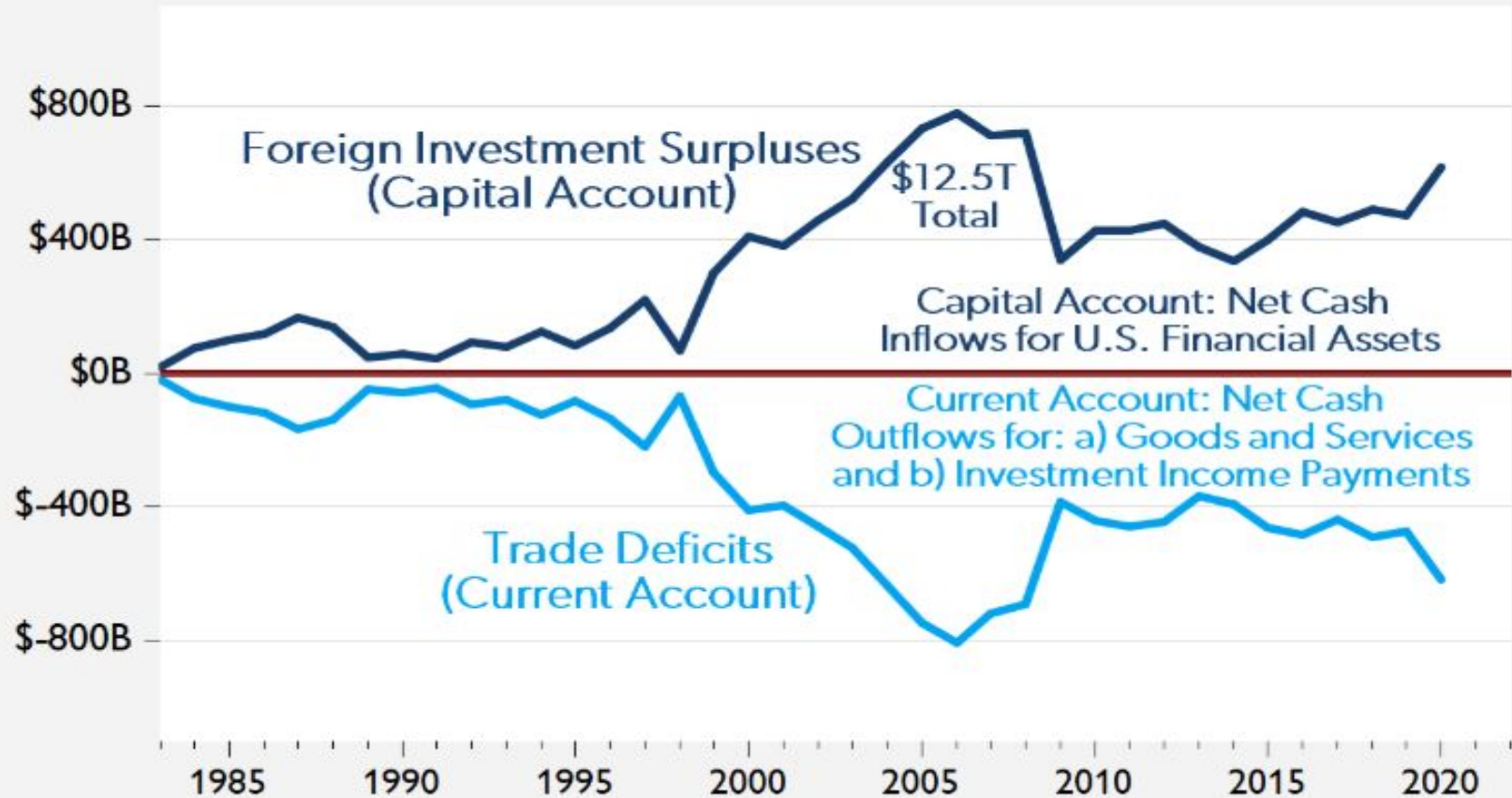
Trade

Current Account Deficit to GDP

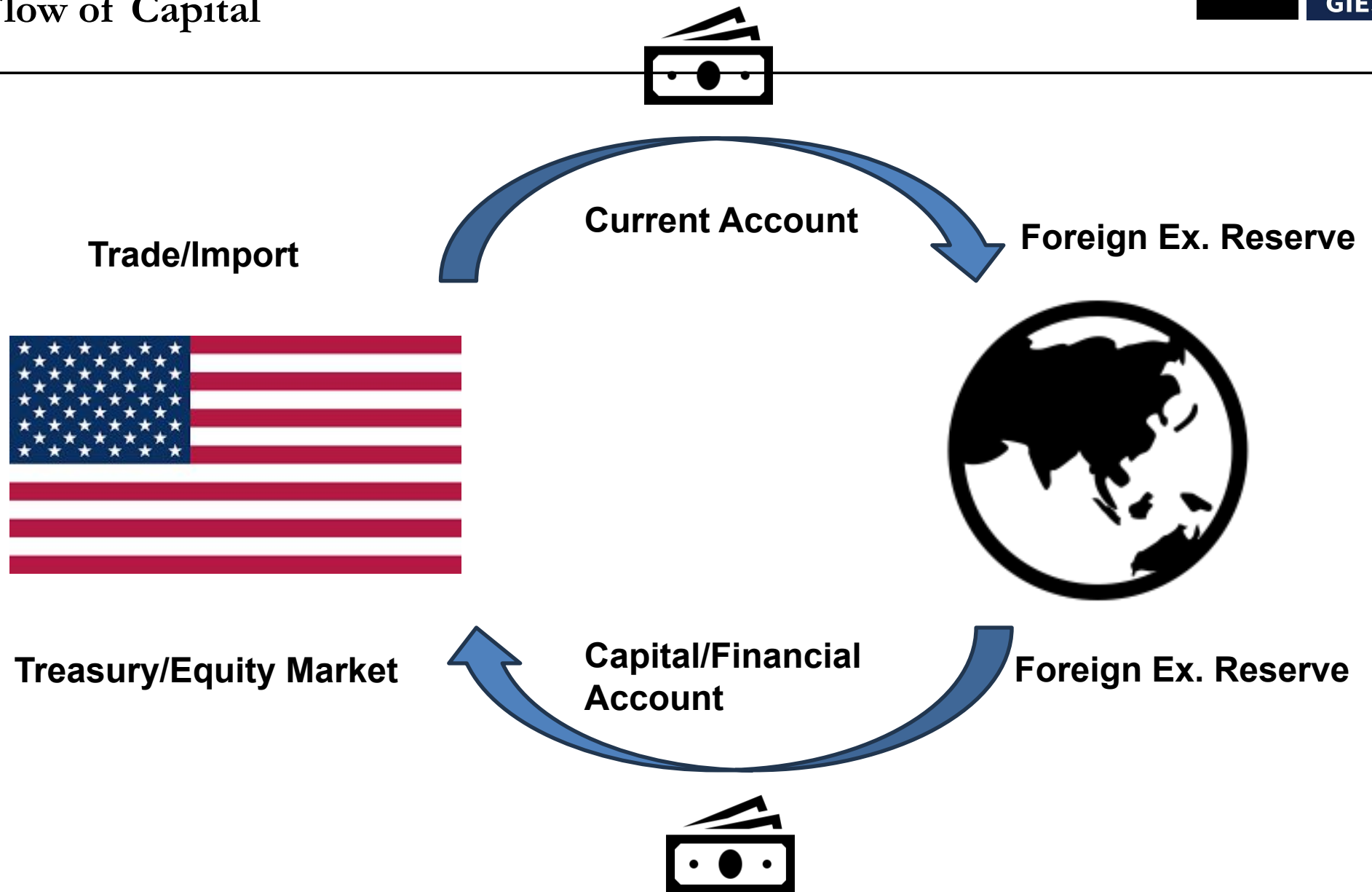
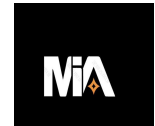


Balance of Payments

U.S. Balance of Payments, 1983 to 2020

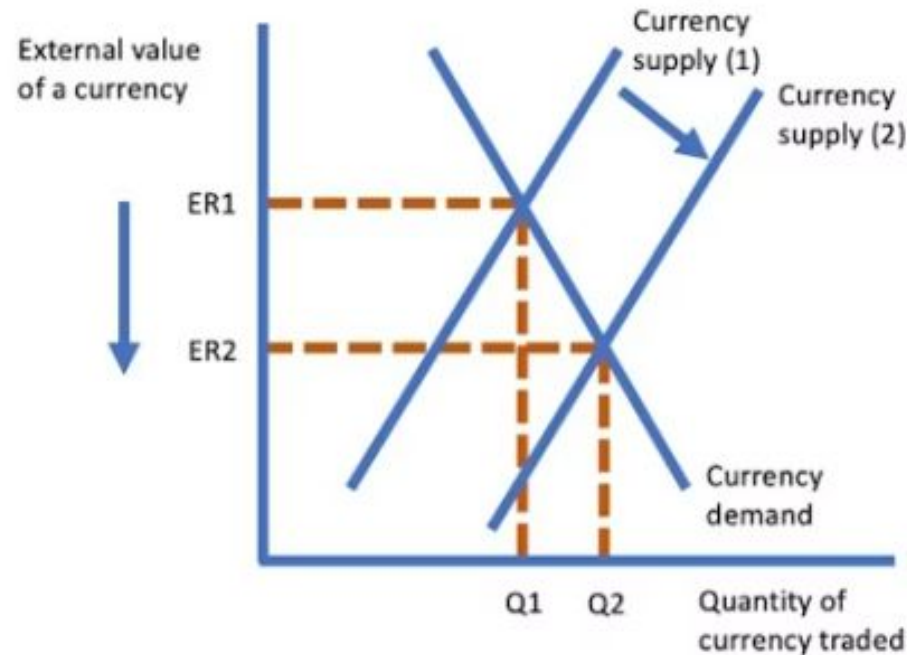


Flow of Capital

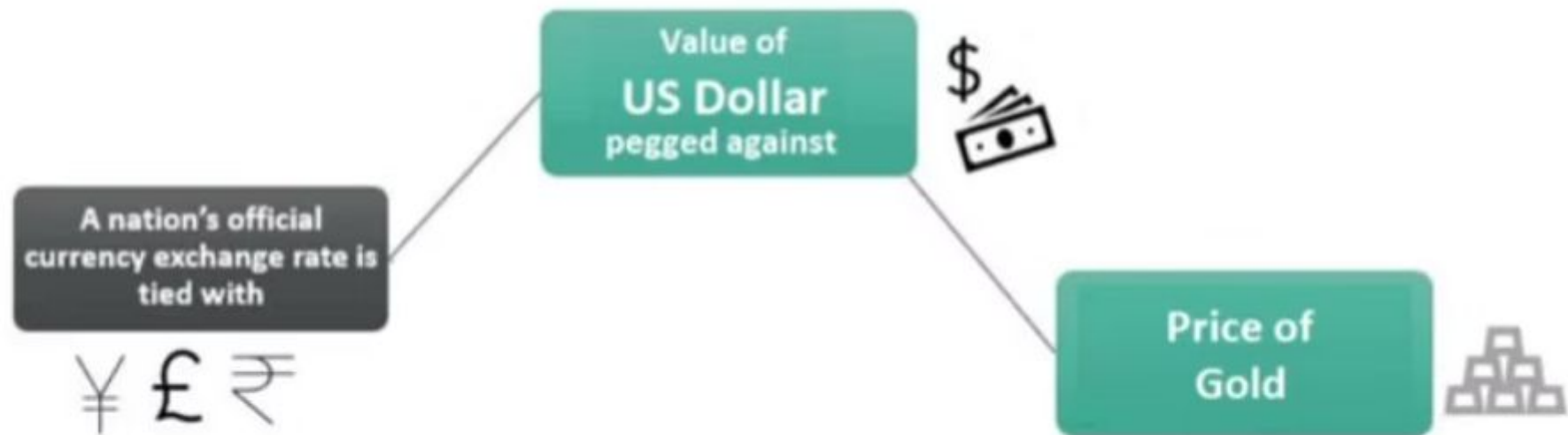


Currency Value Self-Adjust Mechanism when Trade Deficit Rise

Examine how an increase in a country's current account deficit might cause a change in the external value of their currency.

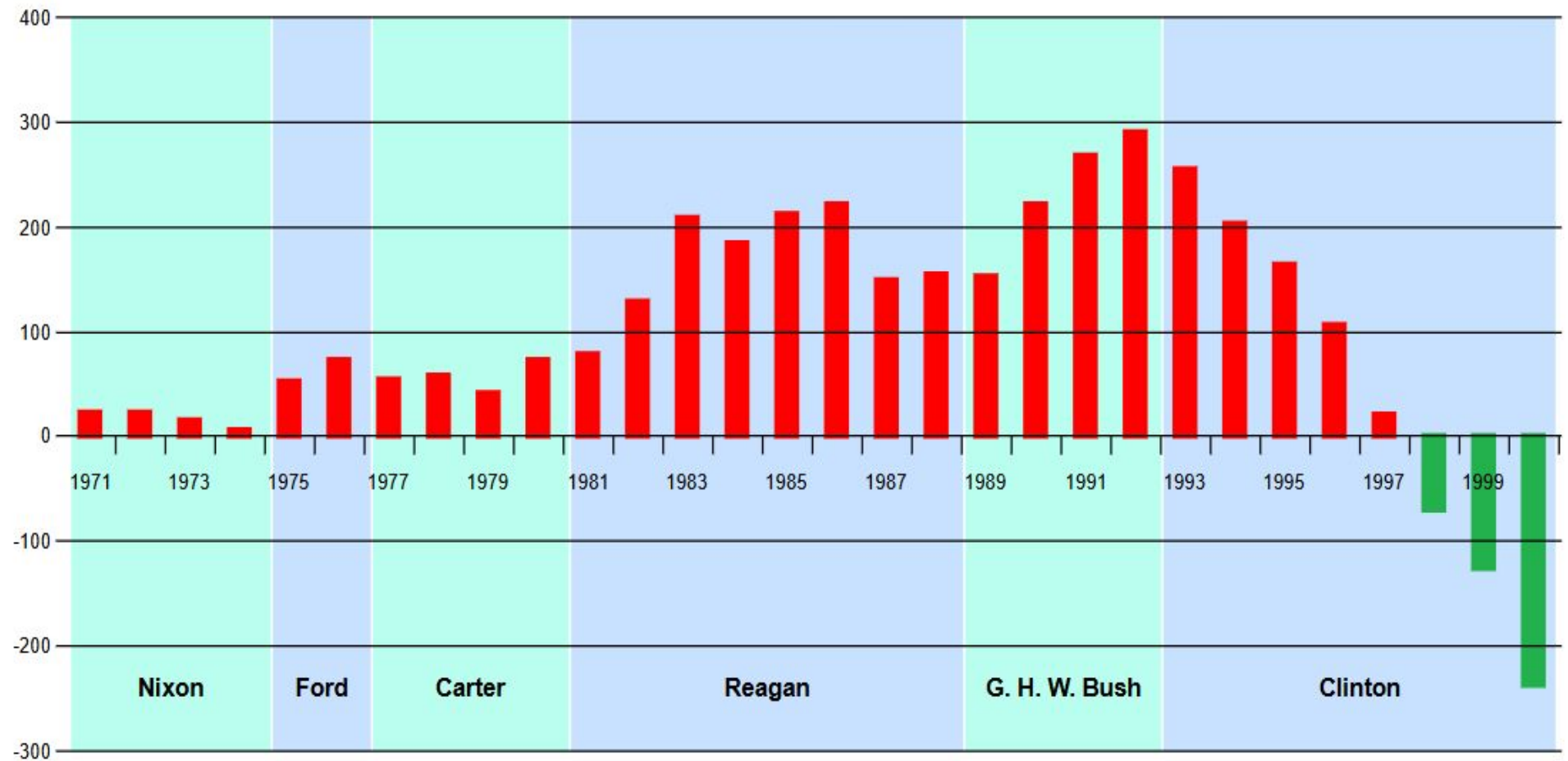
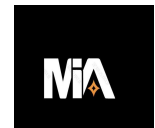


Bretton Woods System

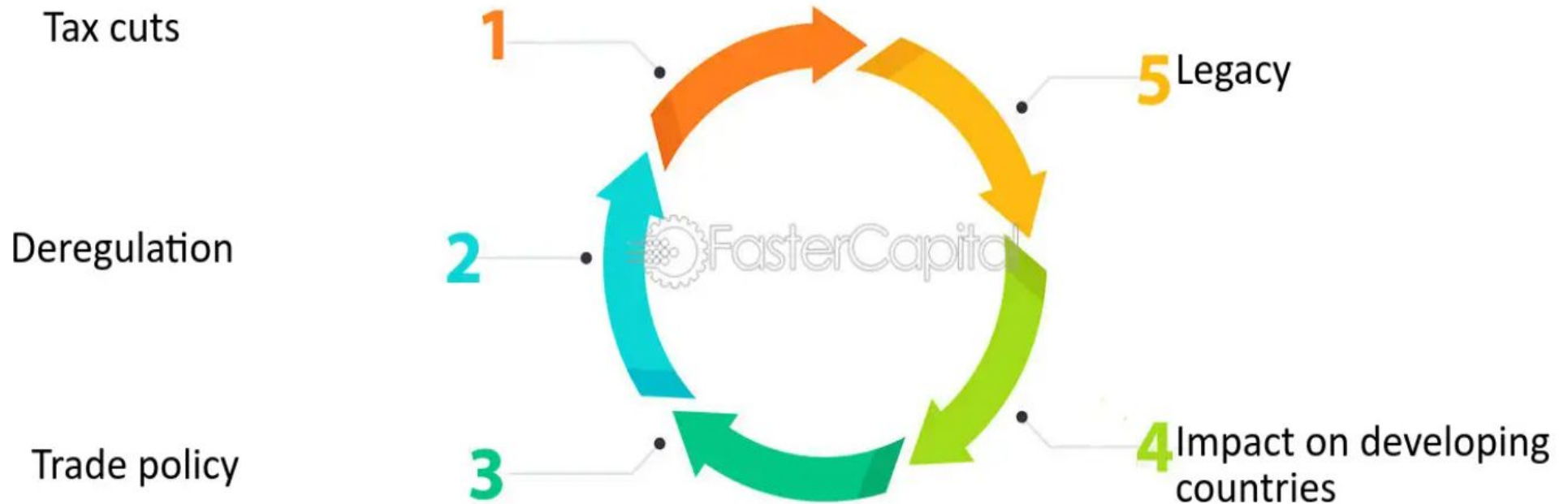


Source: WallStreetMojo

U.S. Trade Deficit After Bretton Wood

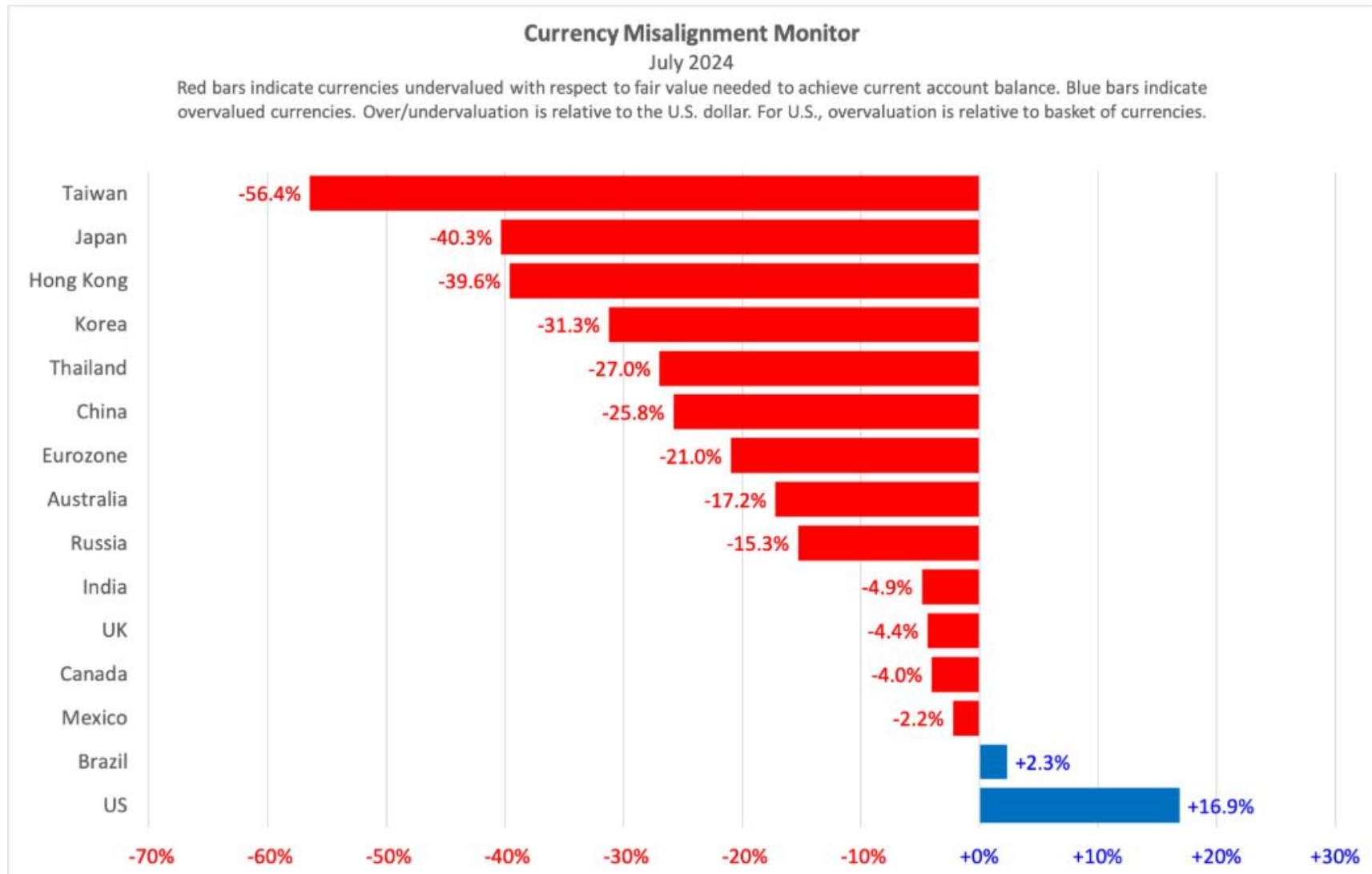
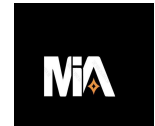


Reaganomics and the Global Economy



1. The relevance of Reaganomics in the 21st century: Reaganomics is still relevant today, as many policymakers continue to advocate for supply-side economics. The idea that lower taxes and less regulation lead to economic growth is still a widely debated topic. However, some argue that the [current economic climate](#) is different from the 1980s, and that Reaganomics may not be as effective in today's globalized economy.
 2. Lessons learned from Reaganomics: One important lesson learned from Reaganomics is that tax cuts must be accompanied by spending cuts to avoid increasing the national debt. Another lesson is that the benefits of [economic growth must be shared](#) more equitably to avoid income inequality. For example, during the Reagan era, the top 1% of earners saw their incomes increase by 10%, while the bottom 20% saw their incomes decrease by 4%.
 3. The impact of Reaganomics on the economy: Reaganomics had a significant impact on the US economy, leading to a [period of economic growth](#) and job creation. However, it also led to an increase in income inequality and a growing national debt. Today, policymakers continue to debate the merits of supply-side economics and the best ways to promote economic growth while ensuring that the benefits are shared by all.
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U.S. Dollar Overvalued as the World Reserve Currency



KEY TAKEAWAYS

- Robert Triffin believed the dollar could not survive as the world's reserve currency without requiring the United States to run ever-growing deficits.
 - A popular reserve currency lifts its exchange rate, which hurts the currency-issuing country's exports, leading to a trade deficit.
 - A country that issues a reserve currency must balance its interests with the responsibility to make monetary decisions that benefit other countries.
 - Another reserve currency replacing the dollar would increase borrowing costs, which could impact the United States' ability to repay debt.
 - A new international monetary system could potentially help countries maintain a reserve currency status.
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Discussion



Attendance

